AUTOMATED CI/CD PIPELINE WITH AWS CODEPIPELINE

Overview:

To automate the build, test and deployment of an application using CI/CD pipeline with AWS codepipeline.

Description:

- Using the **CI/CD pipeline with AWS codepipeline**, to build an application using **AWS codebuild**, and to built the code the code is taken from **AWS CodeCommit** which is a version control system and then the build code is stored in an **Artifact** which is Amazon S3, then the stored code is deployed on **AWS Codedeploy** which is Amazon EC2 and to build, test and deploy an application automatically the code is pushed to AWS Codepipeline in which CI/CD means Codecommit, Codebuild and Codedeploy is done automatically.

Prerequisites:

We must have some basic knowledge on:

1. CI/CD (Continuous Integration and Continuous Delivery) - A CI/CD pipeline is a set of automated processes that enable developers to continuously integrate and deliver code changes.

CI (Continuous Integration) - This extends CI by automatically testing and merging code changes into a shared repository multiple times a day, that includes

1. Code Commits : Developers commit code changes frequently.

2. Automated builds : The system automatically compiles the code to ensure it builds correctly.

3. Automated tests : The system runs tests on the build to ensure code quality and functionality.

- By adopting continuous Integration, is the best practice of automating the integration of code changes from multiple contributors into a single software project.
- It is a primary DevOps best practice, allowing developers to frequently merge code changes into a central repository where builds and tests then run.

CD (Continuous Delivery) - is a software development practice where code changes are automatically prepared for a release to production.

- It builds on continuous integration by deploying all the code changes to a testing environment and or a production environment after the build stage.
- This means that the code changes from different developers are automatically tested and deployed to ensure that they are ready to be released to user at any time, that includes

1. **Automation of Deployments:** Automated deployment pipelines ensure that every change is automatically built, tested, and prepared for a release.
2. **Frequent and Reliable Releases:** With CD, you can release software in shorter cycles, ensuring new features, updates, and fixes are delivered to users quickly and reliably.
3. **Feedback Loops:** Rapid feedback on the quality and stability of the software changes.
4. **Reduction of Risks:** Smaller, more frequent releases reduce the risk of large, disruptive deployments.

➢ By adopting continuous delivery, teams can ensure that their software is always in a deployable state, leading to faster and more reliable software releases.

- The CI/CD pipeline aims to reduce manual errors, provide feedback faster, and allow for more reliable and frequent releases, it consists of various stages such as source code control, built automation, automated testing and deployment.

2. AWS CodePipeline - AWS CodePipeline is a fully managed continuous integration and continuous delivery (CI/CD) service that enables you to automate your release pipelines for fast and reliable application and infrastructure updates. It allows you to model, visualize, and automate the steps required to release your software, enabling rapid and continuous delivery.

Features of AWS CodePipeline:

1.Automated Workflow: AWS CodePipeline automates the steps required to release your software, from source code to production. This includes building, testing, and deploying applications.

2.Integration with AWS Services: Seamlessly integrates with other AWS services such as AWS CodeBuild, AWS CodeDeploy, and AWS Lambda, as well as third-party tools like GitHub, Jenkins, and Bitbucket.

3.Customizable Pipelines: Define custom workflows with various stages and actions, enabling you to create pipelines that meet your specific requirements.

4.Scalability: As a fully managed service, AWS CodePipeline automatically scales to handle the number of pipelines and the volume of changes, ensuring that your release process can grow with your development needs.

5.Real-time Updates: Provides real-time status updates for each stage of the pipeline, allowing you to monitor and manage the release process effectively.

6.Security and Compliance: Integrates with AWS Identity and Access Management (IAM) to provide fine-grained access control and compliance with industry standards and best practices.

3. AWS CodeCommit – AWS Codecommit is a Version control service, which is fully managed by AWS cloud service provider, which is a secure, highly scalable, managed source control service that makes it easier for teams to collaborate on code. AWS CodeCommit eliminates the need for you to operate your own source control system or worry about scaling its infrastructure.

4. AWS CodeBuild – AWS CodeBuild is a Code building service in the cloud, it allows users to build code in the cloud, which is a fully managed Continuous Integration service that compiles the source code, runs tests and produces software packages that are ready to deploy and CodeBuild eliminates the need to provision, manage, and scale your own build servers.

5. AWS CodeArtifact - AWS CodeArtifact is a fully managed artifact repository service that makes it easy for organizations to securely store, publish, and share software packages used in their software development process.

6. Amazon S3(Simple Storage Service) – Amazon S3 which is a simple storage service which is provided by Amazon Web Service which is a Cloud service provider, S3 is a Scalable storage service which is object level storage that offers

scalability, data availability, security, and performance. We can use Amazon S3 to store and retrieve any amount of data at any time, from anywhere.

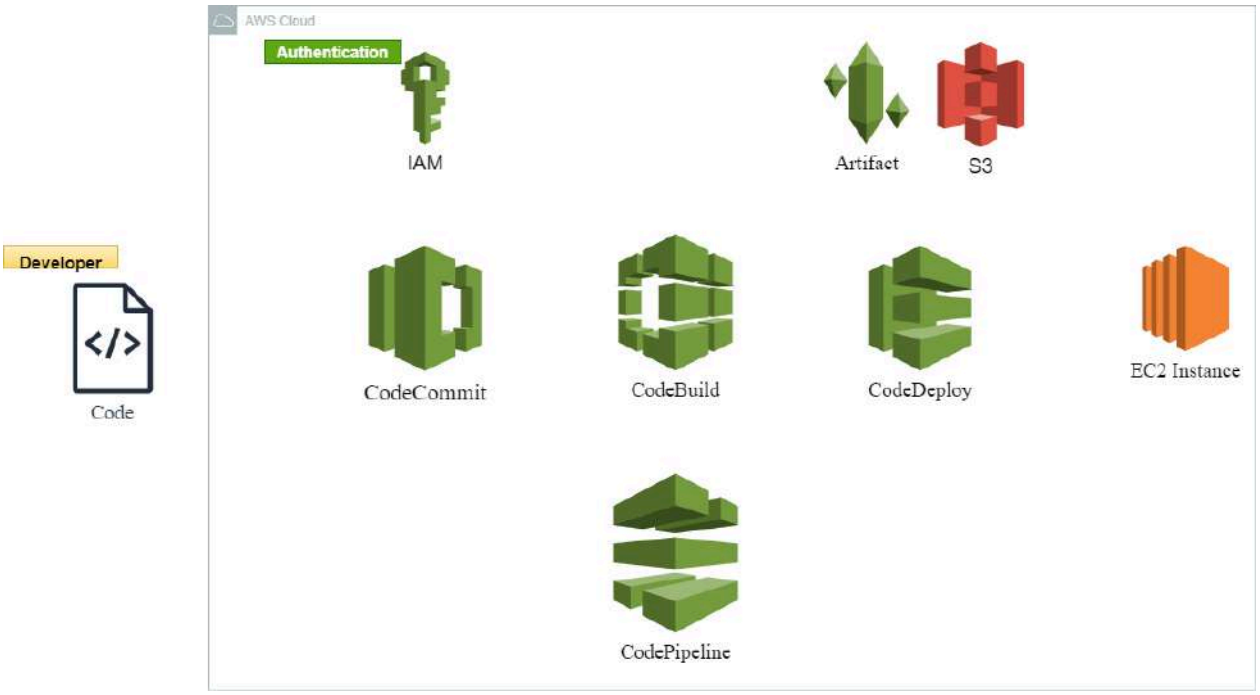
7. AWS CodeDeploy - AWS CodeDeploy is a deployment service that automates the deployment of applications to various computing services such as EC2 Instance, ECS (Elastic Container Service), AWS Lambda and even on premises servers.

AWS CodeDeploy makes it easier for you to rapidly release new features, helps you avoid downtime during deployment, and handles the complexity of updating your applications.

8. Amazon EC2 (Elastic Compute Cloud) - Web service which provides scalable computing capacity in the cloud.

Instance - Virtual computing environments.

Architecture:



Data Flow:

a. Firstly, a user must create an IAM USER from AWS ROOT ACCOUNT , because, IAM which is an Authentication and Authorization service provided by AWS that is an identity created within an AWS account that has permission to interact with AWS resources.

b. When a Developer writes the code the code is pushed to AWS CodeCommit which is a version control system AWS CodeCommit like GITHUB for AWS.

c. Then from CodeCommit the code is pushed to AWS CodeBuild in which committed code is built.

d. The build code is packed and stored in AWS CodeArtifact which is a repository service that makes it easy for organizations to securely store, publish, and share software packages used in their software development process such as Amazon S3.

e. After storing the code the code must be deployed to make sure that the application is reached to all the users who wanted to access it, to do that the code is deployed on AWS CodeDeploy such as EC2 Instance in this case.

f. If we use AWS CodePipeline, every stage is done automatically i.e codecommit, codebuild, code artifact and codedeploy are these stages of CI/CD is done automatically.

g. AWS CodePipeline is a continuous delivery service you can use to model, visualize, and automate the steps required to release your software.

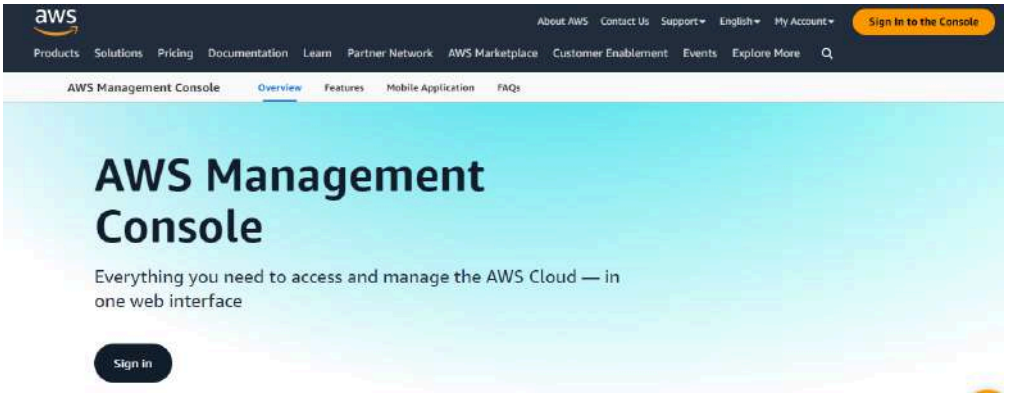
h. You can quickly model and configure the different stages of a software release process.

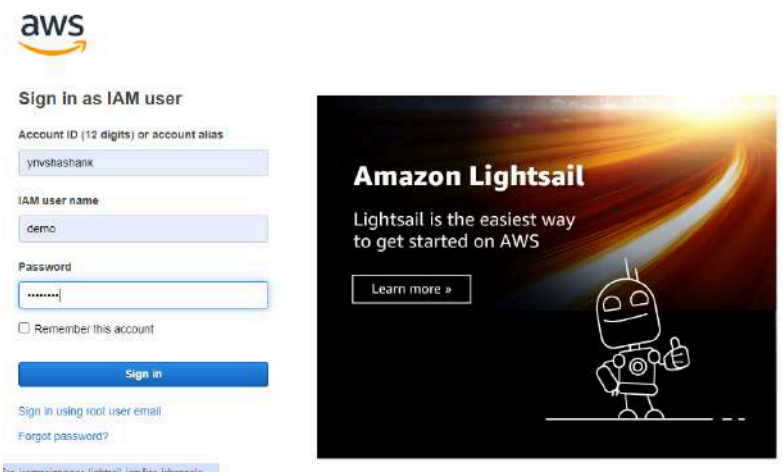
i. CodePipeline automates the steps required to release your software changes continuously.

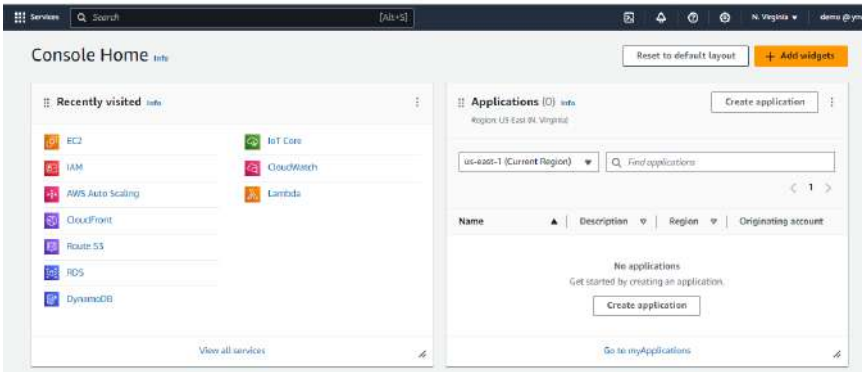
Steps:

Step 1: Login into AWS Console

1.1 Create an account and login into it by providing the credentials.

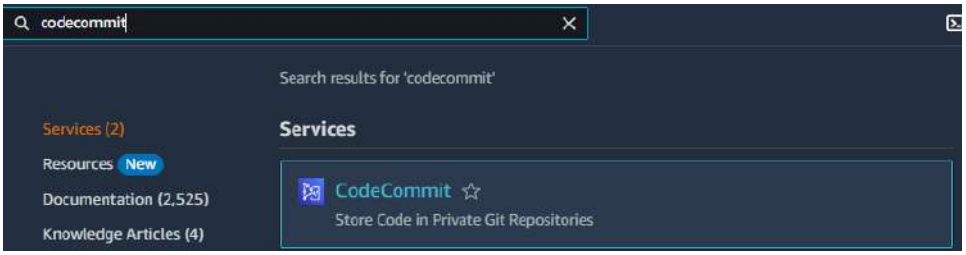




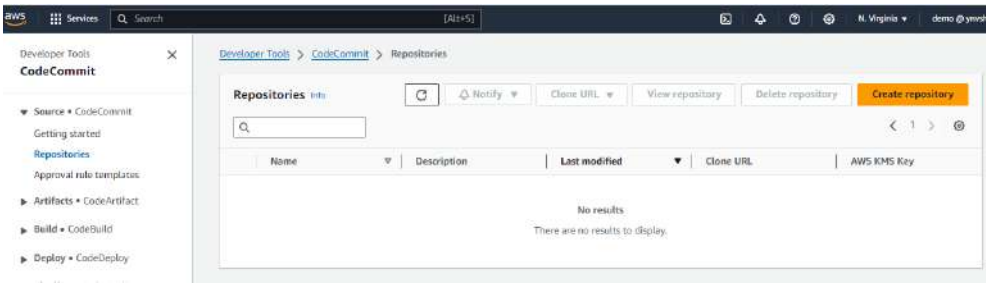


Step 2: To manage the source code using the AWS CodeCommit service in Console

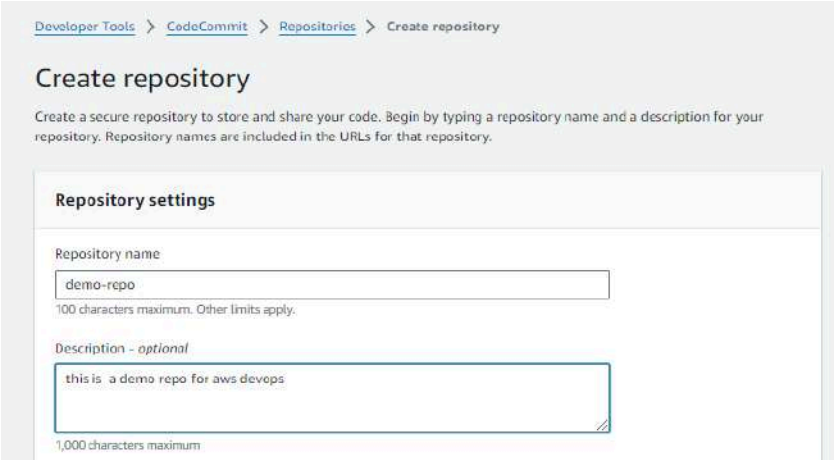
2.1 In Dashboard, search for **AWS CodeCommit,** click on **CodeCommit** to open it.



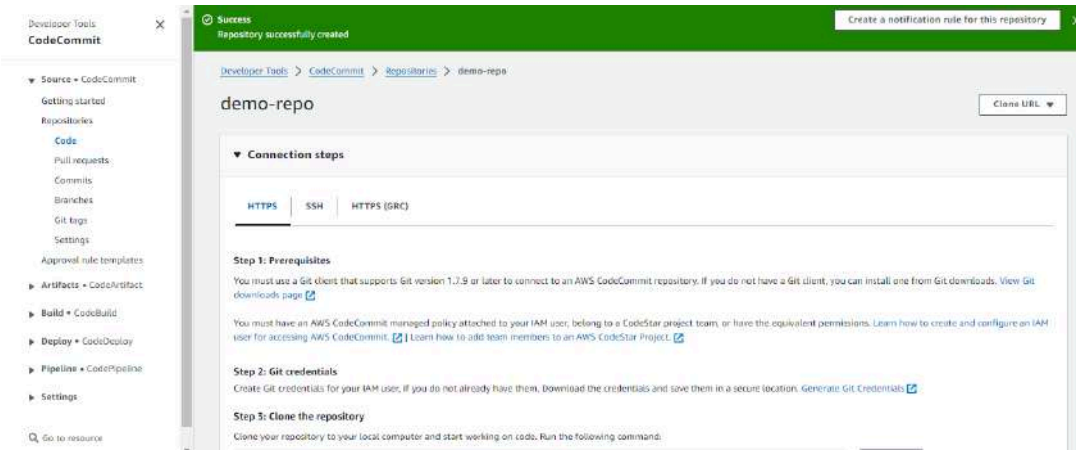
2.2 Click on **Repository**, then click on **Create repository** to configure.



2.3 Type the name of the repository in the **Repository name** which you prefer for naming the codecommit repository and then give any **Description** to it and then click on **Create**.

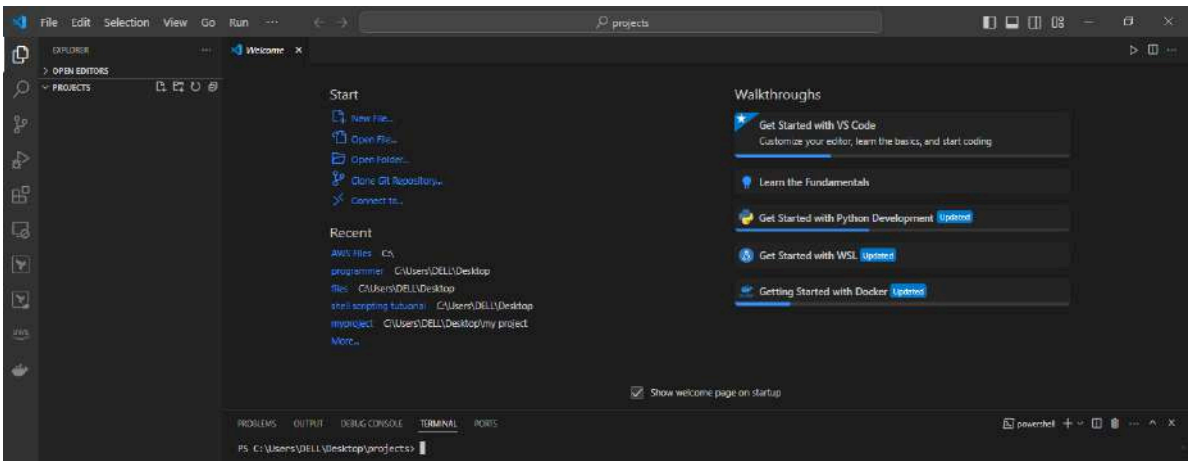


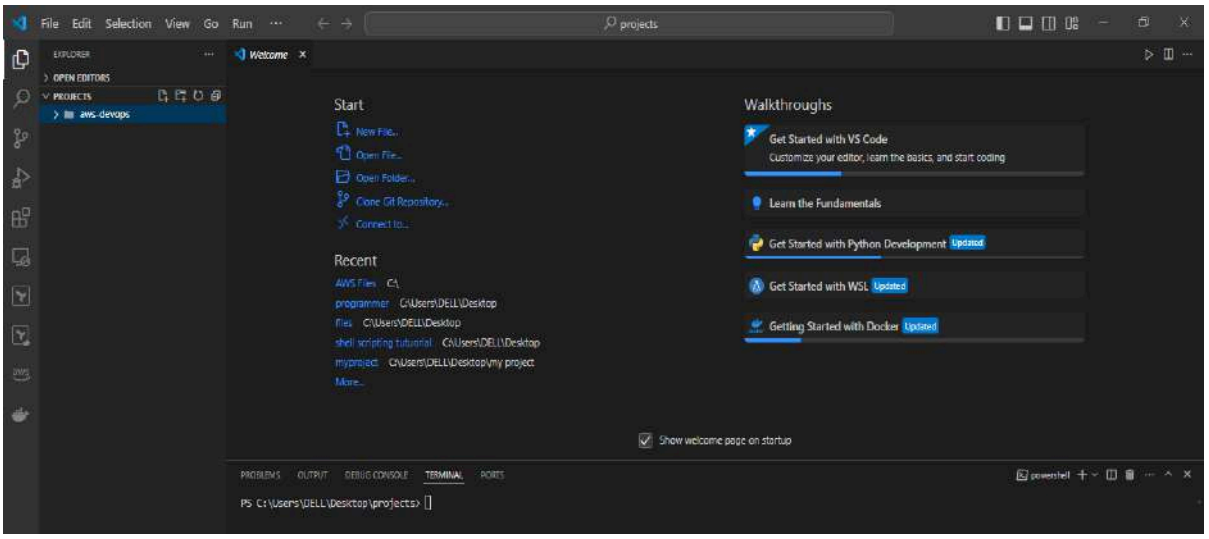
2.4 After **Repository successfully created**, Click On Clone URL, then Click on **Clone HTTPS,** then a **URL** will be generated which is **git-codecommit repo,** which is used to clone the repository locally.

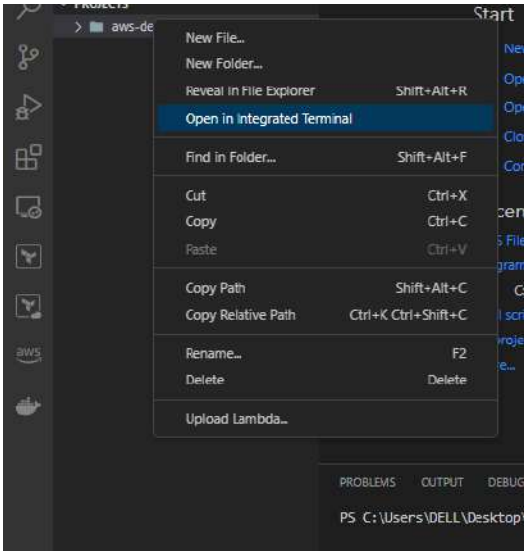


Step 3: managing the source code with AWS CodeCommit

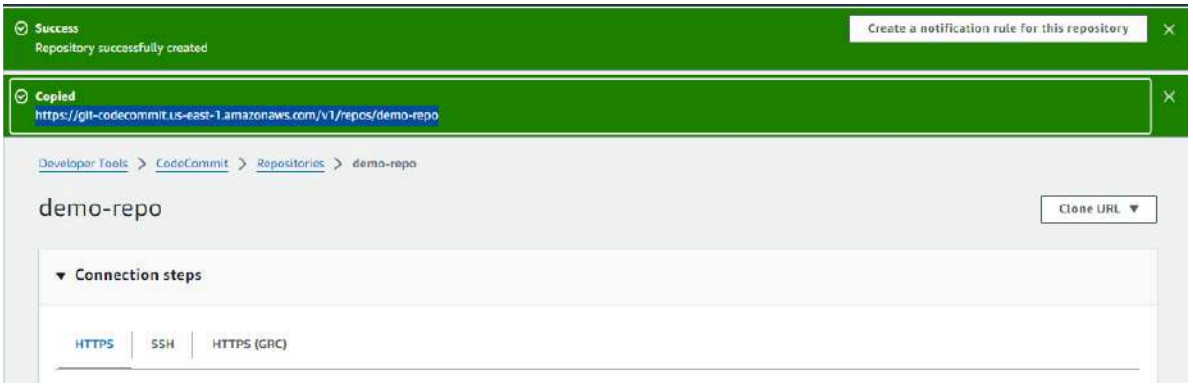
3.1 To write or to build the code for an application use any terminal, in my case I am using **VS Code**, open the VS Code and create a folder in it, and name any for it, in my case **AWS-DEVOPS** and open the created folder as **open as the integrated terminal.**





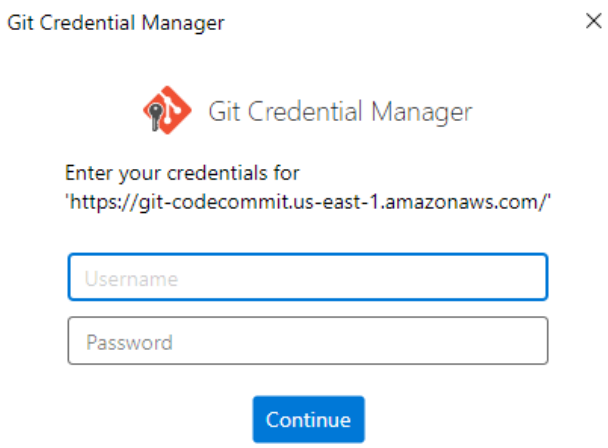


3.2 To integrate your repository, clone the aws-codecommit-repository using the URL which is generated and Clone that URL in the VS Code then Git Credential Manager will be popping up.



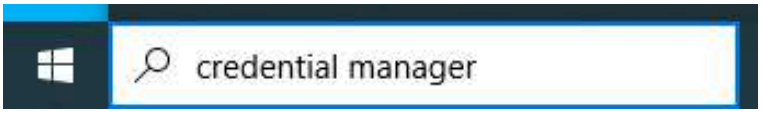

```
PS C:\Users\DELL\Desktop\projects\aws-devops> git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
```

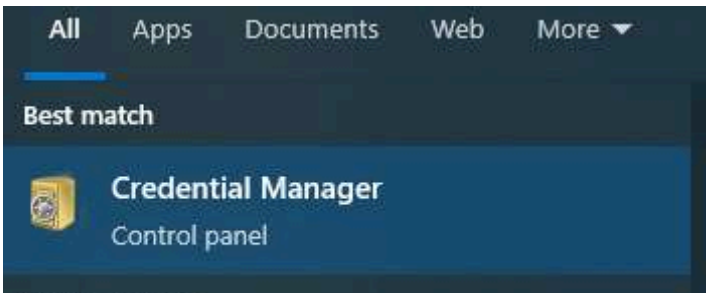
```
PS C:\Users\DELL\Desktop\projects\aws-devops> git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
Cloning into 'demo-repo'...
```

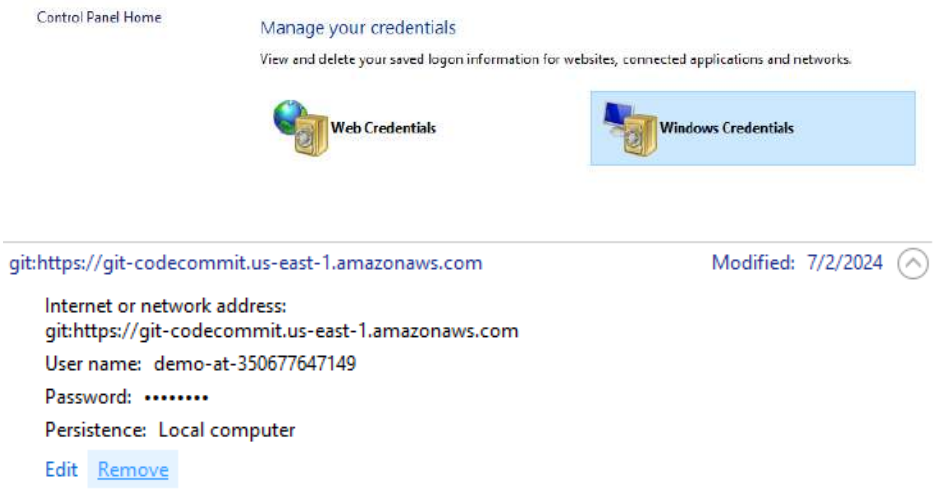


Troubleshooting:

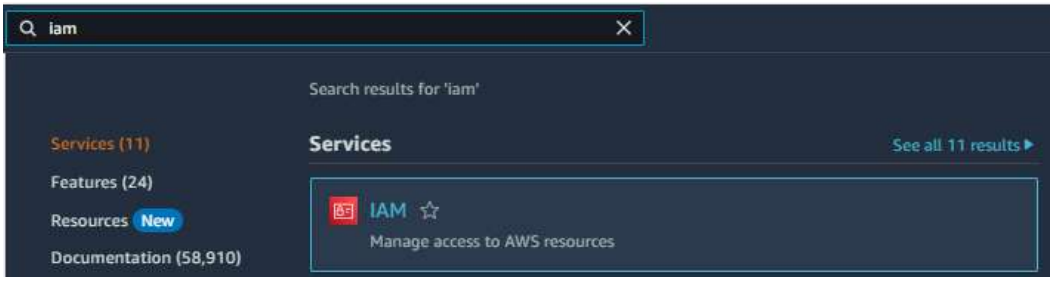
- If **Git Credential Manager** is not popping up , then remove the previous Git credentials by searching **Credential manager** on your system windows search bar, then Click on **Credential Manager** and then select **windows credentials** and remove the earlier **Git Credentials**, so then **Git Credential Manager** will be popping up.

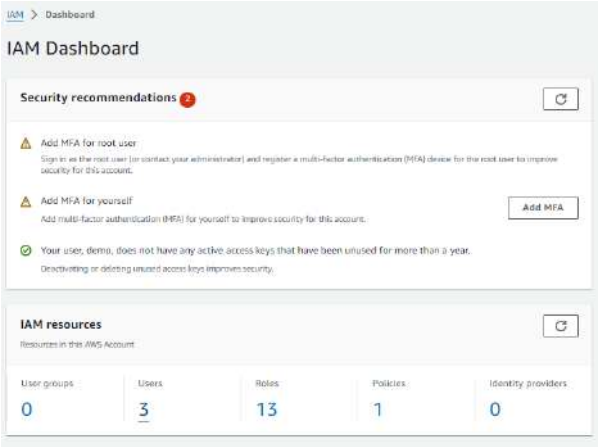




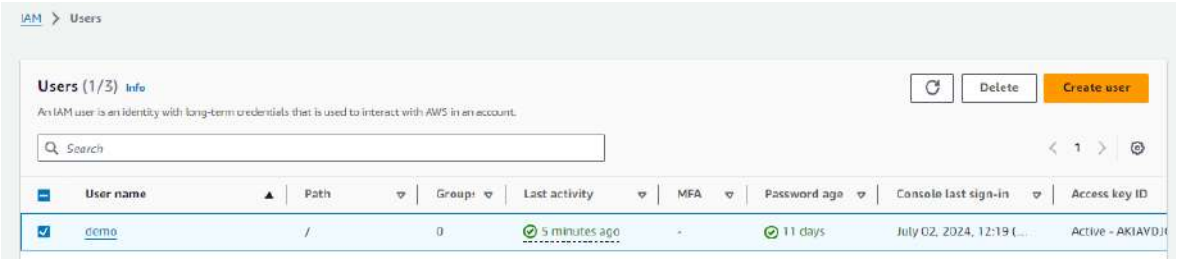


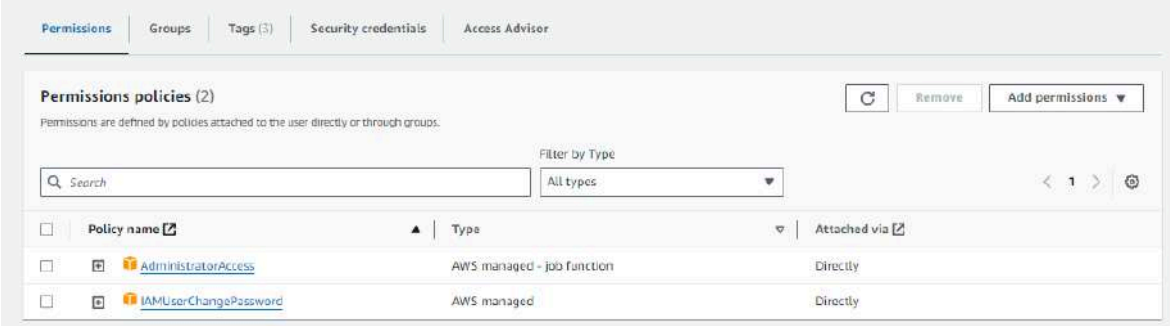
3.3 To access Git, firstly we have to have permission to access it, for that we have to attach **aws codecommit full access** policy for the user and then only we have to generate the credentials which are Username and password for this search IAM in the AWS console, then the IAM Dashboard will be opened , now choose Users.

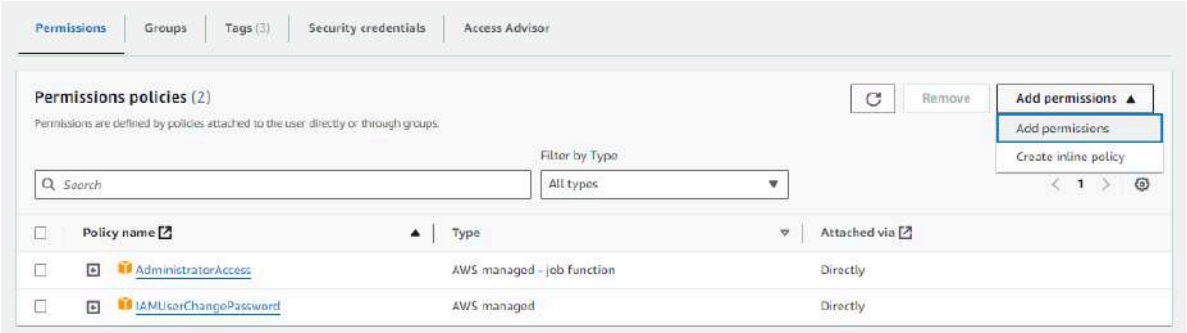


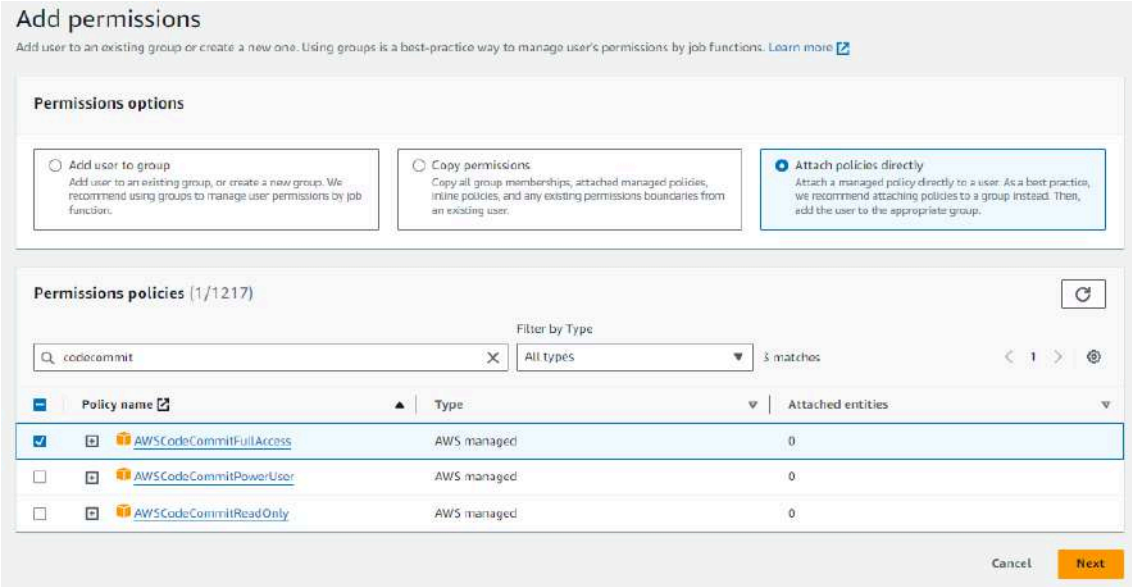


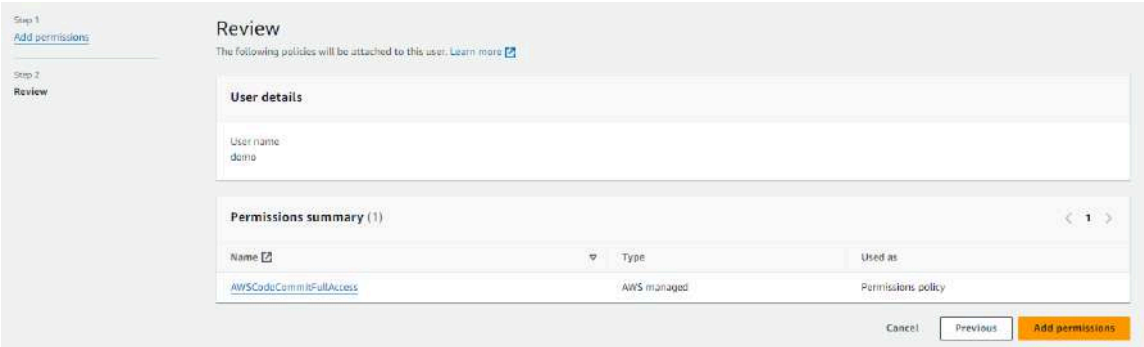
3.4 Now click on the **user** name which you are using and then go to **Permissions**, now click on **add permissions** in Permissions Option select **Attach Policy Directly** and in **Permissions Policies** section search for **AWSCodecommitFullAccess** and then click on **Next** and now Click on **Add Permissions** then you will be notifying **1 Policy added**.











3.5 Now we have given permission to access the AWS CodeCommit Repository by attaching the policy, then we have to create credential to access AWS CodeCommit Repository with the help of git for that we need go to IAM, now click on the **user** name which you are using and then go to Security credentials and go to **HTTPS Git credentials for AWS CodeCommit,** now click on **Generate Credentials** and then **Download credentials** or copy and paste on notepad and then Enter the **username** and **password** in the **Git Credential Manager**, then your AWSCodeCommit will be now cloned locally.

IAM > Dashboard

IAM Dashboard

Security recommendations 2

Add MFA for root user
Sign in as the root user (or contact your administrator) and register a multi-factor authentication (MFA) device for the root user to improve security for this account.

Add MFA for yourself
Add multi-factor authentication (MFA) for yourself to improve security for this account.

Add MFA

Your user, demo, does not have any active access keys that have been unused for more than a year.
Deactivating or deleting unused access keys improves security.

IAM resources
Resources in this AWS Account

User groups	Users	Roles	Policies	Identity providers
0	3	13	1	0

Users (1/3) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Delete | Create user

Search

User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
demo	/	0	7 minutes ago	-	12 days	July 02, 2024, 12:19 (...	Active - AKIAVDJ

Permissions | Groups | Tags (3) | Security credentials | Access Advisor

HTTPS Git credentials for AWS CodeCommit (0)

Actions | Generate credentials

Generate a user name and password you can use to authenticate HTTPS connections to AWS CodeCommit repositories. You can have a maximum of 2 sets of credentials (active or inactive) at a time. Learn more

User name	Created	Status
	No credentials Generate credentials	

Generate credentials

Your new credentials are available.

Save your user name and password or download the credentials file.

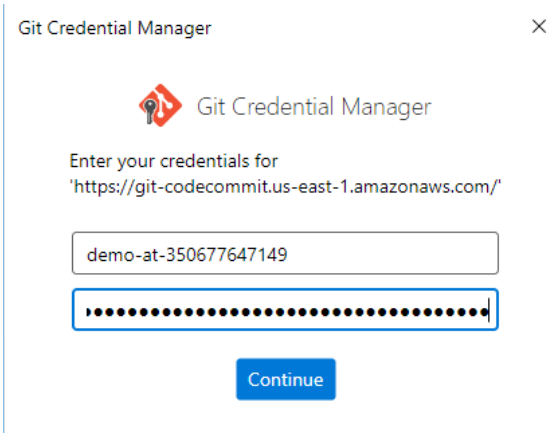
This is the only time you can view the password or download it. You cannot recover it later. However, you can reset your password at any time.

You can use these credentials when connecting from your local computer, or from tools that require a static user name and password. Learn more

User name
demo-at-350677647149

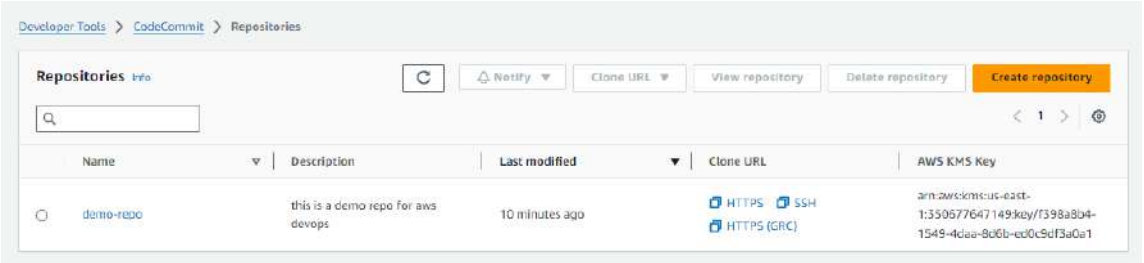
Password
********** Show

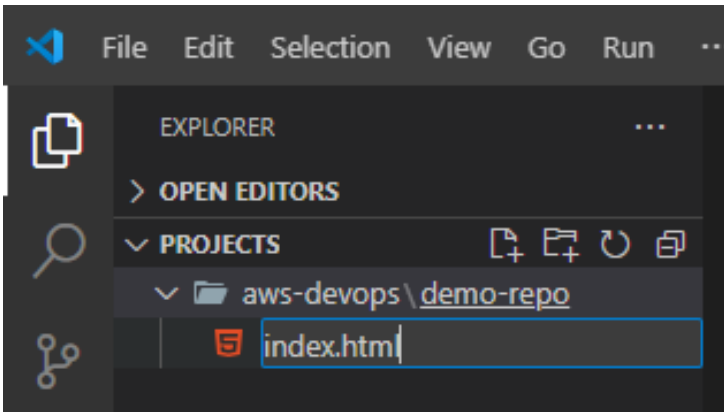
Download credentials | Close

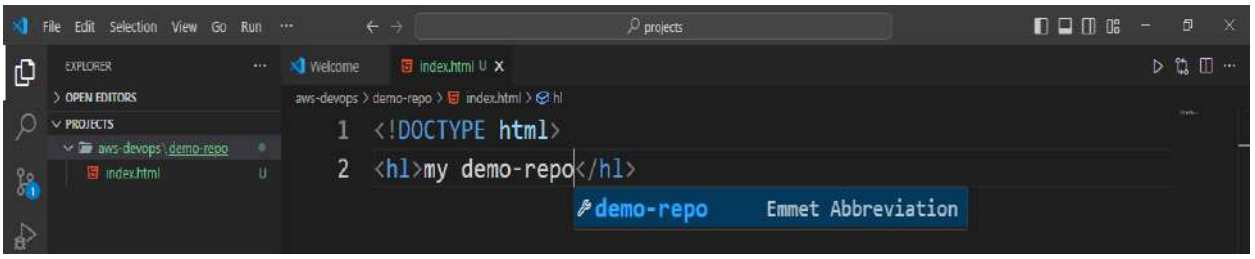



```
PS C:\Users\DELL\Desktop\projects\aws-devops> git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
Cloning into 'demo-repo'...
warning: You appear to have cloned an empty repository.
PS C:\Users\DELL\Desktop\projects\aws-devops>
```

3.6 Now create an index.html file in the repository (demo-repo) and write something in the index.html file.







3.7 Now you have to use some Git commands in the terminal.

- Basic Git Workflow Steps:

1.Create a Repository (Repo):

➢ What it is: A repository is like a folder where all our project files and the history of changes are stored.

How to do it:

[git init]

➢ The above command sets up a new Git repository.

2.Make Changes to Your Files:

➢ What it is: You write or edit your project files, like adding code or removing code.
➢ How to do it: Just use your favorite code editor to make changes(vs code for example)

3.Check the Status:

➢ What it is: This lets you see what changes you've made since your last save.

[git status]

➢ The above command shows which files have been changed or new.

4.Add Changes:

➢ What it is: where you gather all the changes you want to save.

[git add filename]

➢ Replace filename with the name of the file you changed.

NOTE: To add all changes, use:

[git add .]

5.Commit Your Changes:

➢ What it is: Committing is like writing a note in your notebook about the changes you made. Each commit saves a snapshot of your project.

[git commit -m "Write a short message about what you changed"]

6.Push Changes to a Remote Repository:

- What it is: If you're working with others or want to back up your work online, you push your changes to a remote repository (like on GitHub).

[git push origin main]

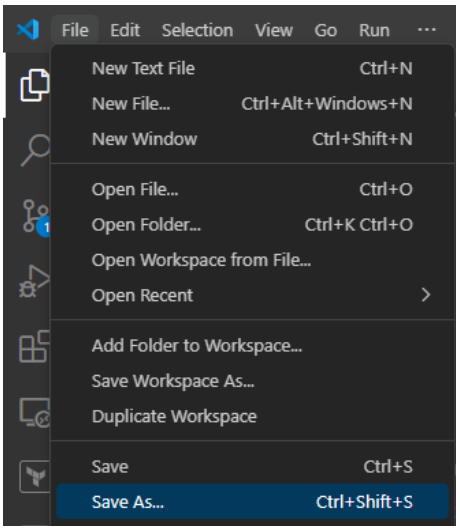
- 'origin' is the name of the remote repository.
- 'main' is the name of the branch you're working on.

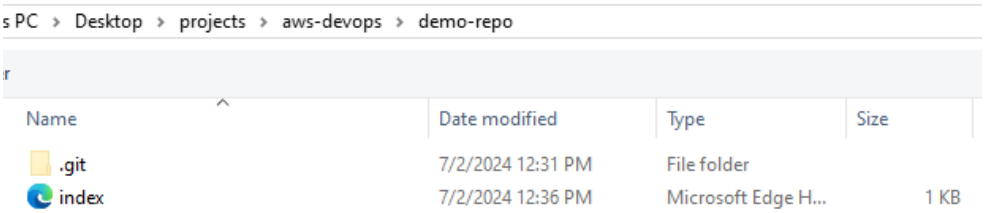
1. Git init

```
PS C:\Users\DELL\Desktop\projects\aws-devops> git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
Cloning into 'demo-repo'...
warning: You appear to have cloned an empty repository.
PS C:\Users\DELL\Desktop\projects\aws-devops> git init
```

```
PS C:\Users\DELL\Desktop\projects\aws-devops> git init
Initialized empty Git repository in C:/Users/DELL/Desktop/projects/aws-devops/.git/
PS C:\Users\DELL\Desktop\projects\aws-devops>
```

1. After Git init, you have save the code in the **.git** folder.





- **ls** is used to list the files in the repository.

```
PS C:\Users\DELL\Desktop\projects\aws-devops> ls


    Directory: C:\Users\DELL\Desktop\projects\aws-devops


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          7/2/2024  12:33 PM                demo-repo


PS C:\Users\DELL\Desktop\projects\aws-devops>
```

3.8 Now you have to change the folder to aws-repo in which we have written an intex.html file.

```
PS C:\Users\DELL\Desktop\projects\aws-devops> cd demo-repo
```

- After changing the folder do **ls,** which is used to list the files in the repository.

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> ls


    Directory: C:\Users\DELL\Desktop\projects\aws-devops\demo-repo


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          7/2/2024  12:36 PM             38 index.html


PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

- To check the status use Git Status.

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

- You can see no commits yet, that means we have not added or made any changes to the code.
- To see what changes you have made use **Git add .** , then check the status.

```
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git add .
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.html

PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

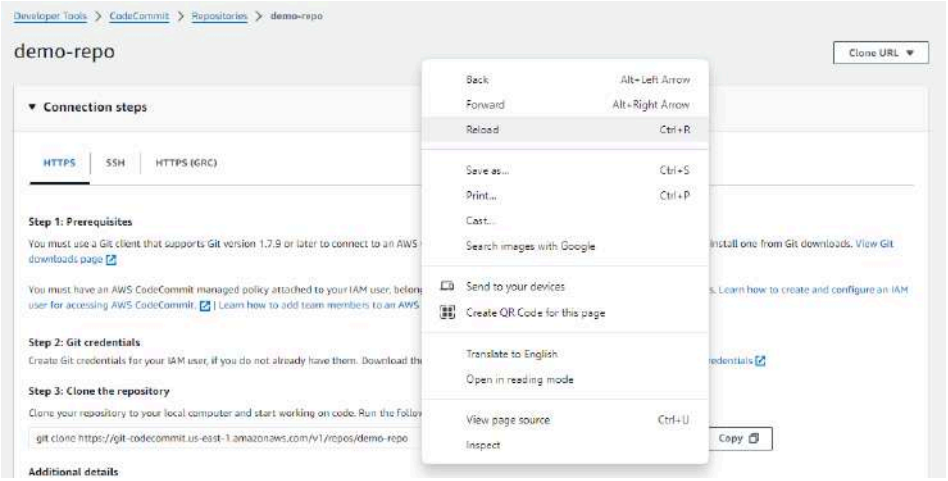
- The above Git status informs us that we have added the changes.
- After adding the changes which we had made, then you have to commit the changes, that means you have tell git that so and so changes we have done, for that use the command **[git commit -m "Write a short message about what you changed"]**

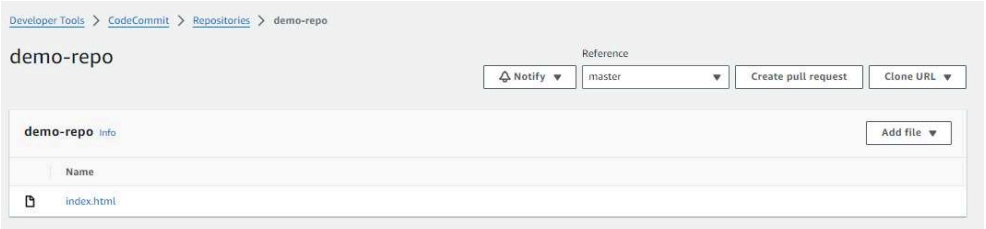
```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git commit -m "my first commit"
[master (root-commit) 876ad6f] my first commit
 1 file changed, 2 insertions(+)
 create mode 100644 index.html
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

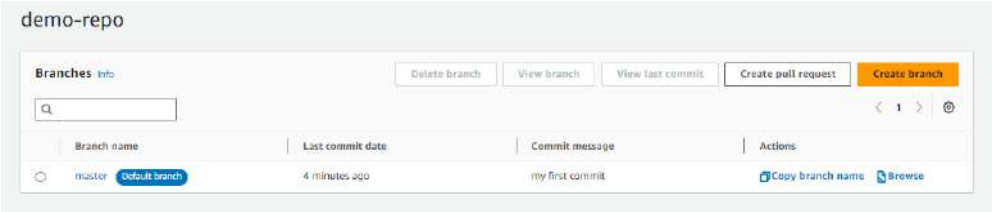
- After committing the changes we have to push the code to master branch to do that we have use command **[git push origin master]**

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git push origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 259 bytes | 86.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
 * [new branch]      master -> master
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

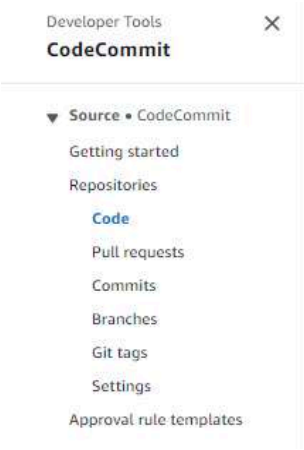
- Now after reloading the repository page, you see the index.html file in that, this is how we connect locally to AWS by using IAM CodeCommit.





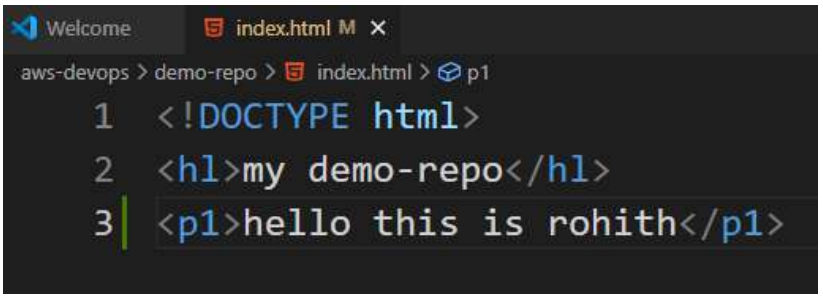


- Like Git, In AWS CodeCommit you will be having Code, Pull requests, Commits, Branches and so on.



3.9 We can also create another branch for example Dev Branch and we can write something in it and use git commands.

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git checkout -b dev
Switched to a new branch 'dev'
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

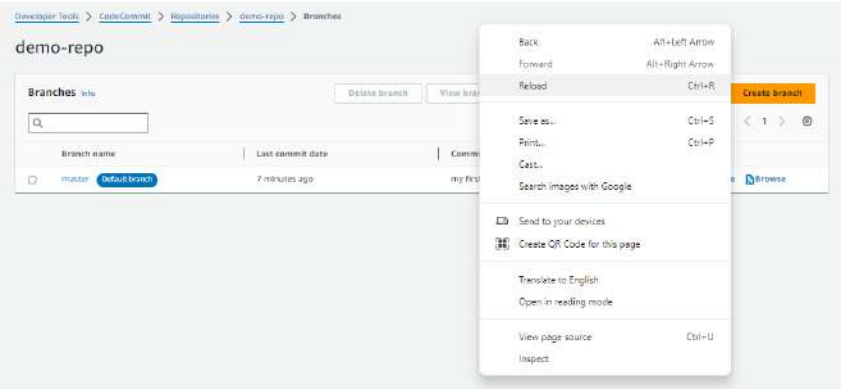


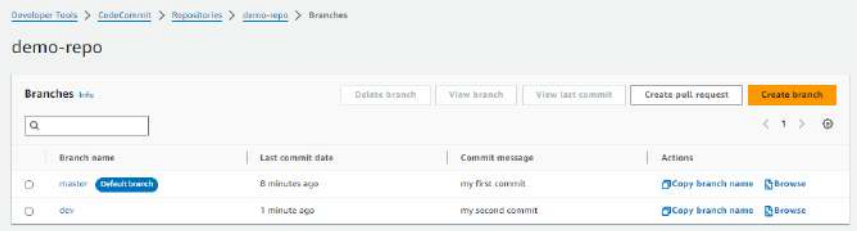

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git add .
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git commit -m "my second commit"
[dev 2632e3c] my second commit
 1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

- Now we have to push it into Dev Branch

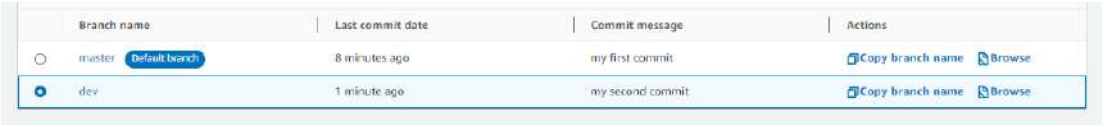
```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git push origin dev
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 105.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
 * [new branch]      dev -> dev
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

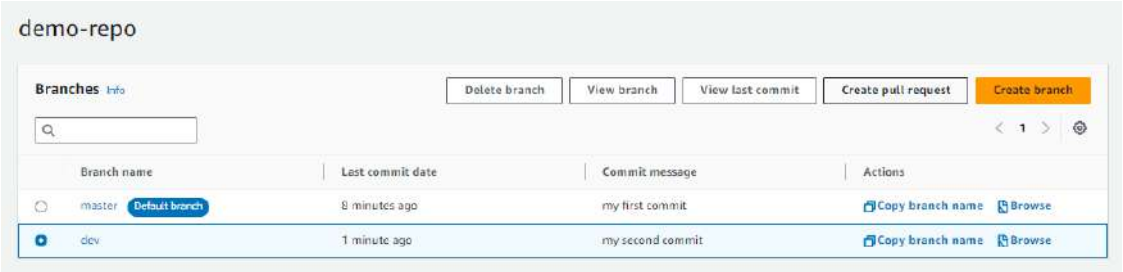
- Now after reloading the repository page, you can see the Dev Branch.



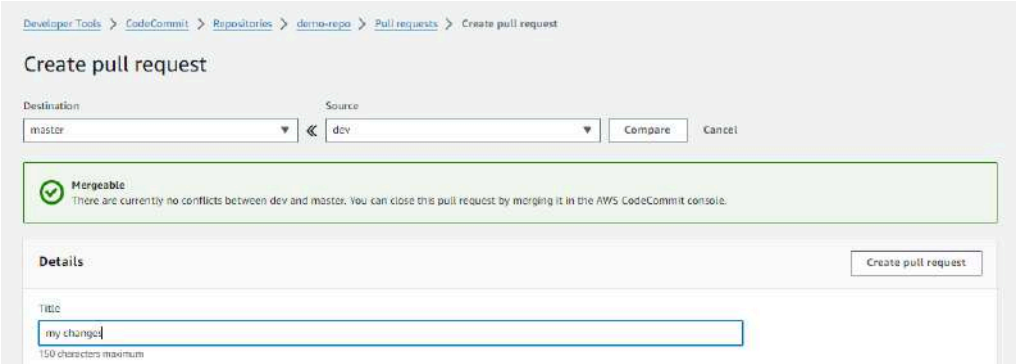


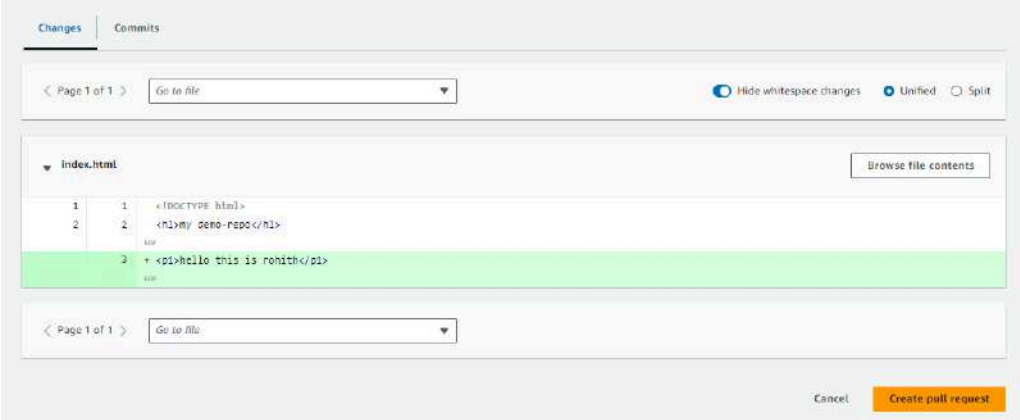
3.10 To merge both the branches(Master and Dev) create a pull request, for this Click on **Dev Branch** and then Click on **Create pull request**.

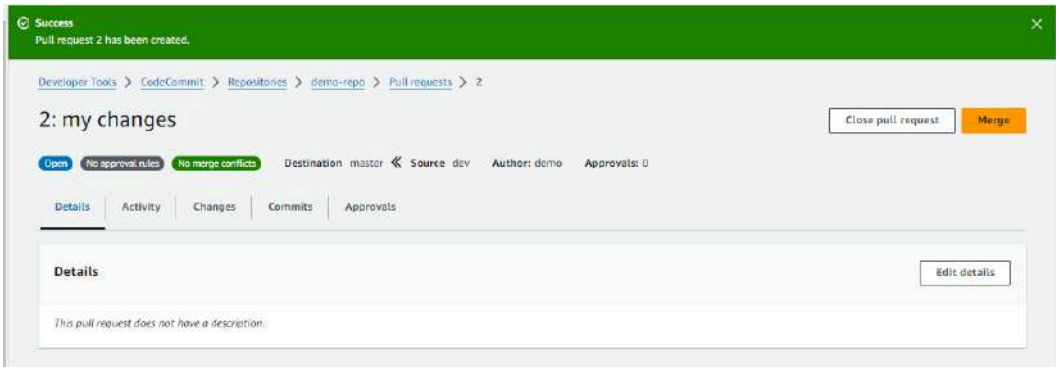


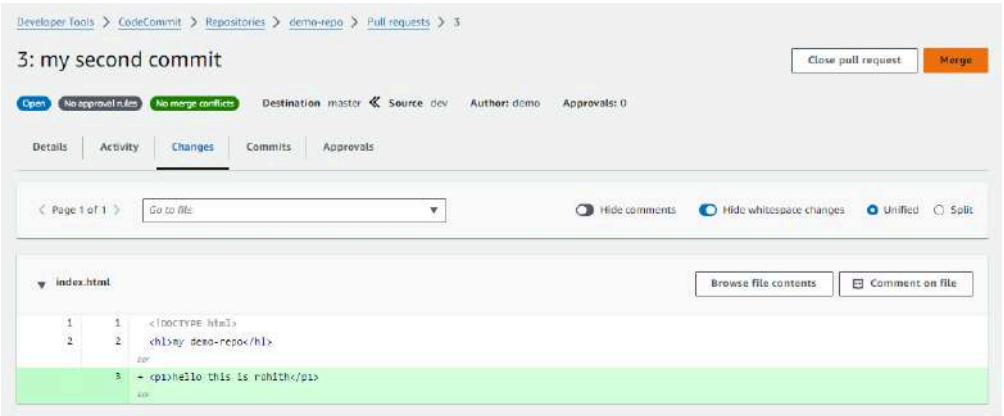


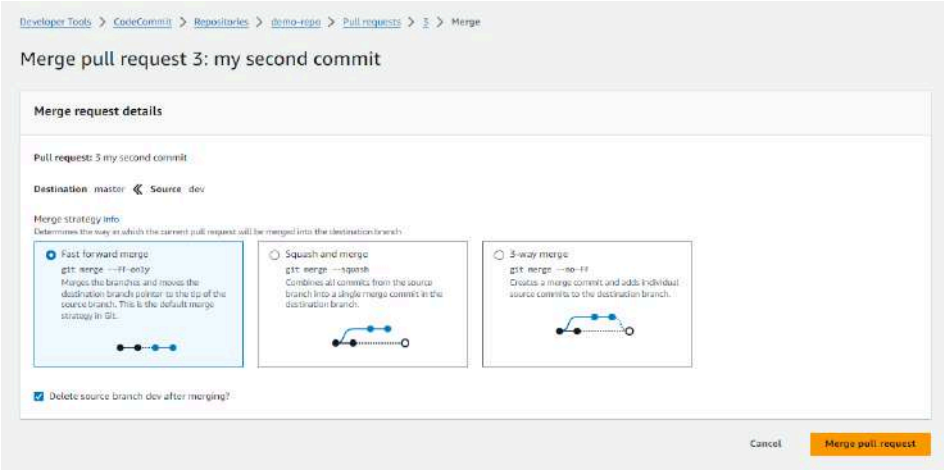
3.11 In Create pull request, give any Title to it and then Click on Create pull request, after pull request have been created, Click on merge to integrate both branches, select in what way you want to merge and Click on Merge pull request.



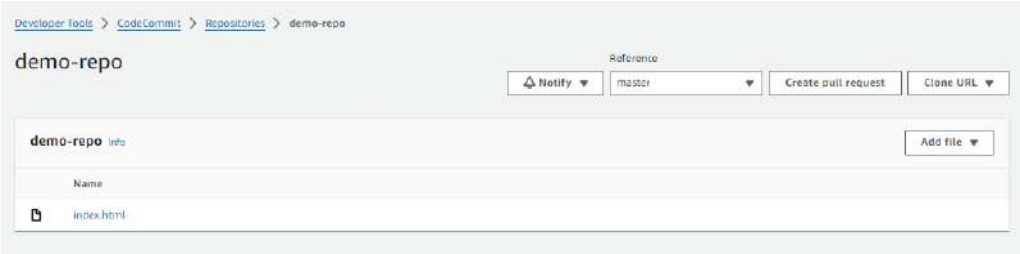




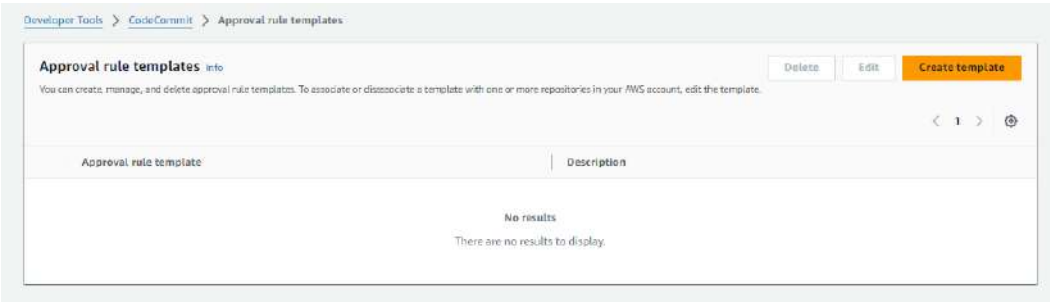


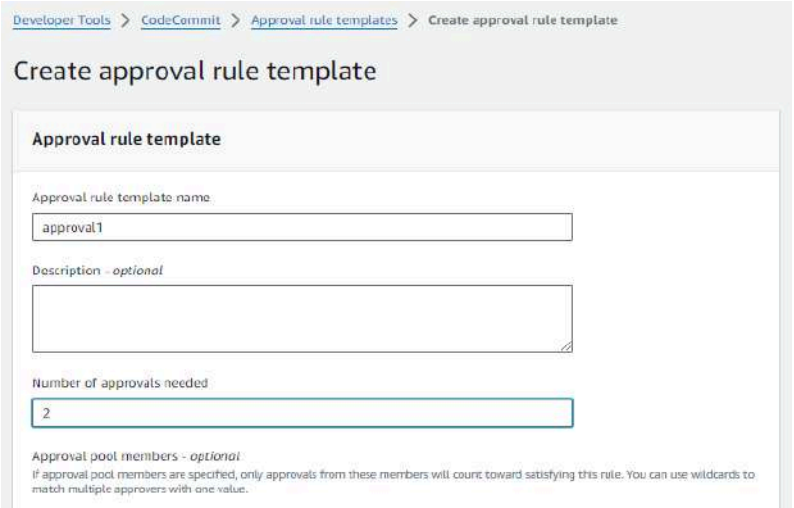


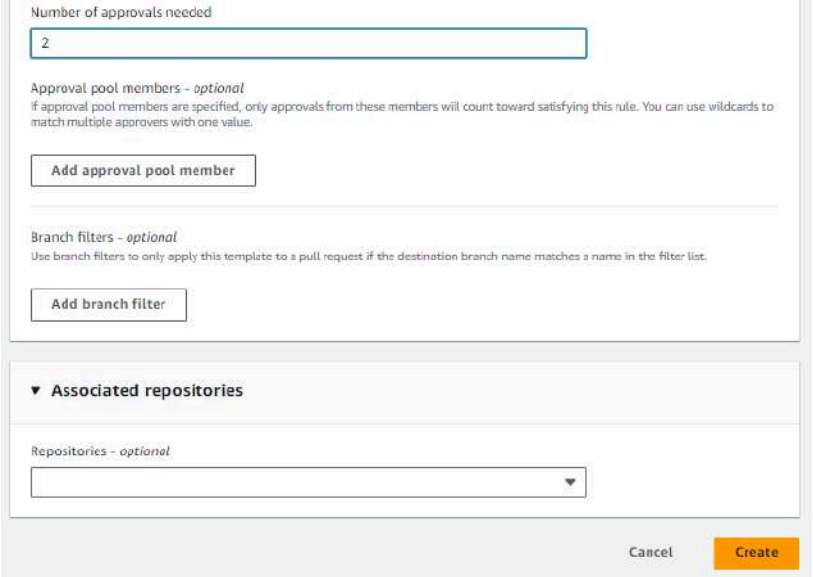
- To check the Merged Branches, Click on the Repository and open it.



NOTE: In this case you are only one who is building the code, if you want you to collaborate with other people and if you want their approval, for this you need Click on **Approval rule template** and Click on **Create template** then give n**umber of approvals needed** and then Click on **Create**.

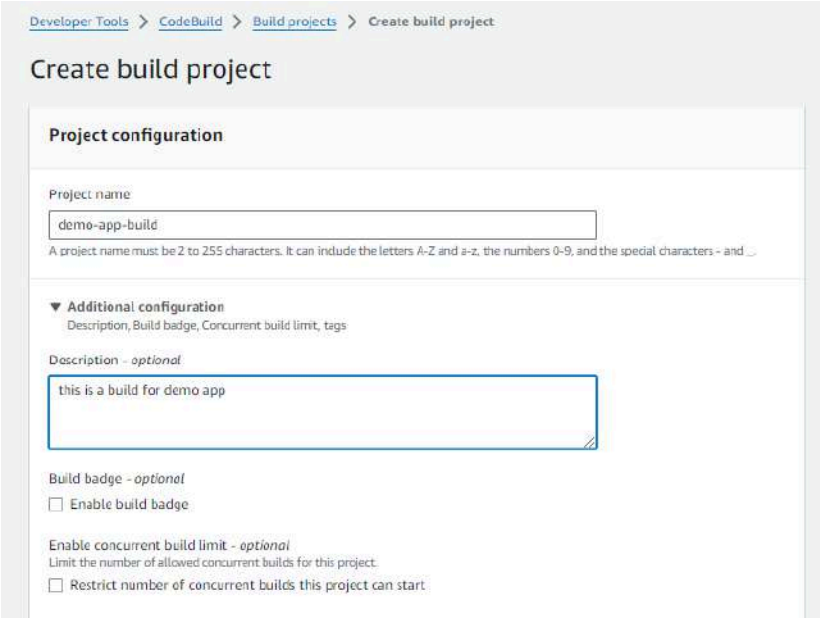




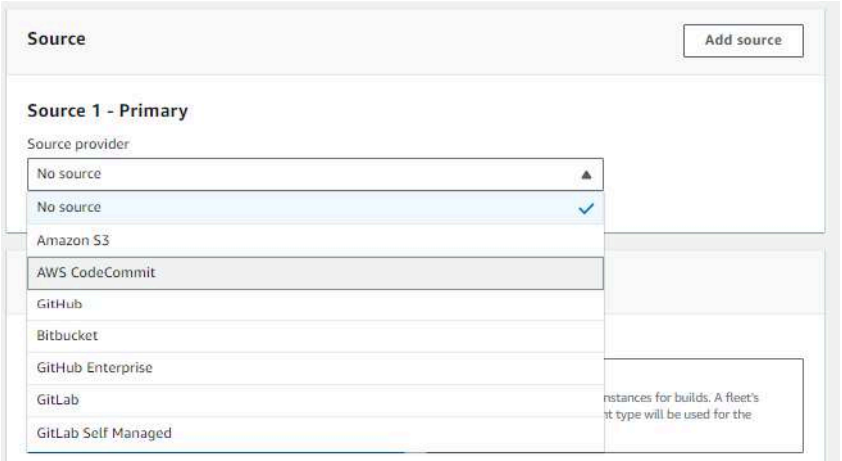


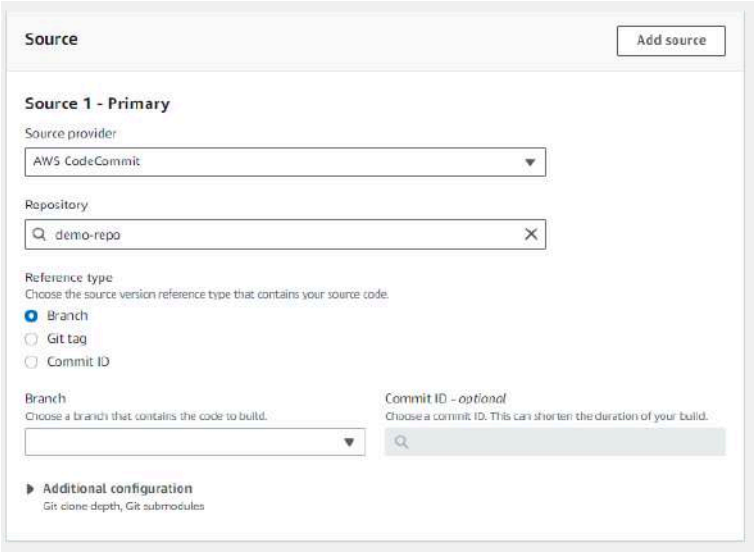
Step 4: Continuous Integration with AWS CodeBuild

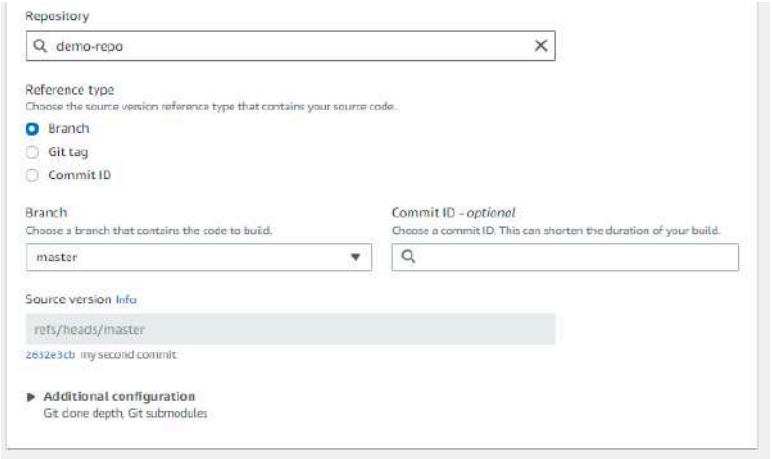
4.1 AWS CodeBuild is like Jenkins, Click on **CodeBuild,** then Choose **Build Projects,** now Click on **Create project,** give the project name.



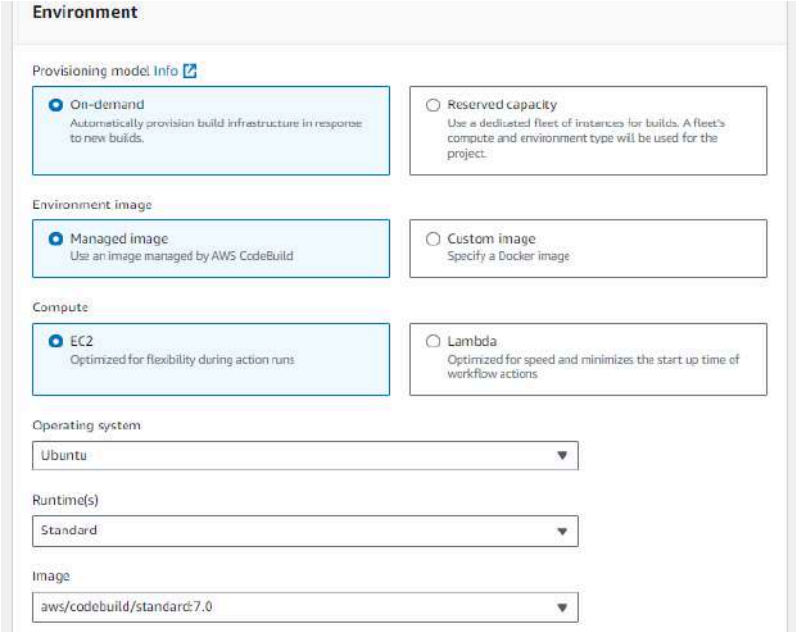
4.2 In Source Section, give the Source Provider, that means from where you are bringing code, i.e AWS CodeCommit and choose the Repository and Branch is Master Branch.



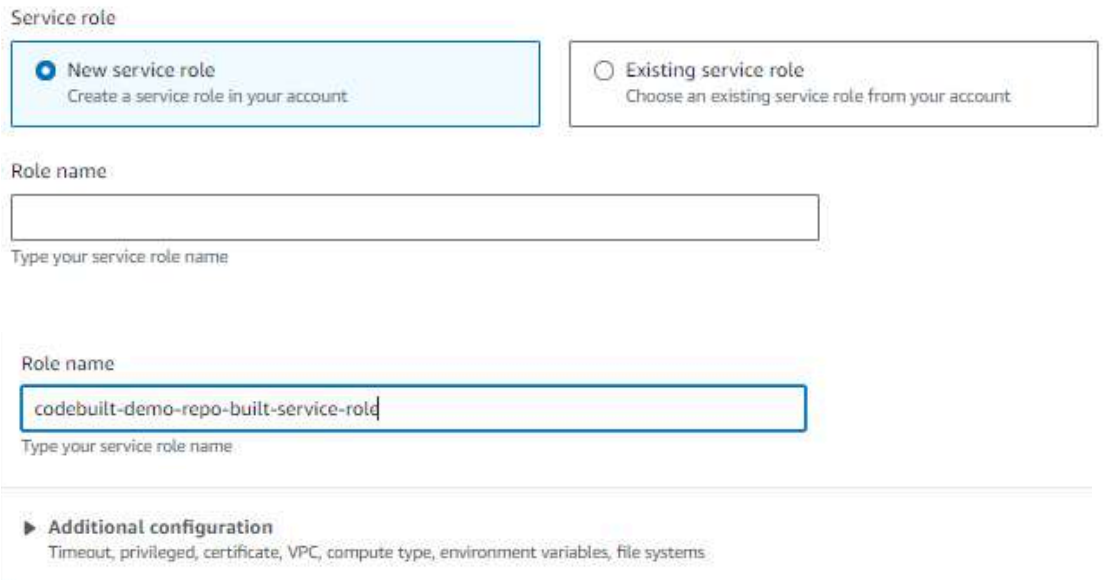




4.3 In the **Environment** Section, which is used to specify in which environment you want to build the code, such or **Linux/Ubuntu**, for this Choose **Managed Image** in **Environment image**, in **Operating system**, Choose what you want to choose, in my case i have chosen **Ubuntu**, choose **Runtime** as **Standard** and then Choose **Image** which is **latest image.**

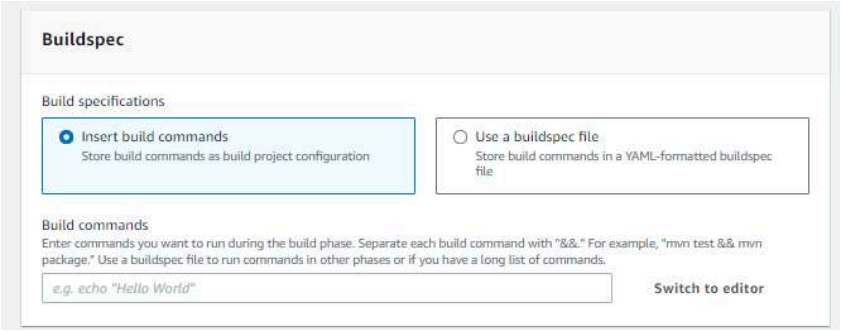


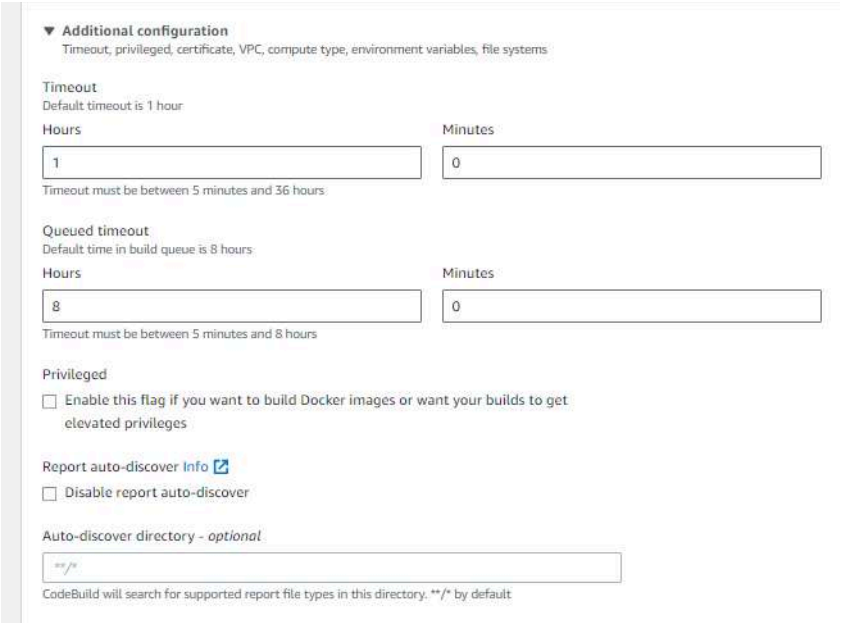
4.4 Now you need to create a new service role,in which Codebuild is to communicate with codecommit and codedeploy. For this you need to create a New service role , enter the Role name.



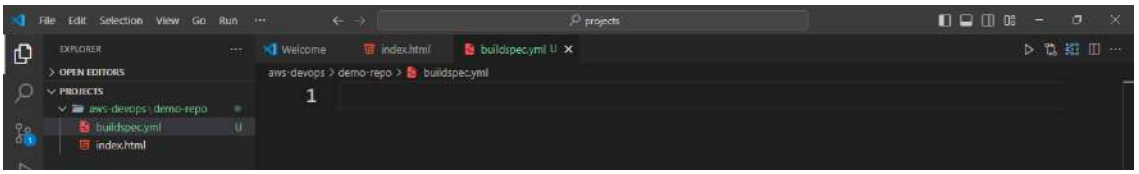
4.5 Now you need a Specification file, that is Buildspec.yml file, so now what is Buildspec, it is a specification file that means, for example we have built a code, for that code you need to specify an application should be installed, that application should be Running or tested and that application code should be Build, all these dependencies are stages or phases in the codeBuild, so all these phases are specified or configured in an file called Buildspec.yml file.

- Buildspec.yml file is created by DevOps Engineer





- Create an Buildspec.yml file in the demo-app folder.



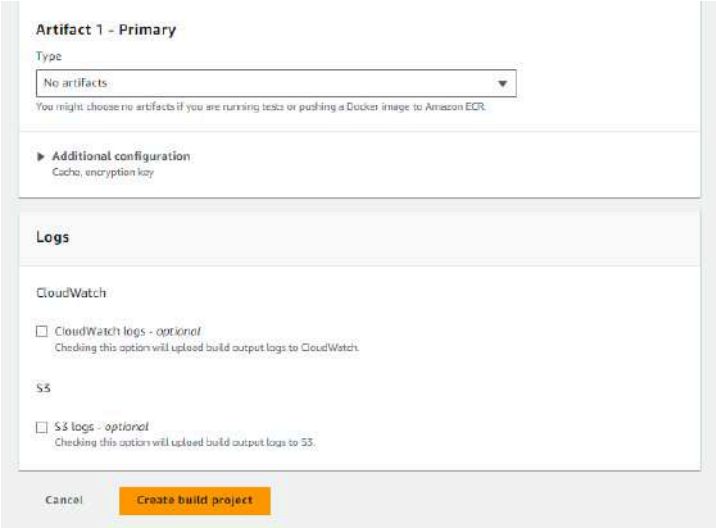
```yaml
version: 0.2

phases:
  install:
    commands:
      - echo Installing NGINX
      - sudo apt-get update
      - sudo apt-get install nginx -y
  build:
    commands:
      - echo Building started on 'date'
      - cp index.html /var/www/html/
  post_build:
    commands:
      - echo configuring NGINX

artifacts:
  files:
    - /var/www/html/index.html
```

- Now perform some actions using the Git commands to push the code to Dev Branch.

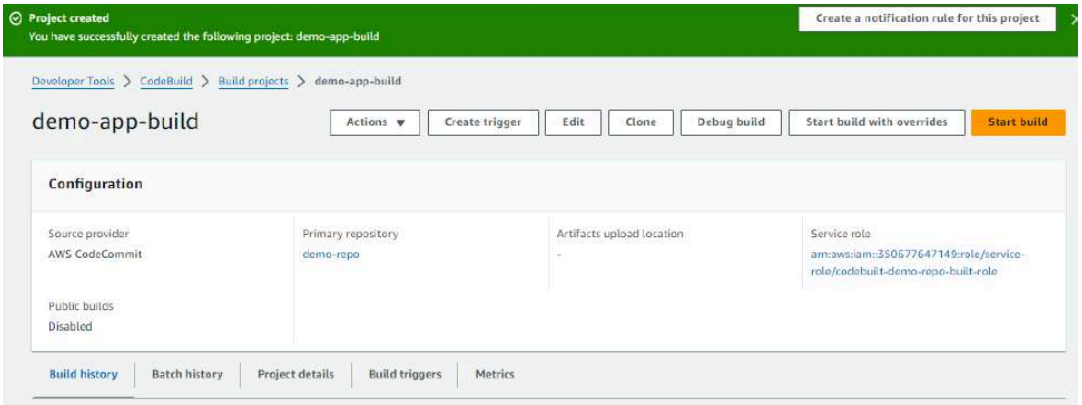
```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git add .
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git commit -m "added buildspec"
[dev 230e8a3] added buildspec
 1 file changed, 19 insertions(+)
 create mode 100644 buildspec.yml
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

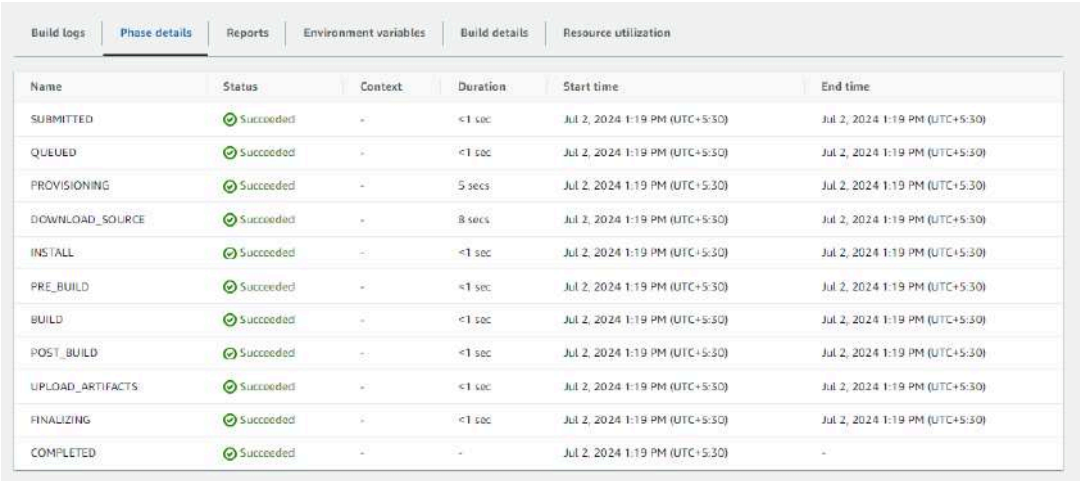
```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git push origin dev
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 468 bytes | 468.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/demo-repo
 * [new branch]      dev -> dev
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

4.6 Leave the Artifact Section as empty, Uncheck Cloudwatch and then Click on Create build project, after Clicking create build project, your project will be created.

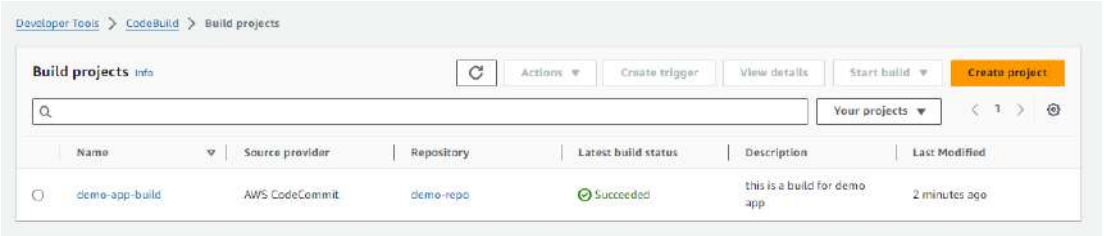


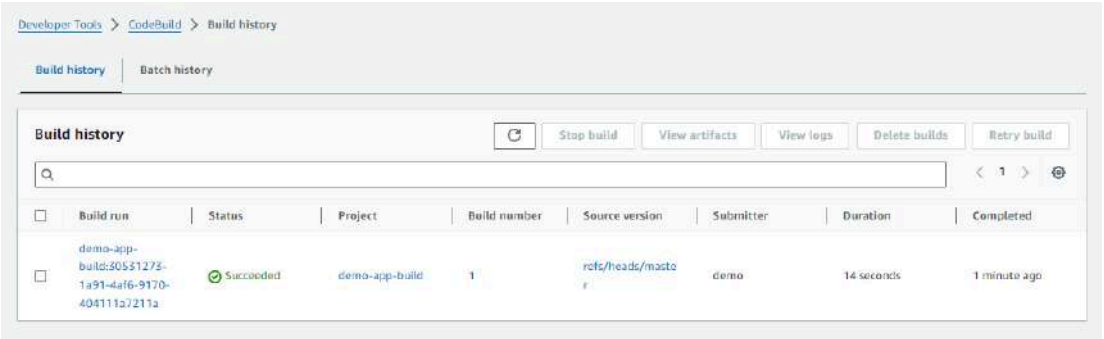
4.7 After the Project was created successfully, to Build the created project click on **Start build,** then your project gets started to build, to look how the project is building Click on **Phase details,** after build is done , you will be notifying build **Succeeded**.





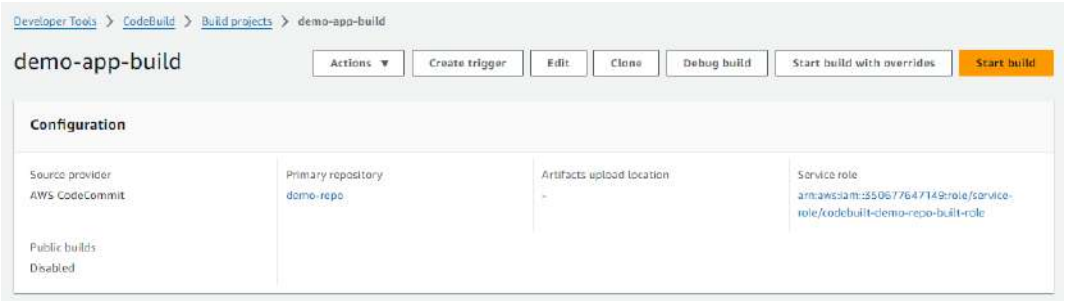
Name	Status	Context	Duration	Start time	End time
SUBMITTED	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
QUEUED	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
PROVISIONING	Succeeded	-	5 secs	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
DOWNLOAD_SOURCE	Succeeded	-	8 secs	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
INSTALL	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
PRE_BUILD	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
BUILD	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
POST_BUILD	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
UPLOAD_ARTIFACTS	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
FINALIZING	Succeeded	-	<1 sec	Jul 2, 2024 1:19 PM (UTC+5:30)	Jul 2, 2024 1:19 PM (UTC+5:30)
COMPLETED	Succeeded	-	-	Jul 2, 2024 1:19 PM (UTC+5:30)	-

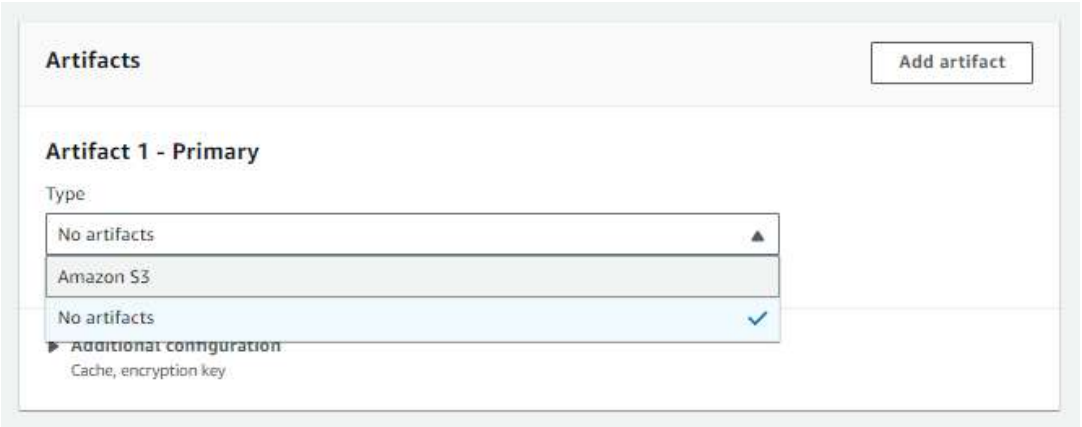




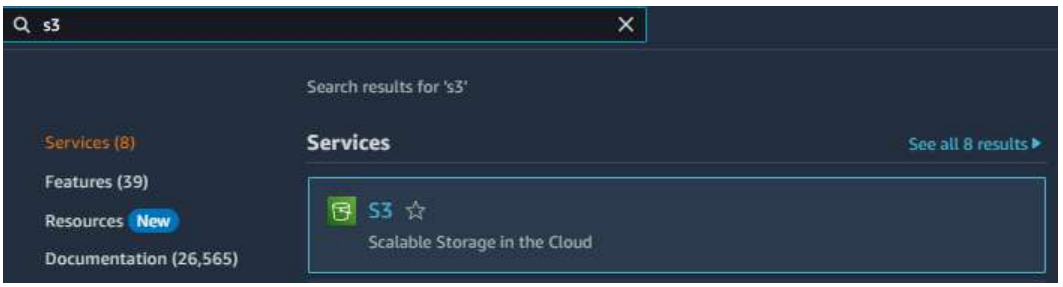
Step 5: Upload the Build code in a specific place, create an Artifact.

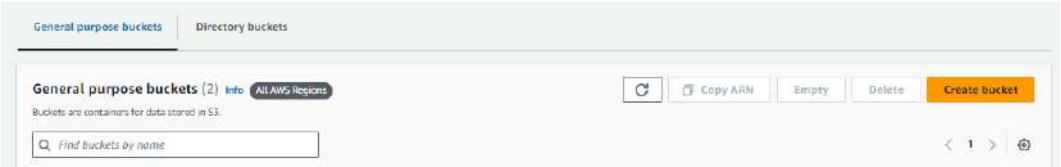
5.1 To keep the build code in a specific place, to do that we need to create an **Artifact**, for that Click on **Edit** , then scroll down to **Artifact**, select **S3 Bucket** in **Type.**

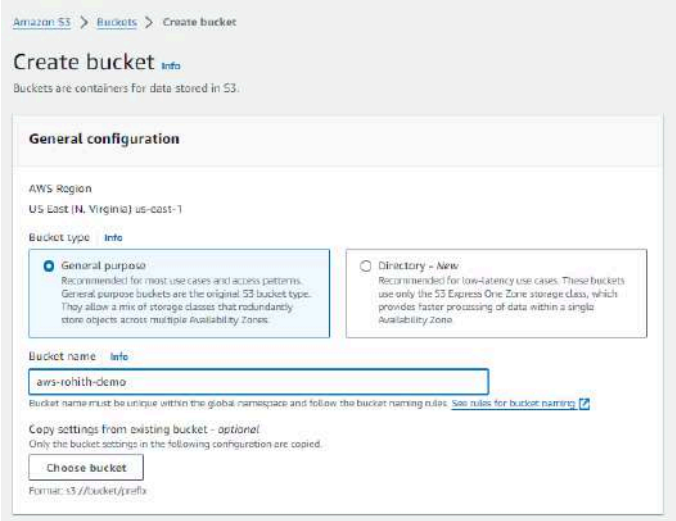


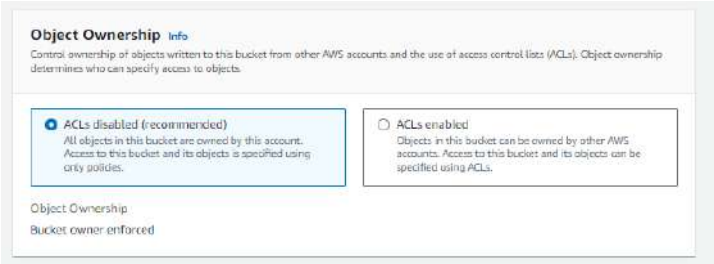


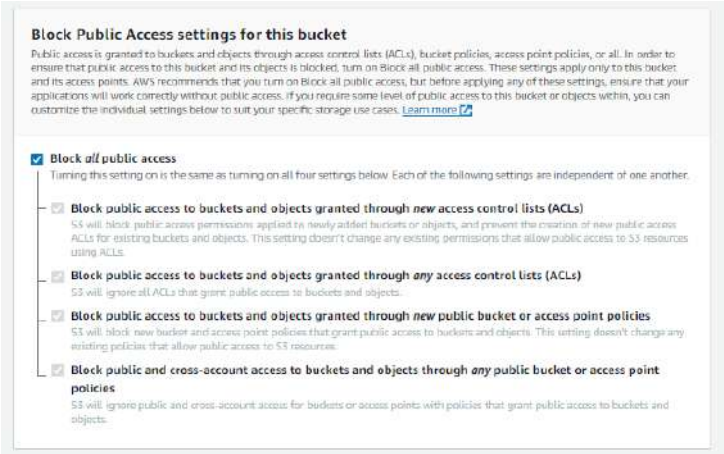
5.2 Create an **S3 Bucket**, for that Search **S3** in the **AWS Console**, Click on **Create bucket**, then give a **Unique name** to the Bucket, now Click on **Create bucket**, then Bucket will be **Created Successfully.**

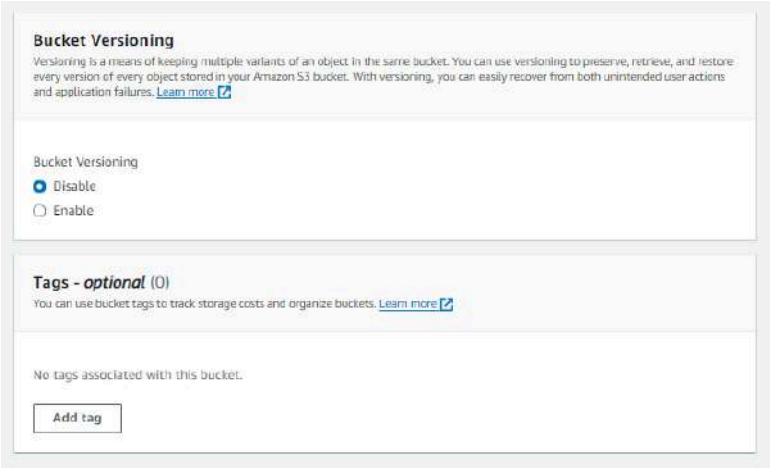

s3
Search results for 's3'
Services (8)
Features (39)
Resources New
Documentation (26,565)
Services
See all 8 results
S3
Scalable Storage in the Cloud

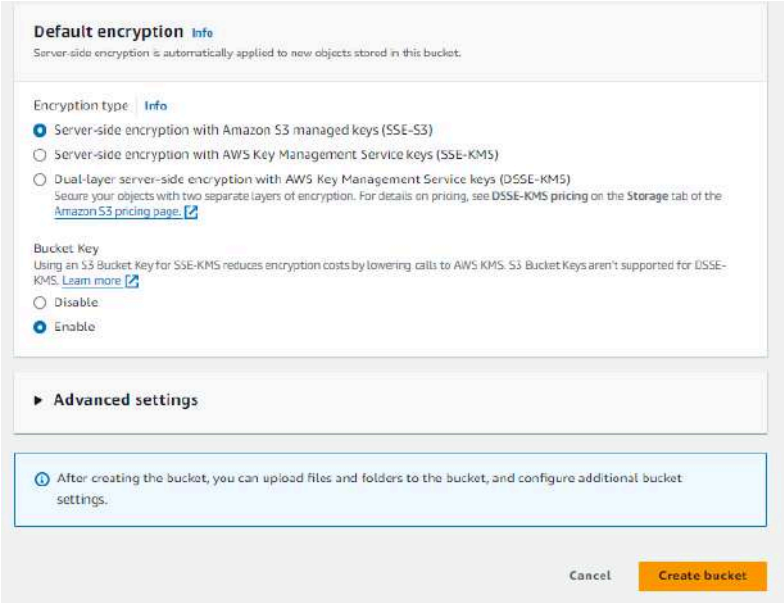

General purpose buckets
Directory buckets
General purpose buckets (2) Info All AWS Regions
Buckets are containers for data stored in S3.
Find buckets by name
Copy ARN
Empty
Delete
Create bucket

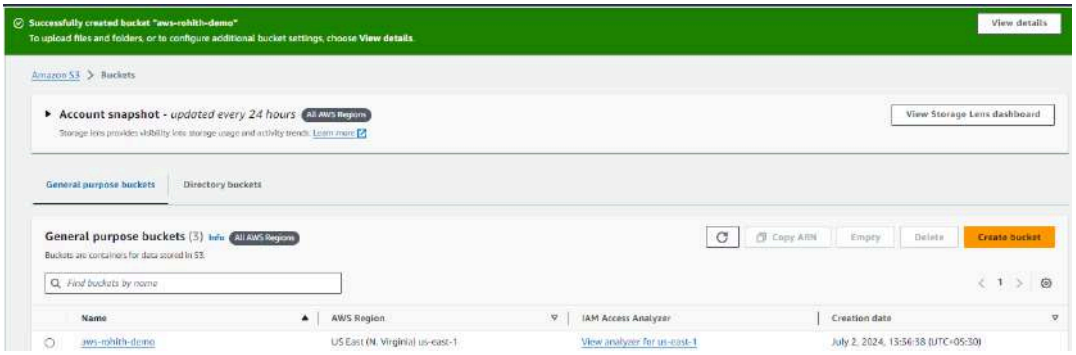

Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory - New
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket name Info
aws-rohith-demo
Bucket name must be unique within the global namespace and follow the bucket naming rules. See rules for bucket naming
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix


Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
Object Ownership
Bucket owner enforced

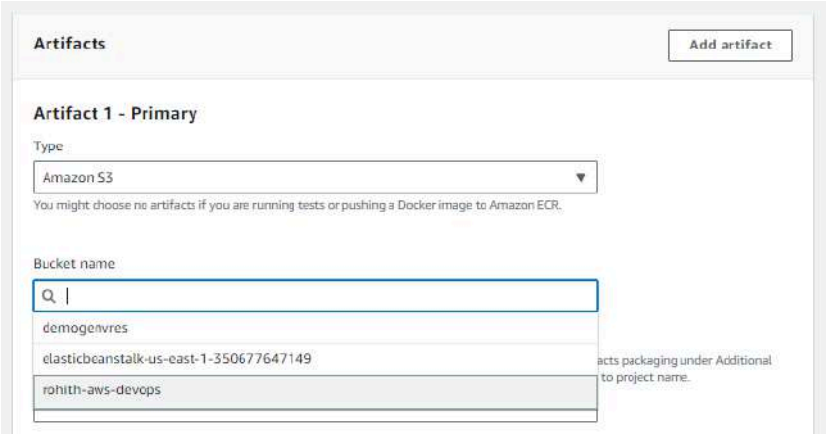

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Block public access to buckets and objects granted through new access control lists (ACLs)
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
Block public access to buckets and objects granted through any access control lists (ACLs)
S3 will ignore all ACLs that grant public access to buckets and objects.
Block public access to buckets and objects granted through new public bucket or access point policies
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
Block public and cross-account access to buckets and objects through any public bucket or access point policies
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

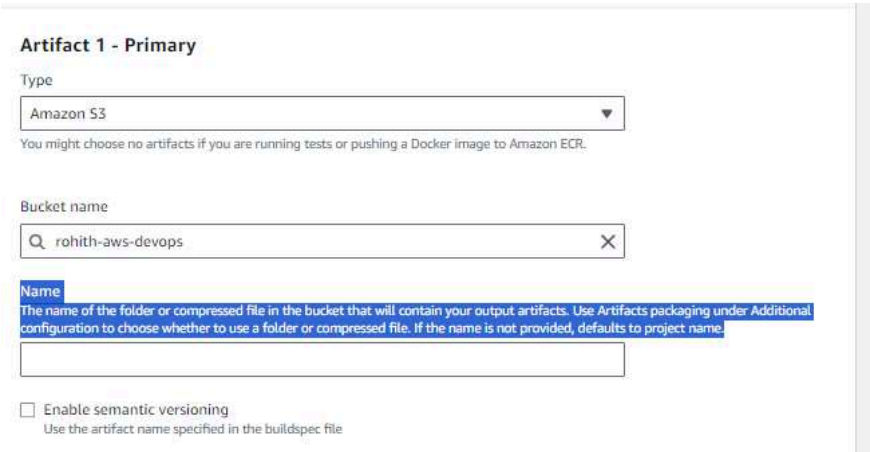


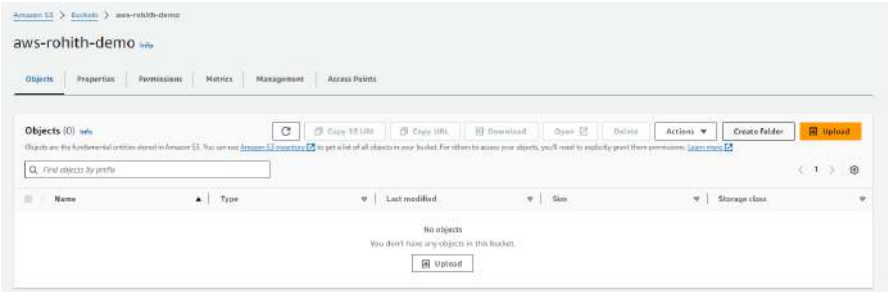




5.3 In the Artifact Section, Select the Bucket name that you have created, then you need to create a folder in the Bucket, where the Build code will be stored, for that you need to go to the S3 Bucket which you have created and in that Bucket Click on Create folder .







5.4 Create a Folder for that, give a **Folder name**, then Click on **Create folder,** then **Successfully folder is created**, after that go inside the folder and select the **folder name** and **paste** it in the **folder name**, then Copy S3 URI and Paste it in the Folder Path and give /artifact.zip and Click on **Update project** .

Amazon S3 > Buckets > aws-rohith-demo > Create folder

Create folder Info

Use folders to group objects in buckets. When you create a folder, S3 creates an object using the name that you specify followed by a slash (/). This object then appears as folder on the console. Learn more

Your bucket policy might block folder creation
If your bucket policy prevents uploading objects without specific tags, metadata, or access control list (ACL) grantees, you will not be able to create a folder using this configuration. Instead, you can use the upload configuration to upload an empty folder and specify the appropriate settings.

Folder

Folder name

code-output-demo-app /

Folder names can't contain "/". See rules for naming

Server-side encryption Info

Server-side encryption protects data at rest.

The following encryption settings apply only to the folder object and not to sub-folder objects.

Folder

Folder name

code-output-demo-app /

Folder names can't contain "/". See rules for naming

Server-side encryption Info

Server-side encryption protects data at rest.

The following encryption settings apply only to the folder object and not to sub-folder objects.

Server-side encryption

- Do not specify an encryption key
  The bucket settings for default encryption are used to encrypt the folder object when storing it in Amazon S3.
- Specify an encryption key
  The specified encryption key is used to encrypt the folder object before storing it in Amazon S3.

If your bucket policy requires objects to be encrypted with a specific encryption key, you must specify the same encryption key when you create a folder. Otherwise, folder creation will fail.

Cancel | Create folder

Successfully created folder "code-output-demo-app".

Amazon S3 > Buckets > aws-rohith-demo

aws-rohith-demo Info

Objects | Properties | Permissions | Metrics | Management | Access Points

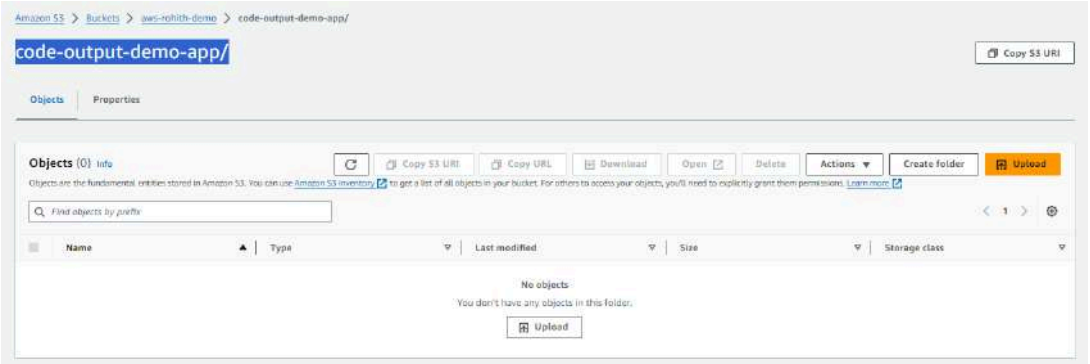
Objects (1) Info

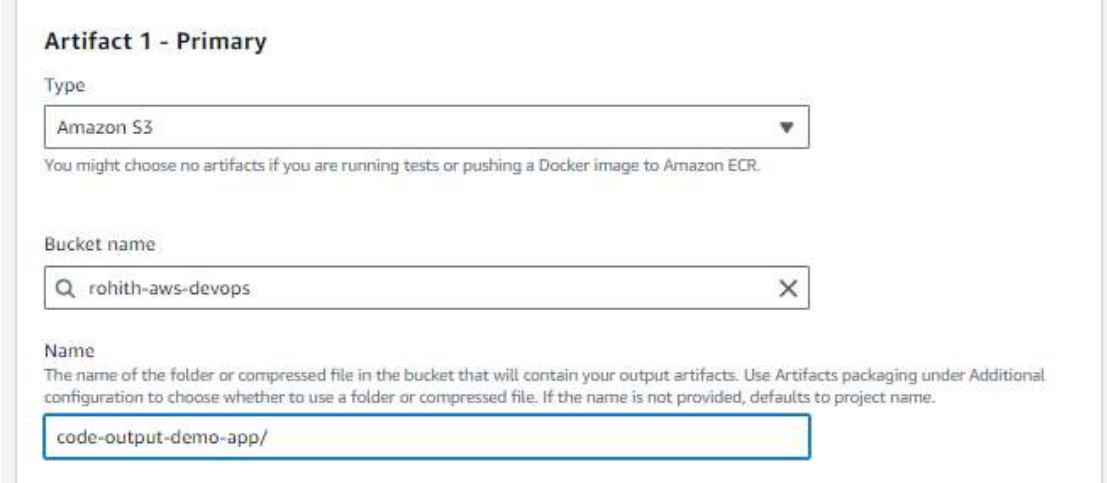
Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

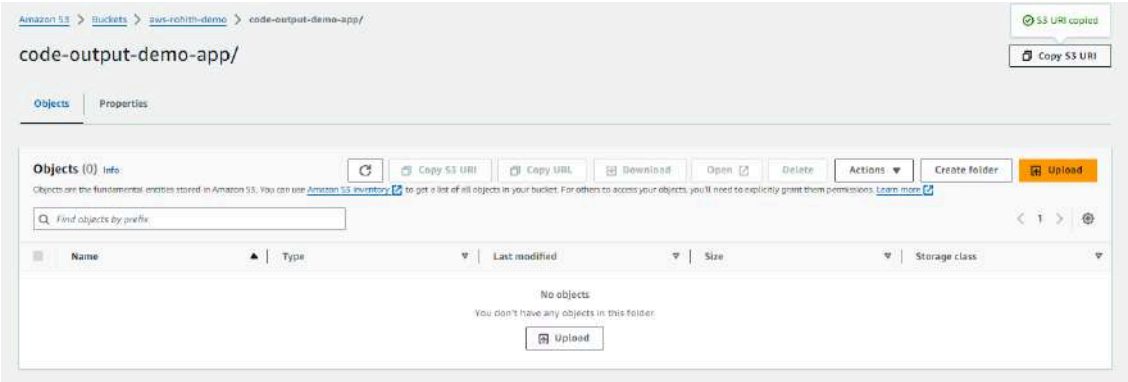
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

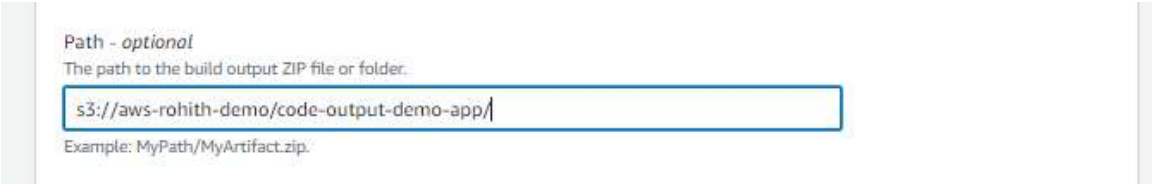
Find objects by prefix

Name	Type	Last modified	Size	Storage class
code-output-demo-app/	Folder	-	-	-

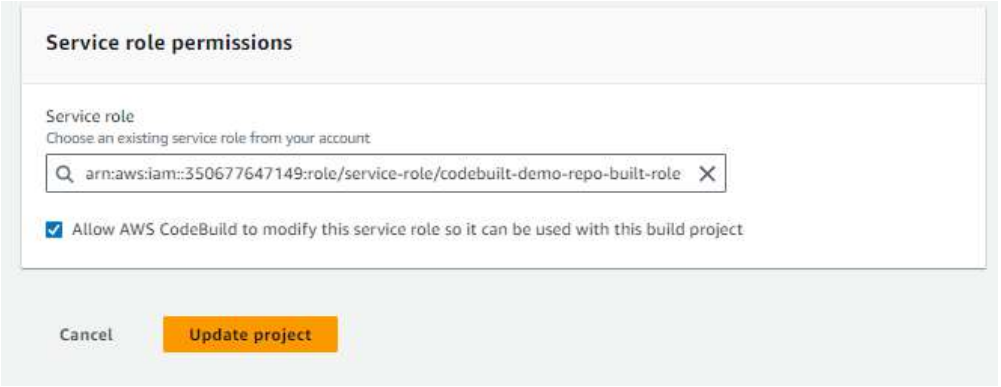

Amazon S3 > Buckets > aws-rohith-demo > code-output-demo-app/
code-output-demo-app/
Copy S3 URI
Objects
Properties
Objects (0) Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this folder.
Upload


Artifact 1 - Primary
Type
Amazon S3
You might choose no artifacts if you are running tests or pushing a Docker image to Amazon ECR.
Bucket name
rohith-aws-devops
Name
The name of the folder or compressed file in the bucket that will contain your output artifacts. Use Artifacts packaging under Additional configuration to choose whether to use a folder or compressed file. If the name is not provided, defaults to project name.
code-output-demo-app/


Amazon S3 > Buckets > aws-rohith-demo > code-output-demo-app/
S3 URI copied
code-output-demo-app/
Copy S3 URI
Objects
Properties
Objects (0) Info
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this folder.
Upload

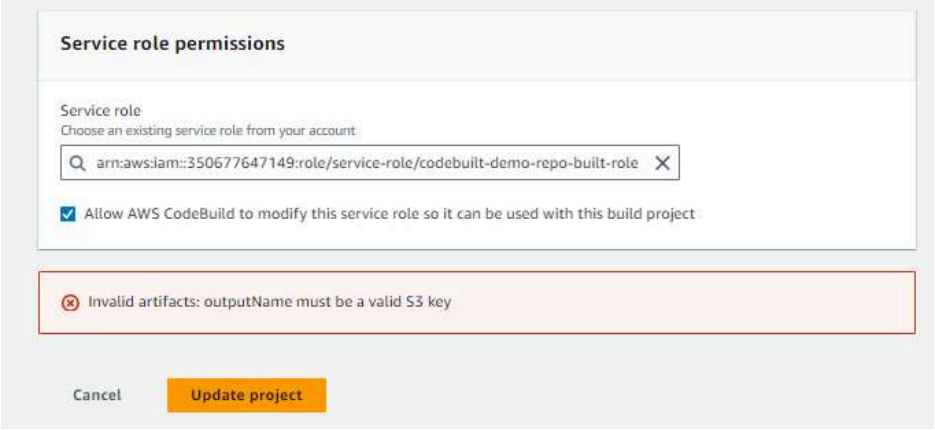

Path - optional
The path to the build output ZIP file or folder.
s3://aws-rohith-demo/code-output-demo-app/
Example: MyPath/MyArtifact.zip.

Path - optional
The path to the build output ZIP file or folder.
s3://aws-rohith-demo/code-output-demo-app/artifact.zip
Example: MyPath/MyArtifact.zip.

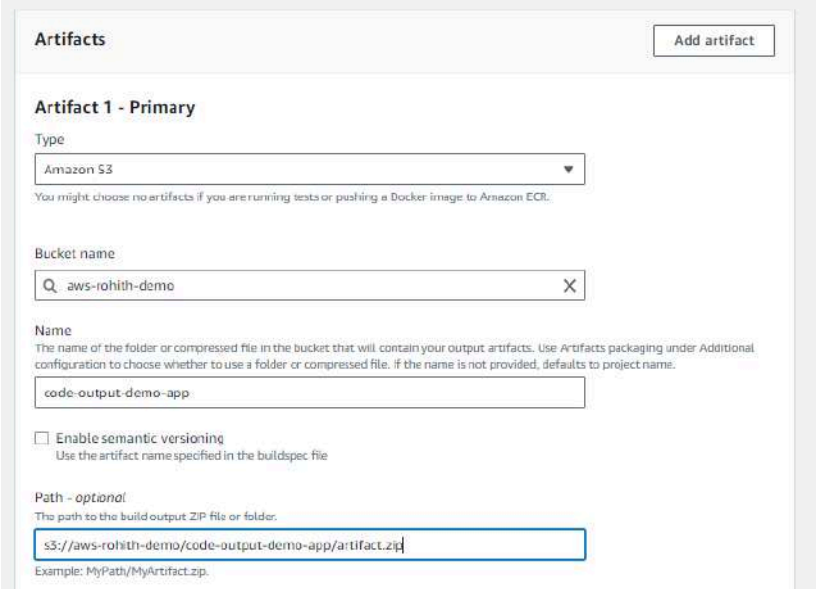


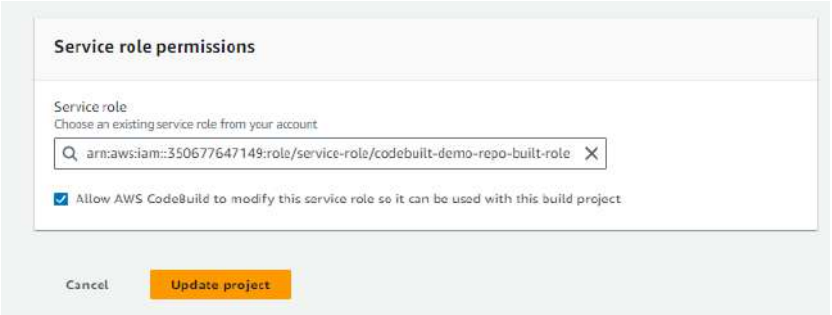
Troubleshooting:

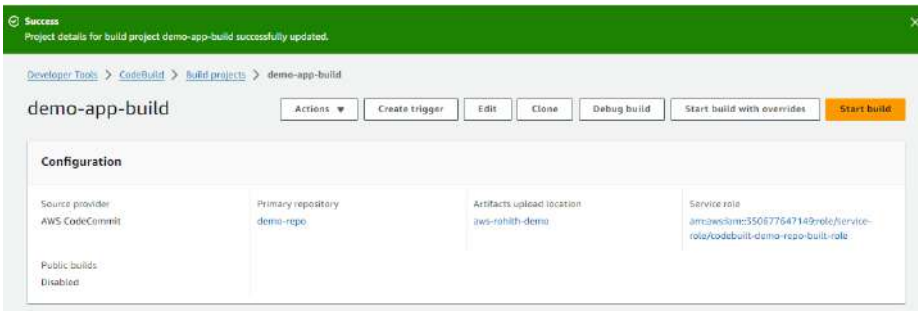
- ➢ The following Error **'invalid artifacts: outputname must be a valid s3 key' ,** there was something went wrong in S3 Bucket Section.



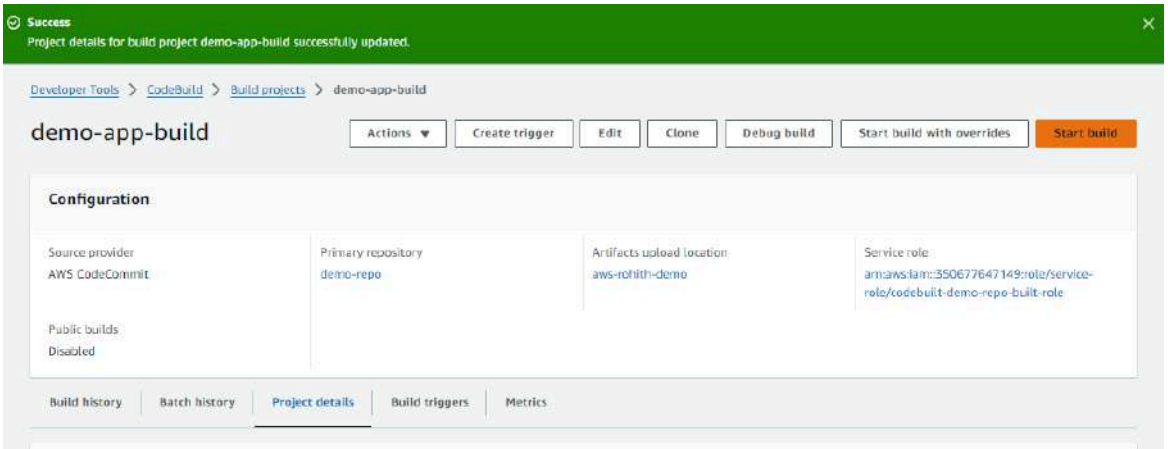
- To resolve the error, remove **/ after the name of the Bucket,** and in the Bucket path after **/ give Artifact.zip,** then your error will be resolved.

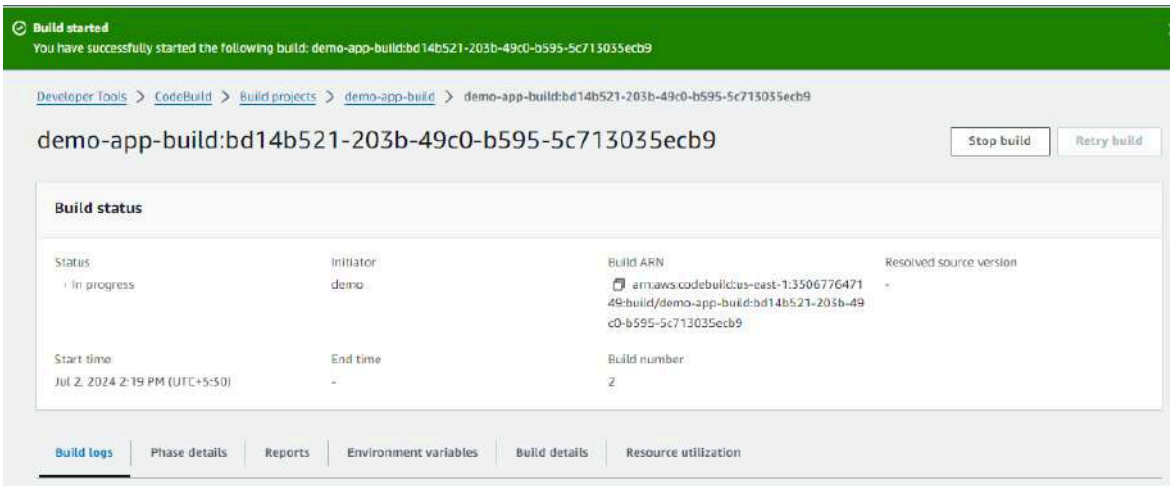


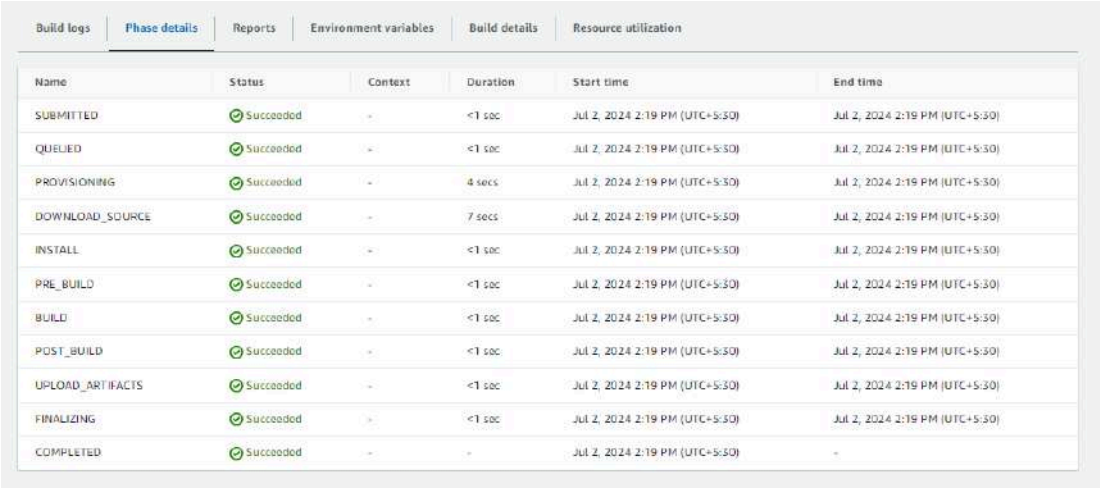


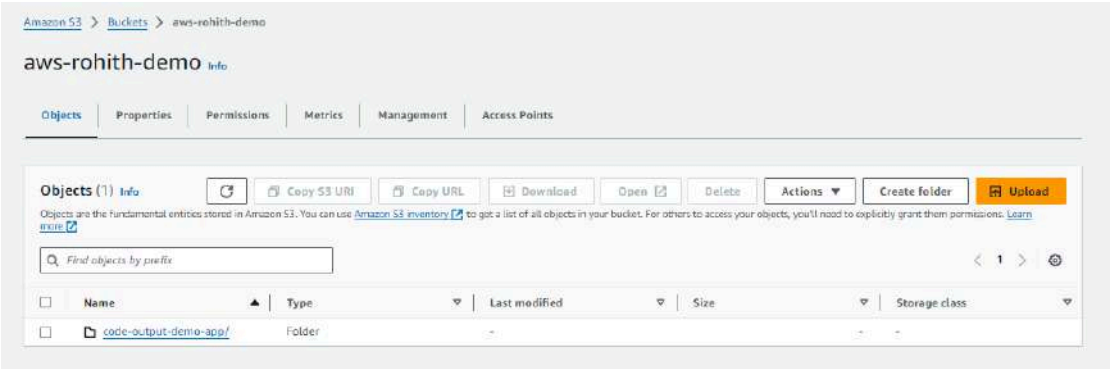


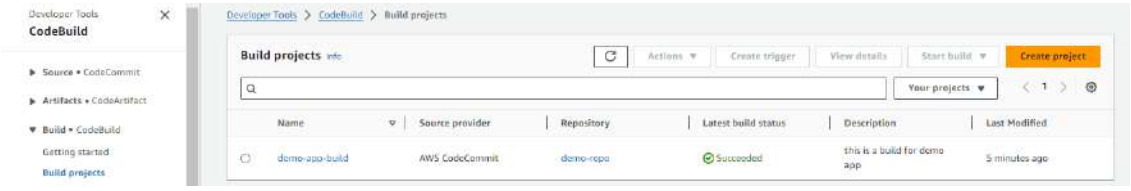
5.5 Now Click on Start Build.





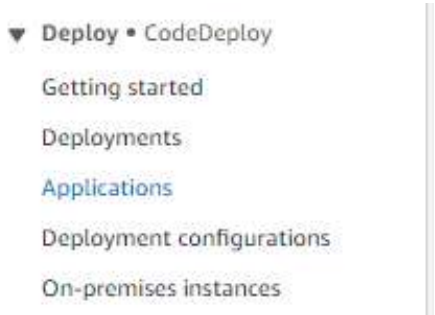


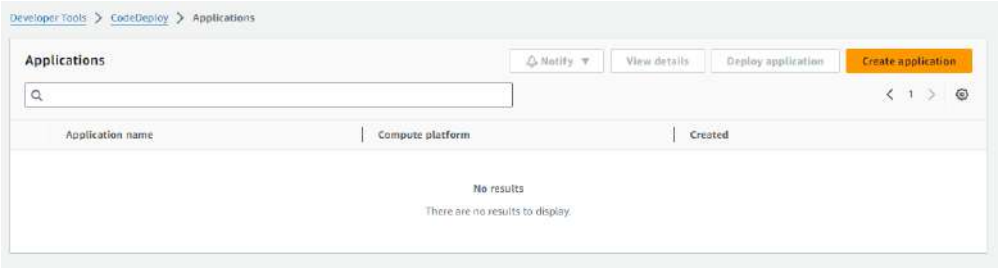




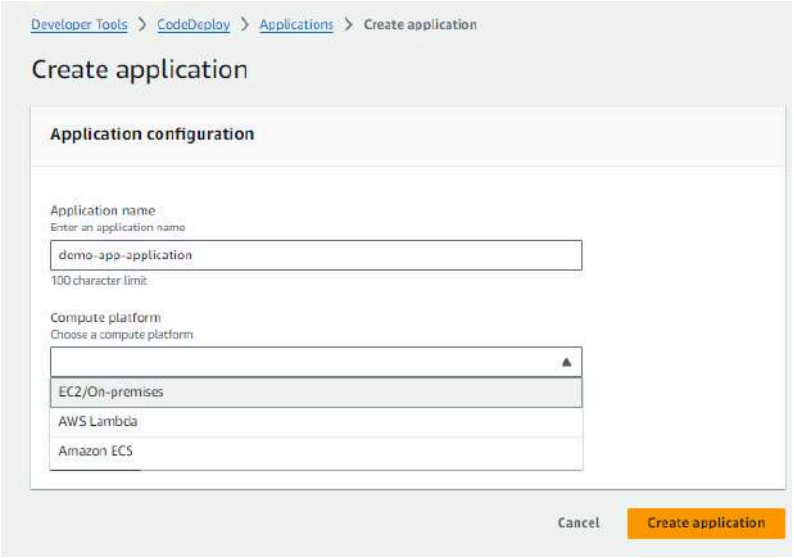
Step 6: Deploying an Application using AWS CodeDeploy

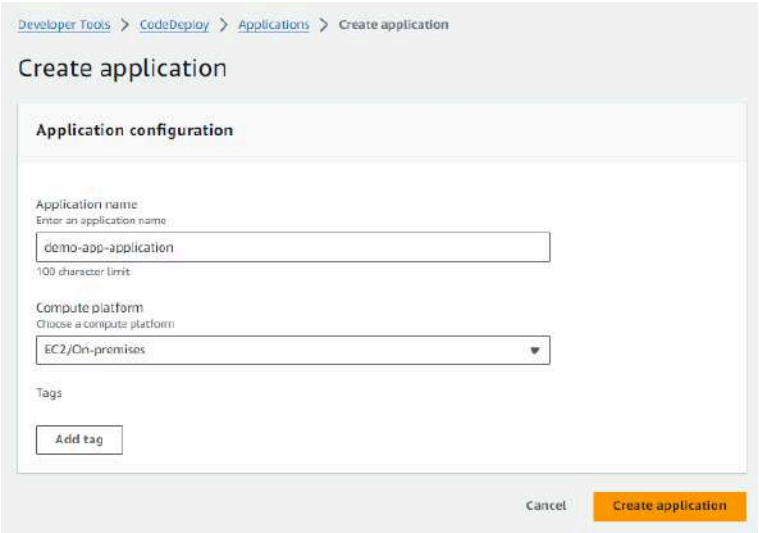
6.1 To Deploy an application, Go to AWS CodeDeploy, Click on Application, then Click on Create application.



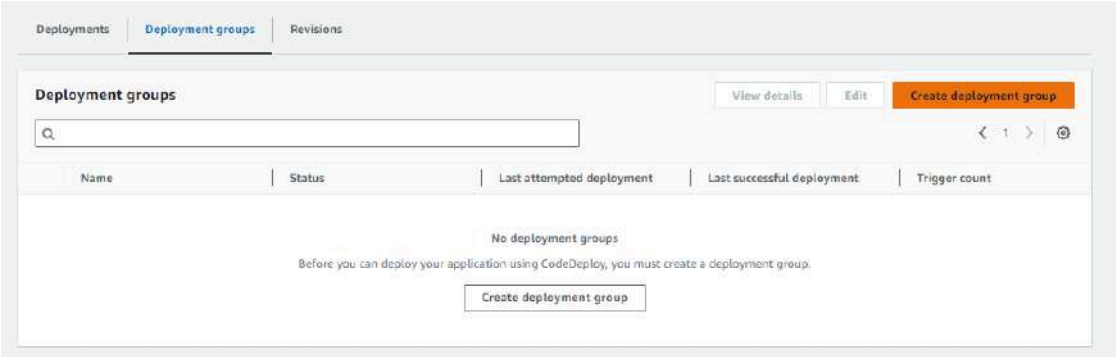


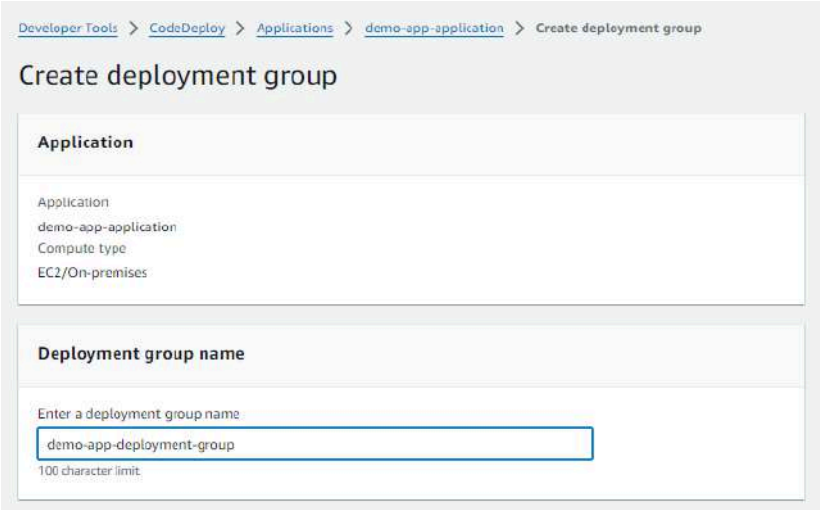
6.2 In **Create application**, Give some name to application and then Choose, on which Computing Service you want to deploy the application, in my case i Choose **EC2 Instance** and now Click on **Create application,** then **application created**.



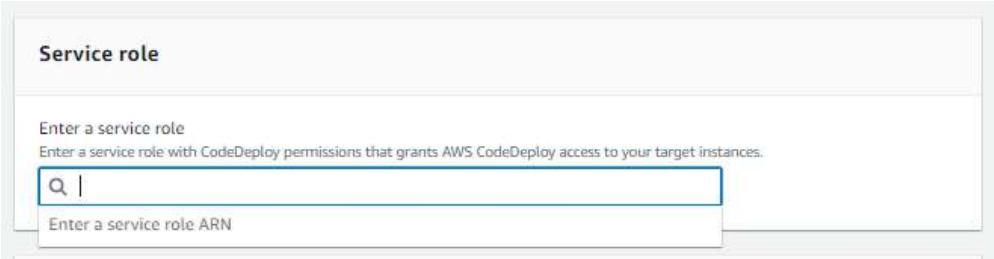


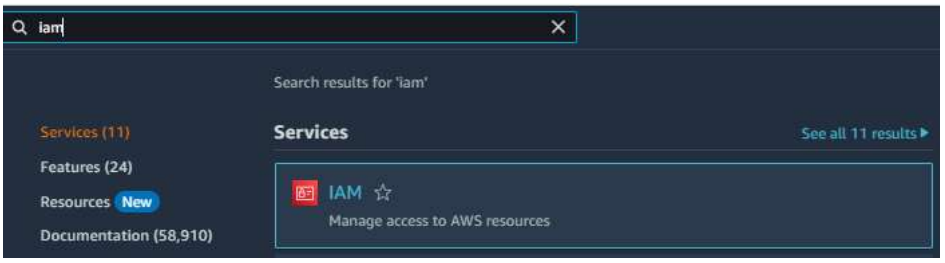
6.3 Now you need to **Create a Deployment Group**, that means if you want to deploy your application on Multiple EC2 servers, you need to create a deployment group, to do that you need to Click on **Create deployment group**, then **Enter a deployment group name.**





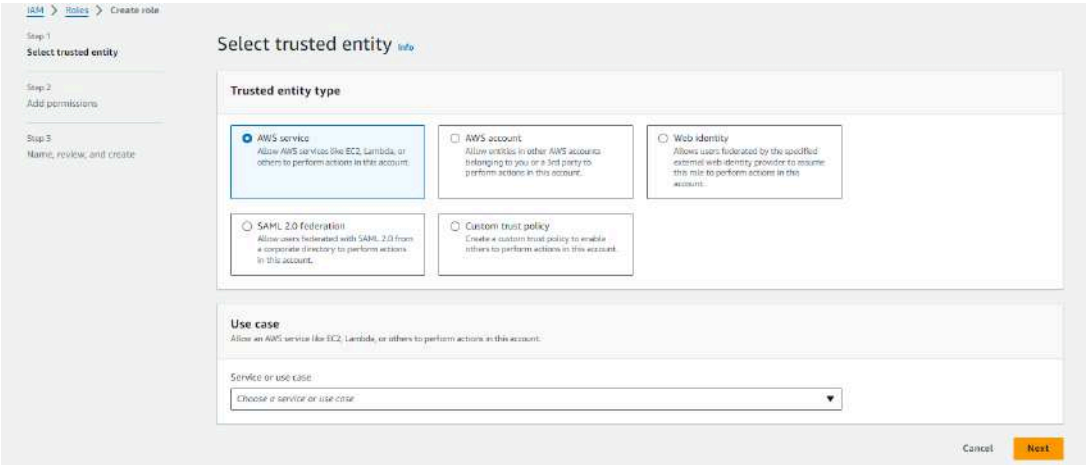
6.4 Now you need to Create a **new service role,** which is used to give **CodeDeploy Permissions that grants AWS CodeDeploy access to your target instances,** to do that you need to go to IAM Service, and select Roles.

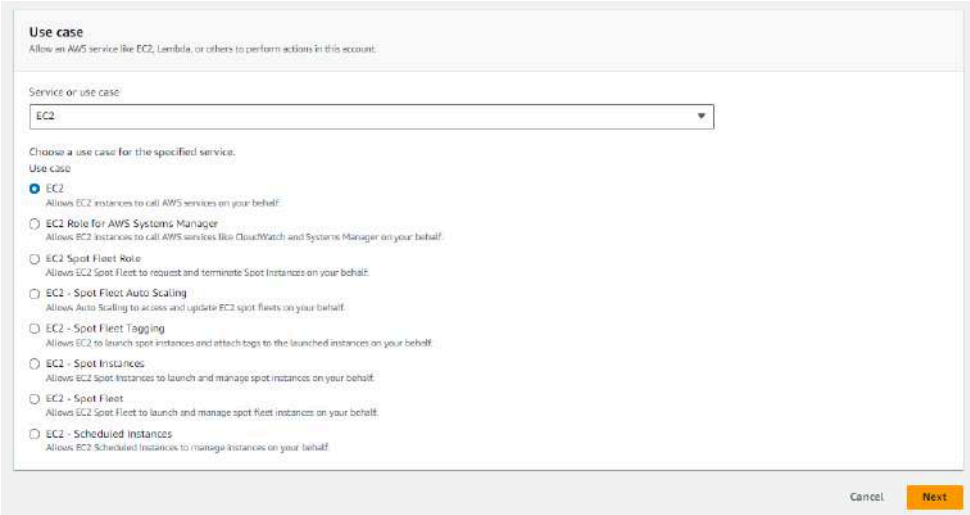




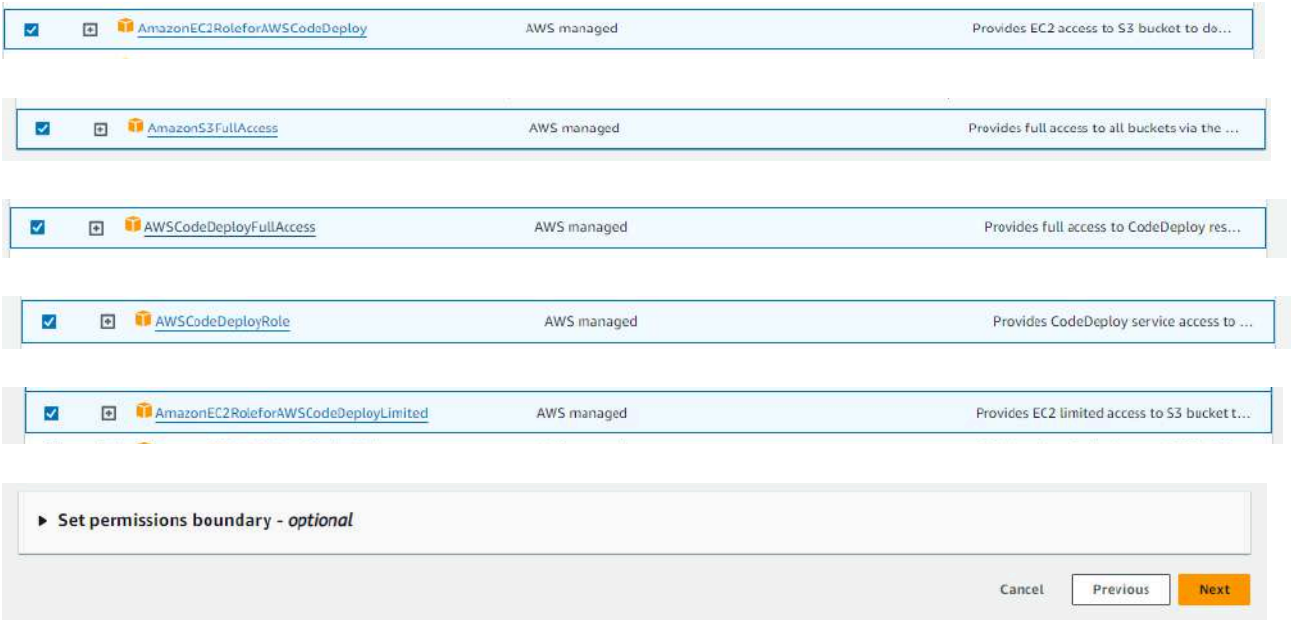
6.5 Select the Roles which are necessary.

Step 1: Select trusted entity, as **AWS service** and in **Use case** Choose Service of use as **EC2** and then Click on **Next**.

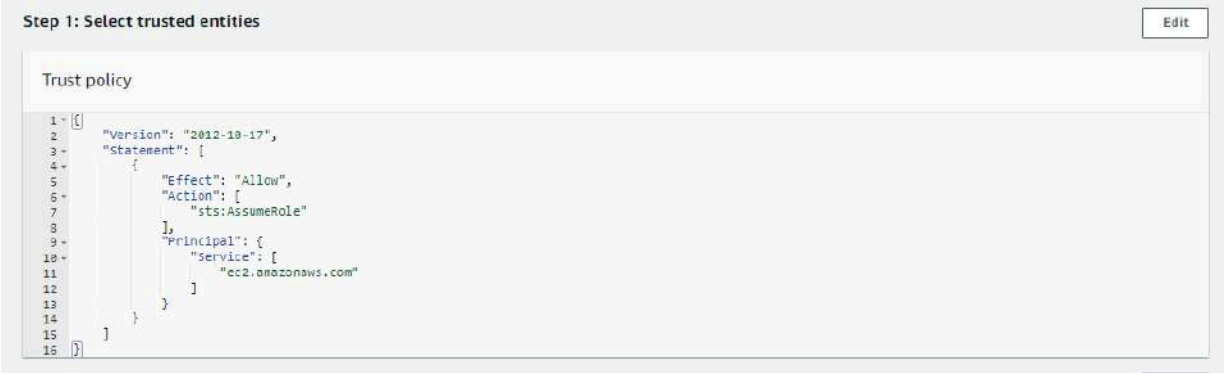


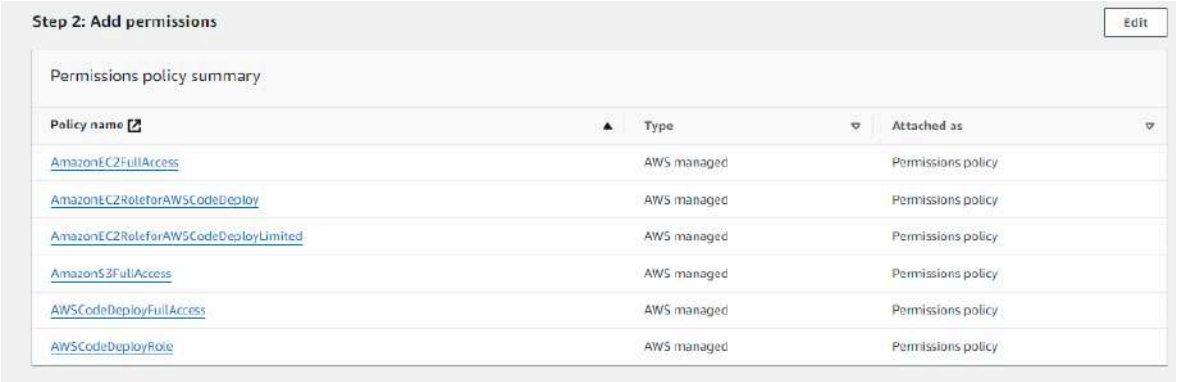


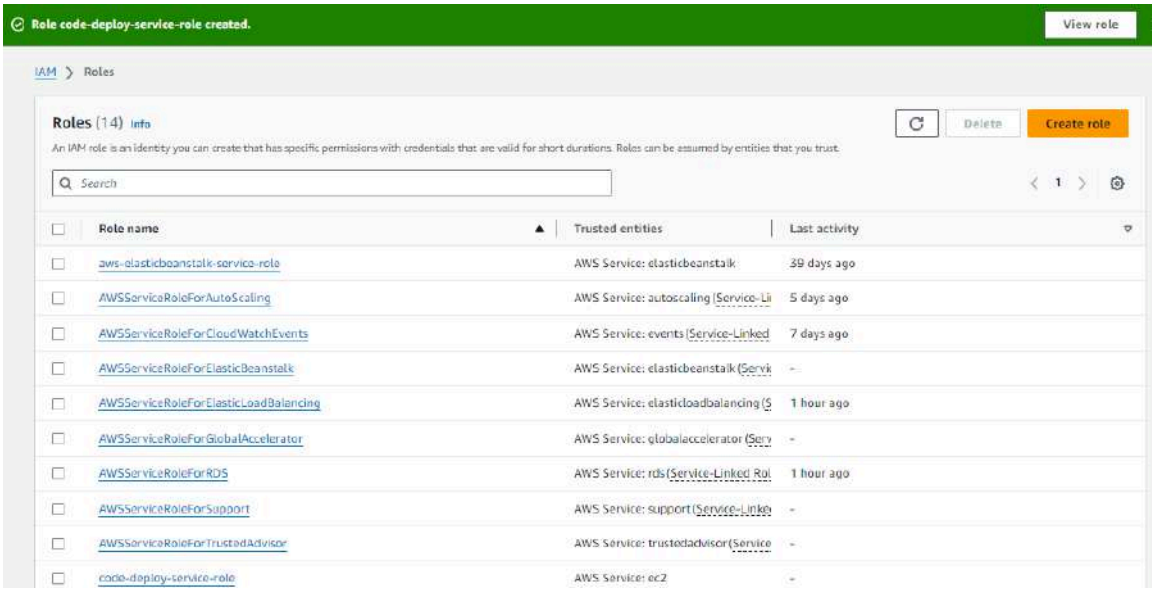
Step 2: Add Permissions, add all the permissions which are necessary, after adding policies Click on **Next.**



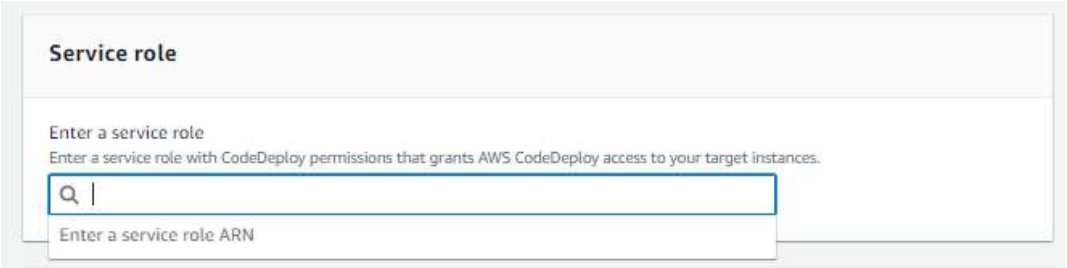
6.6 Once go through all the steps, and then Click on **Create role.**

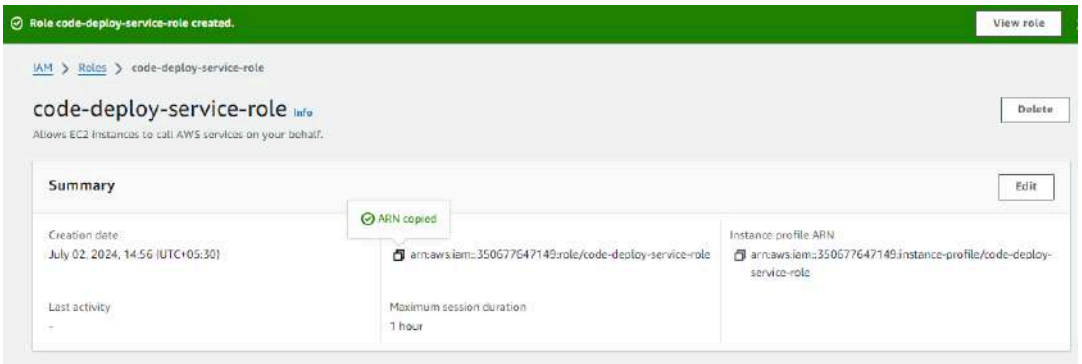


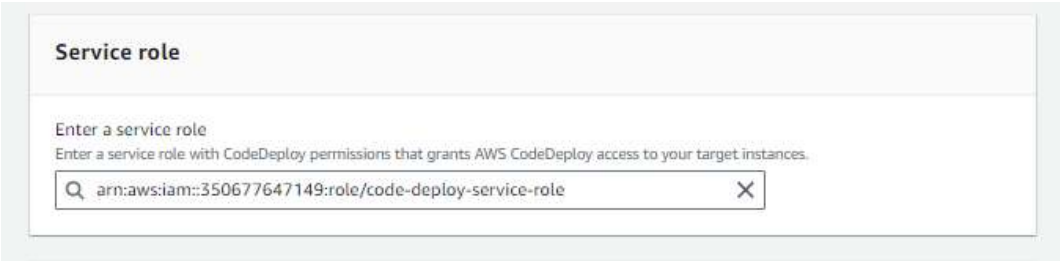




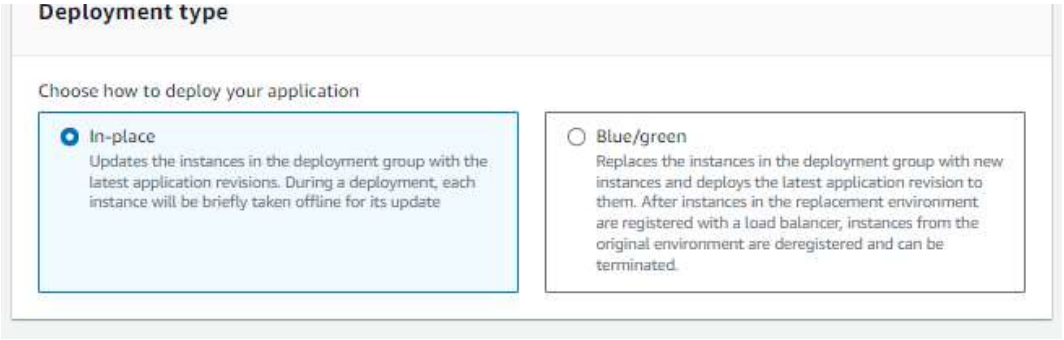
6.7 Now **Select** the created service role in the **Enter a service role name.**

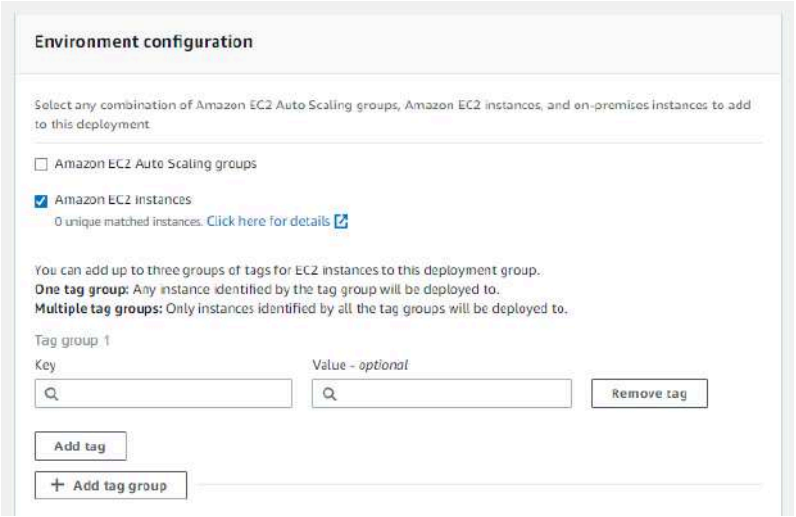




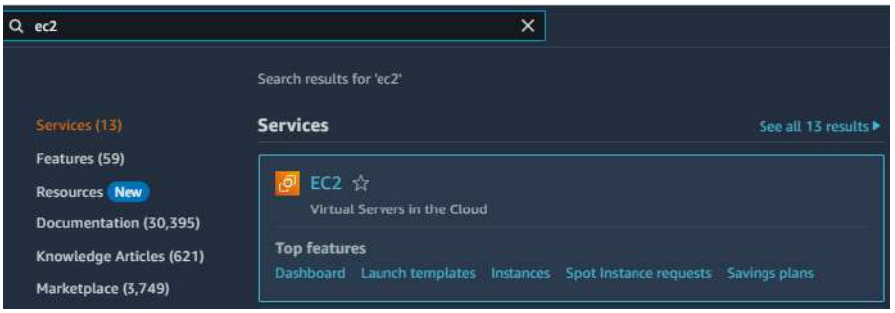


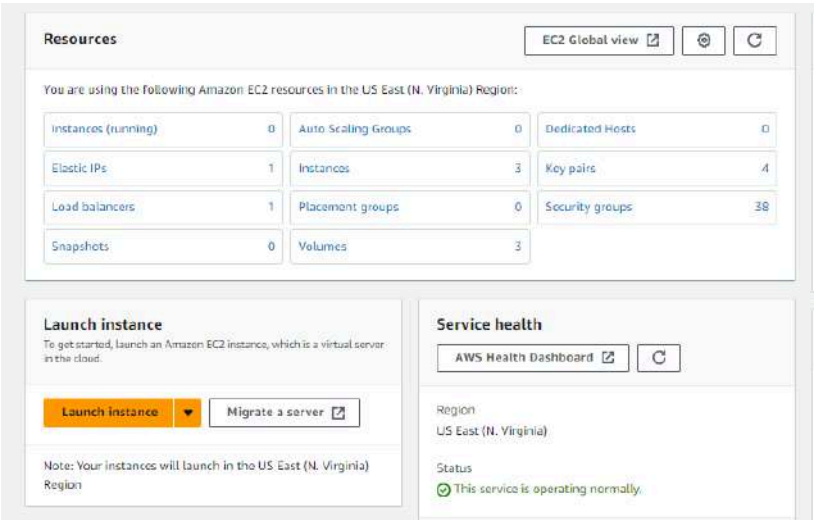
6.8 In **Deployment type**, Select **In-place** and in **Environment configuration,** Select **Amazon EC2 Instance.**



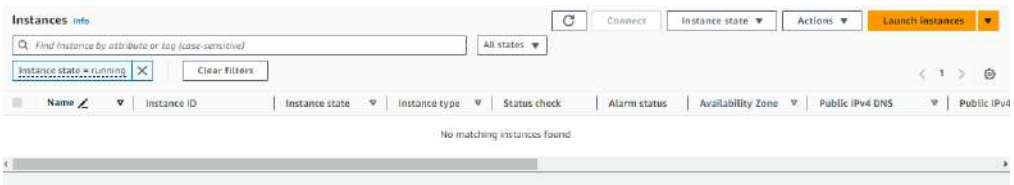


6.9 In Dashboard, search for **EC2 service,** click on **EC2** to open it.

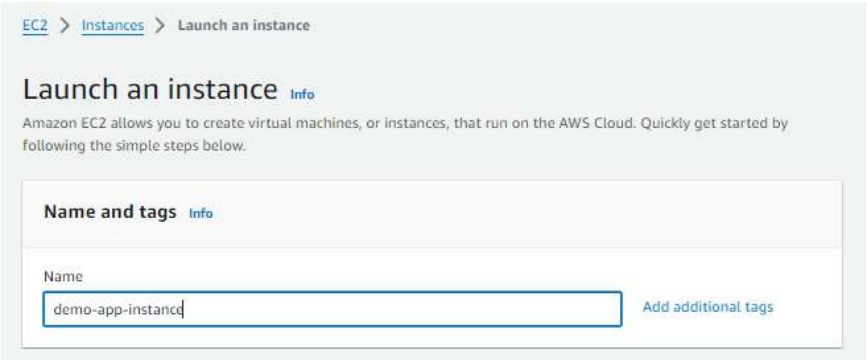




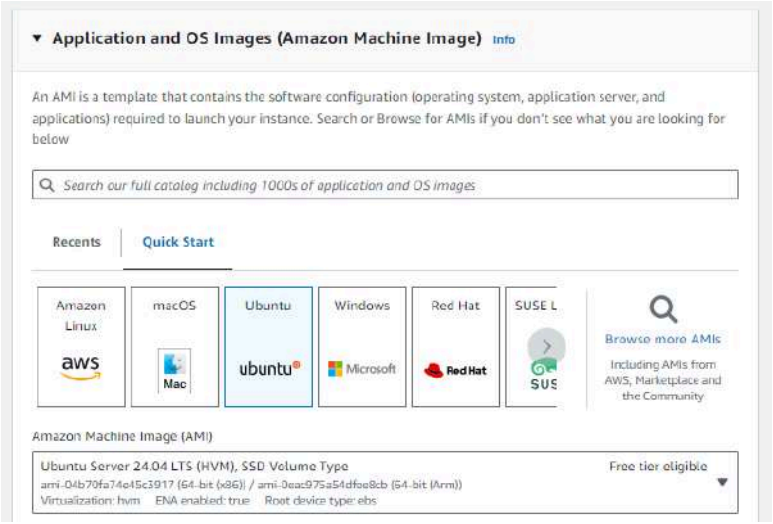
6.10 Click on **Launch instance** to configure.



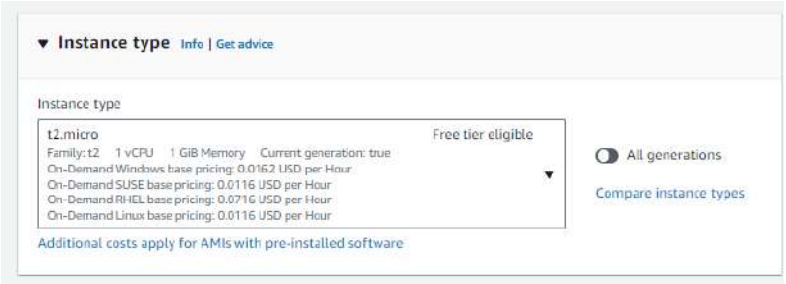
6.11 Type the **Name** which you prefer for naming the instance.



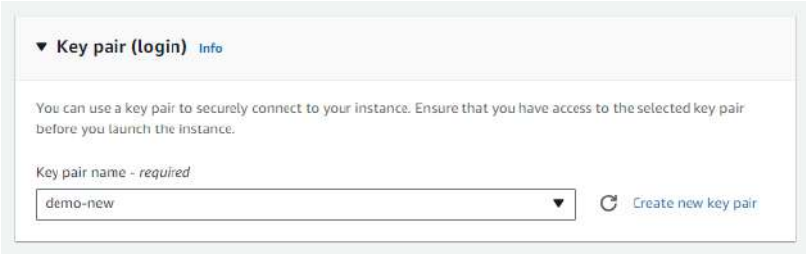
6.12 Choose the **AMI**, so here we are using **Ubuntu**.



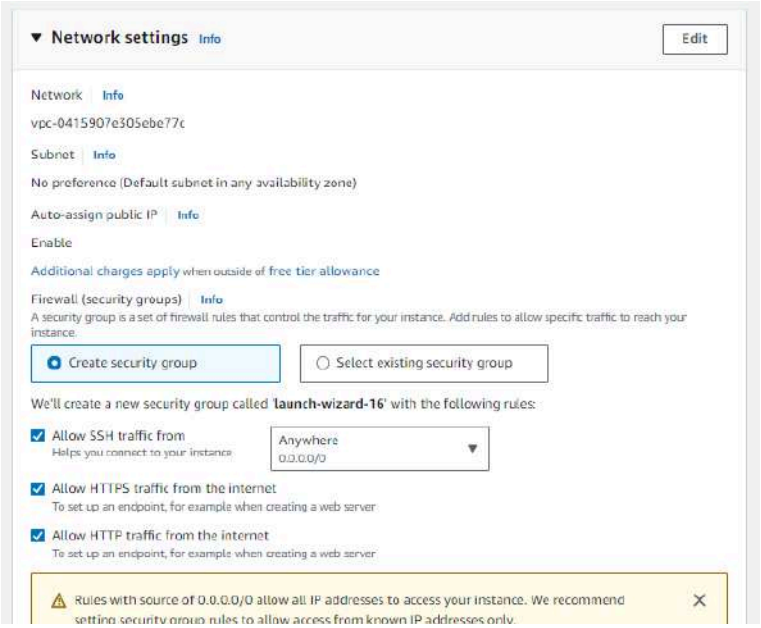
6.13 Select the instance type as **t2.micro**

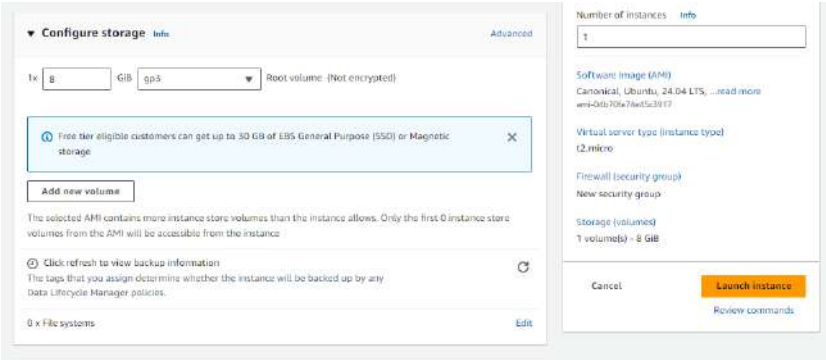


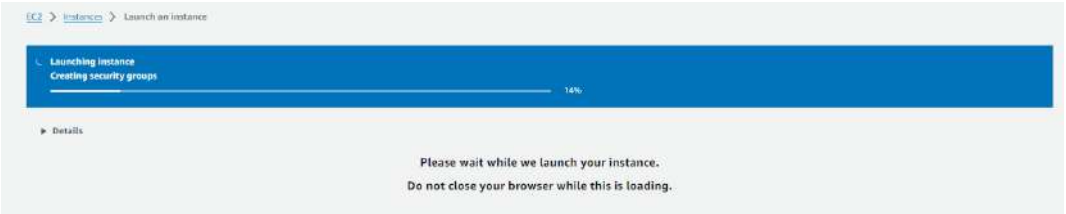
6.14 Create the **key Pair** to do that, Click on Create the **New pair key** for secure connection.



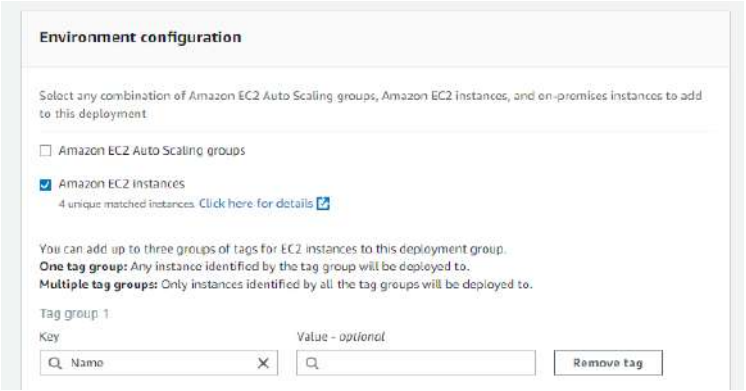
6.15 Under Network Settings, Click on **Create security group**, then Click on **Launch Instance**, then Instance will be **created successfully.**

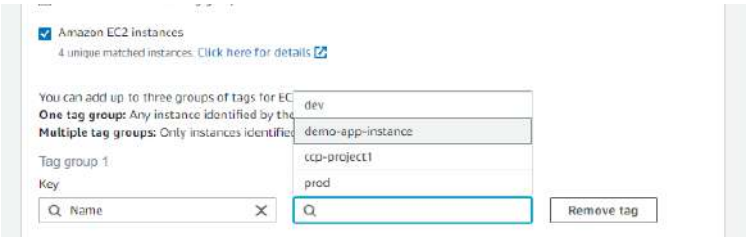


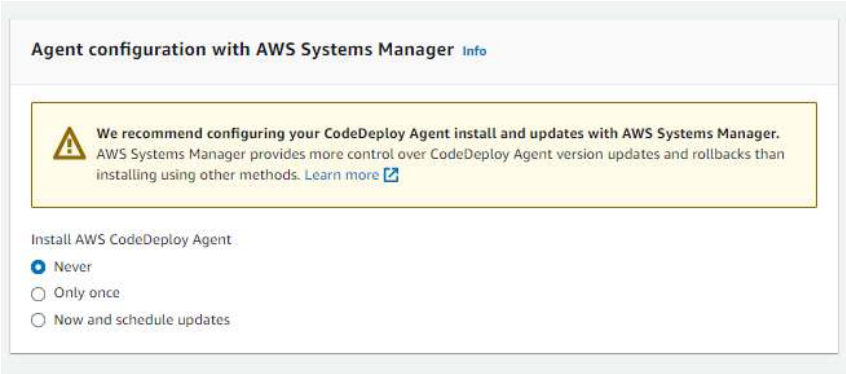


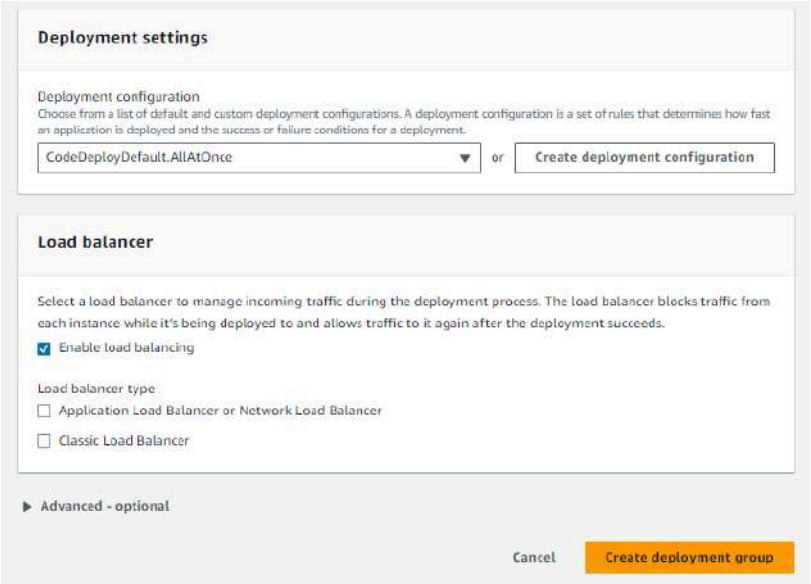


- In value option, Select the created instance



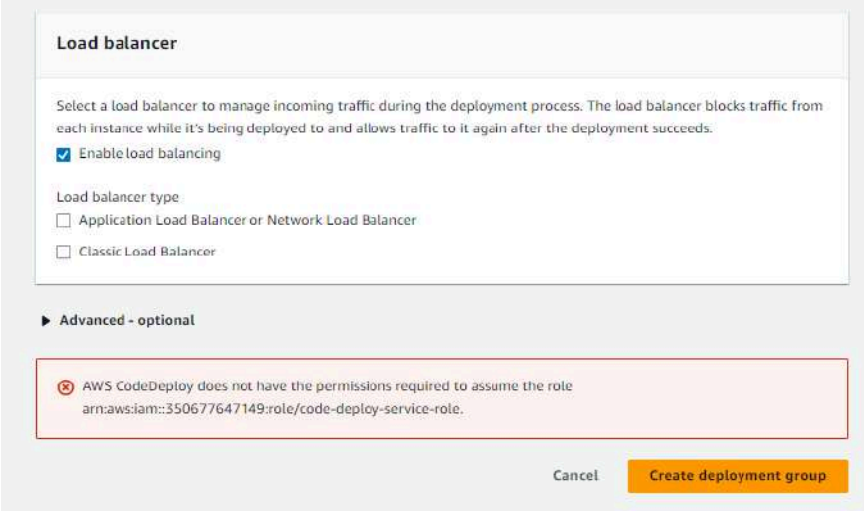






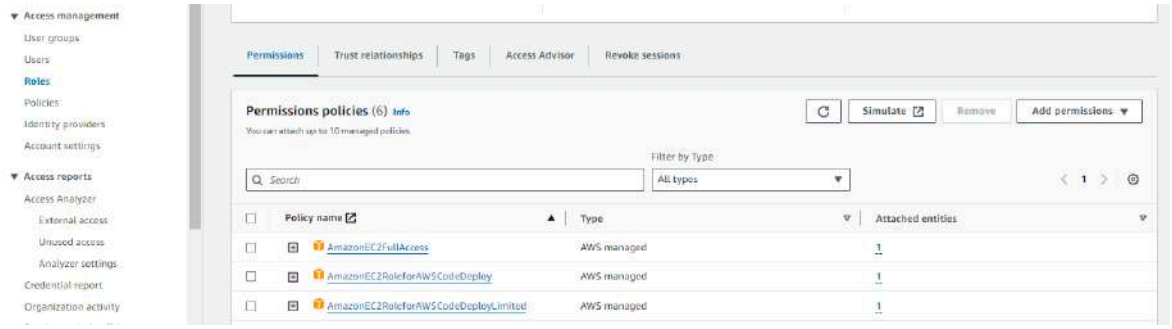
Troubleshooting:

- **AWS CodeDeploy does not have the permissions required to assume the role,** that means You have a role that has the permissions required for the codedeploy to perform the deployment. What you are missing here is, You should have a trust policy defined in the role that allows codedeploy to assume the role, to resolve this error follow the steps below.

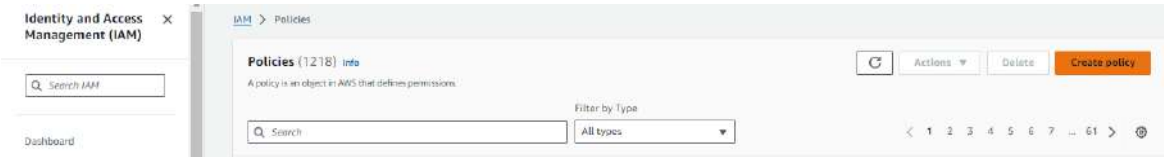


1.Go to the **IAM console** and select the **role** from the roles section.

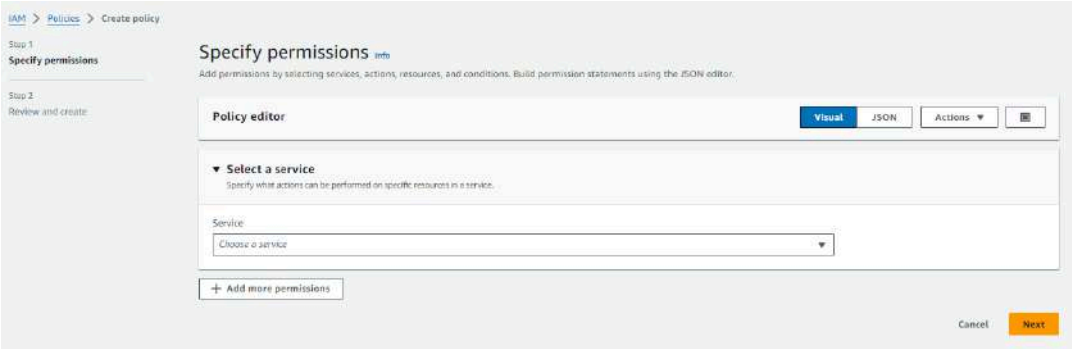
2. Click **Trust relationships**

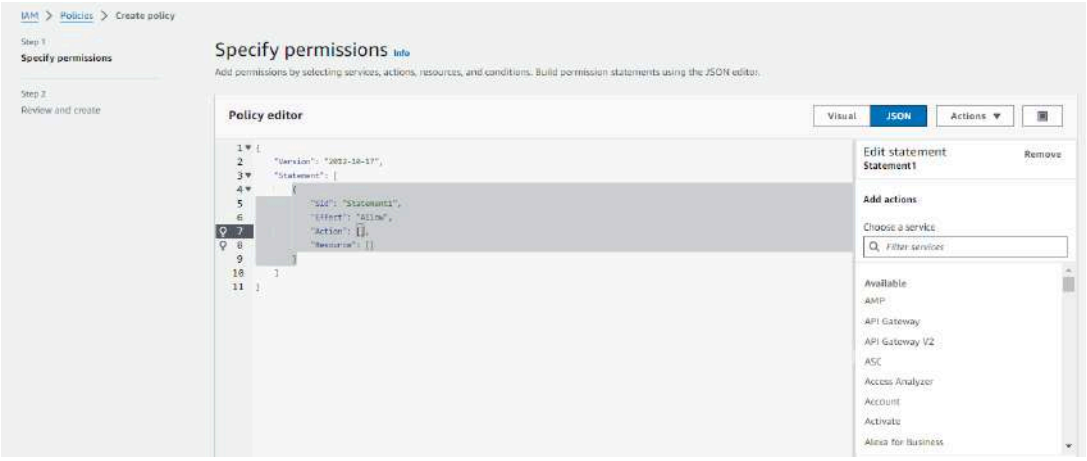


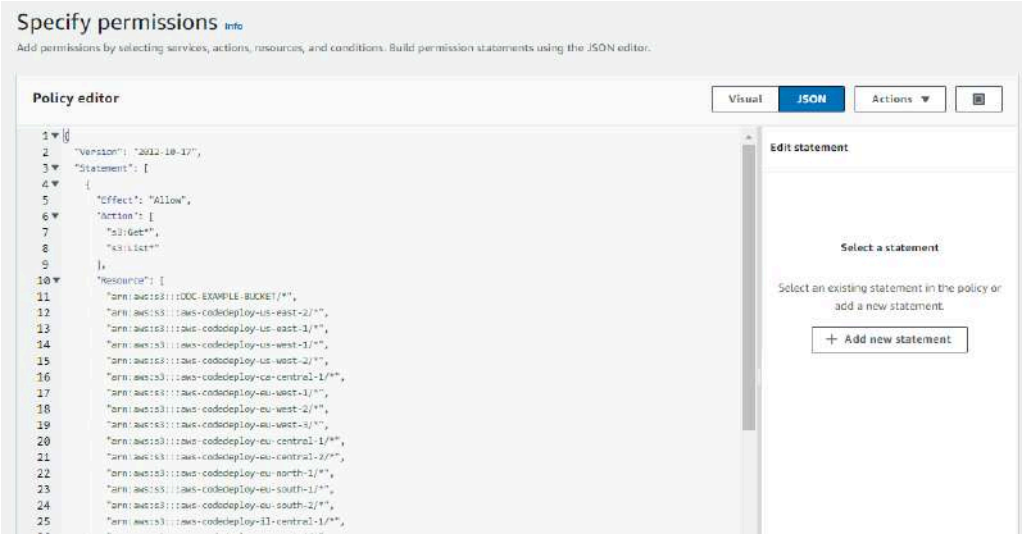
2. Click on **Create policy**

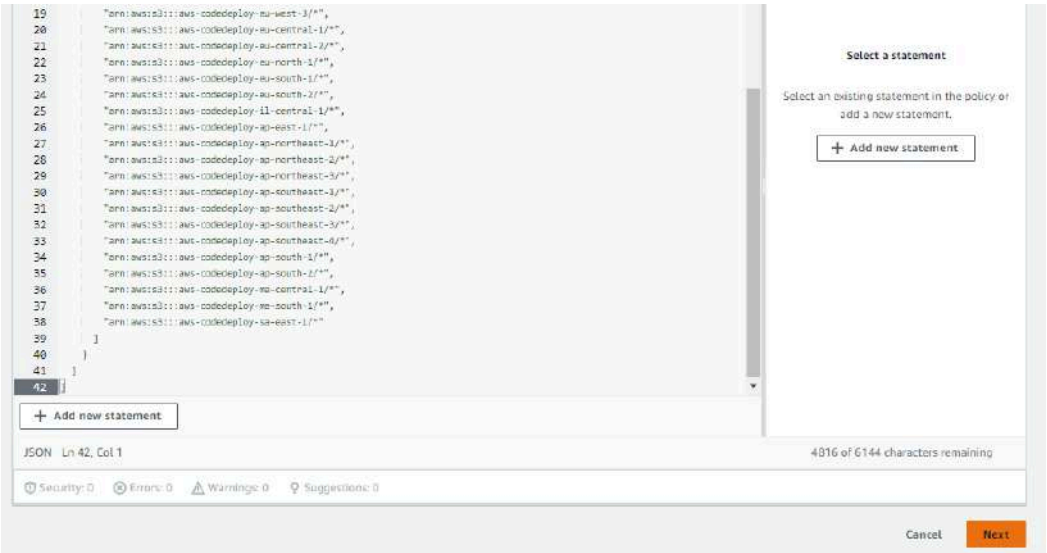


3. Specify permissions

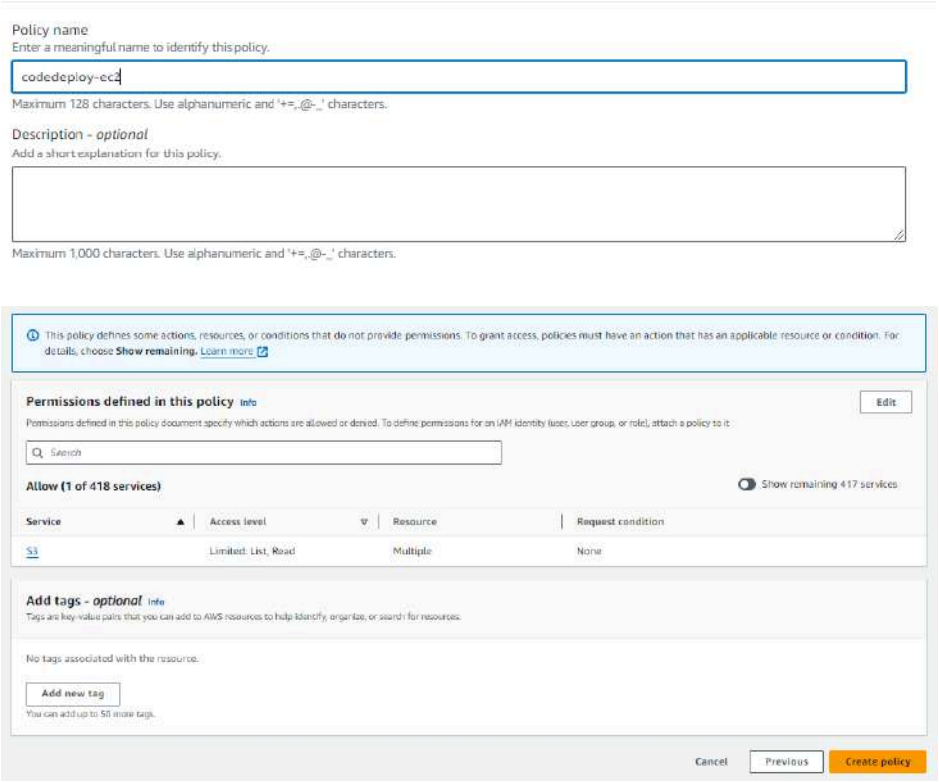




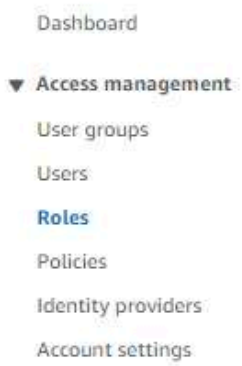




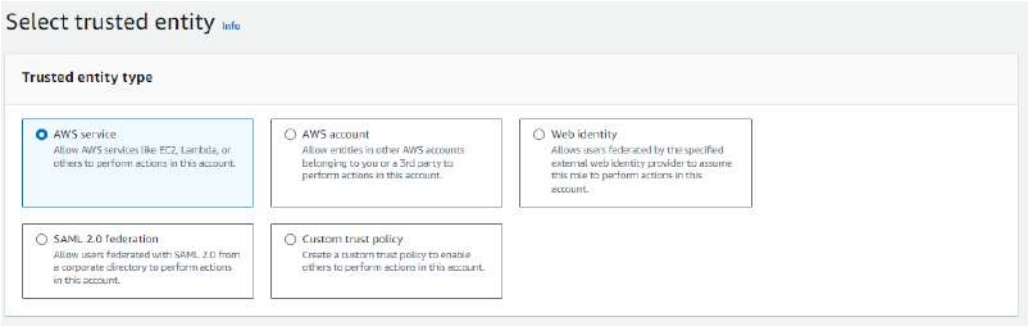
4. In **Review and create**, enter the **policy name.**

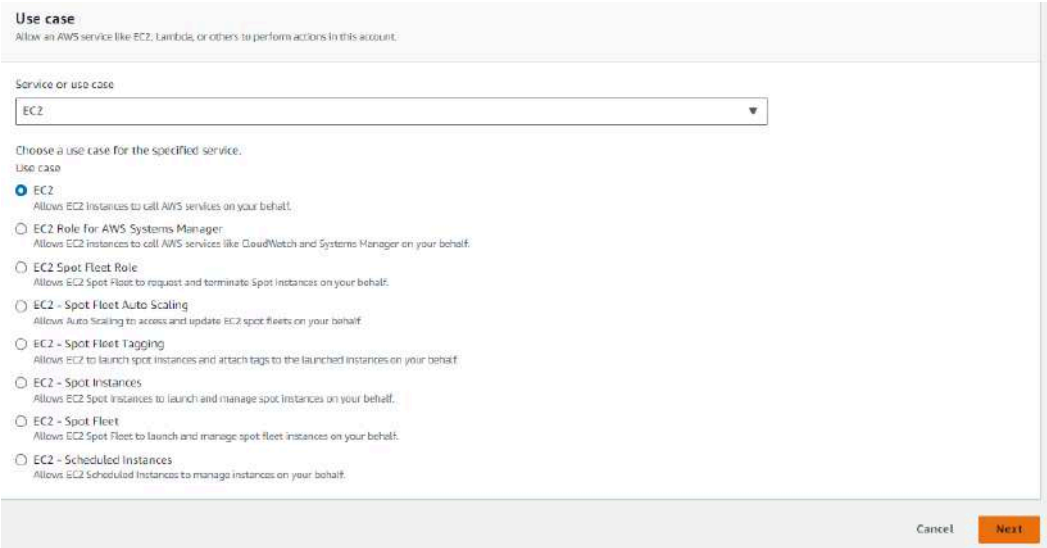


➢ Now Click on **Roles**, then Click on **Create role**.

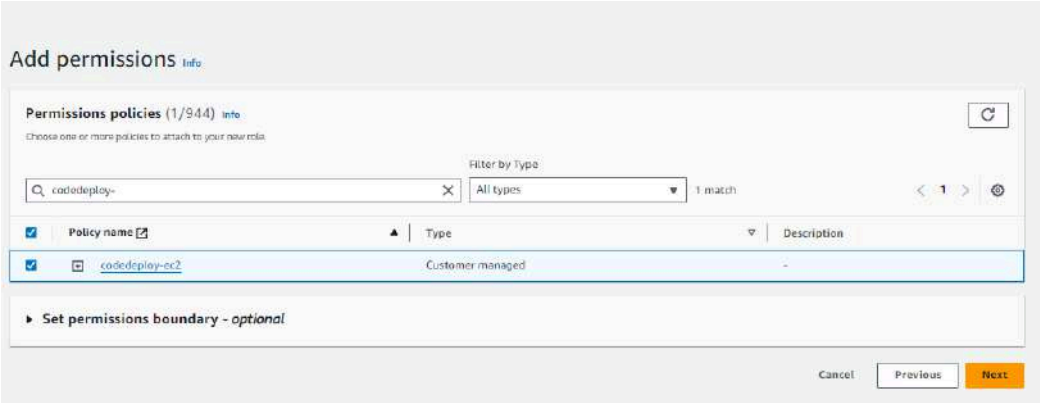


Step 1: In **Select trusted entity**, Click on **AWS Service** and in **Use case** Select **EC2** service, then Click on **Next**.

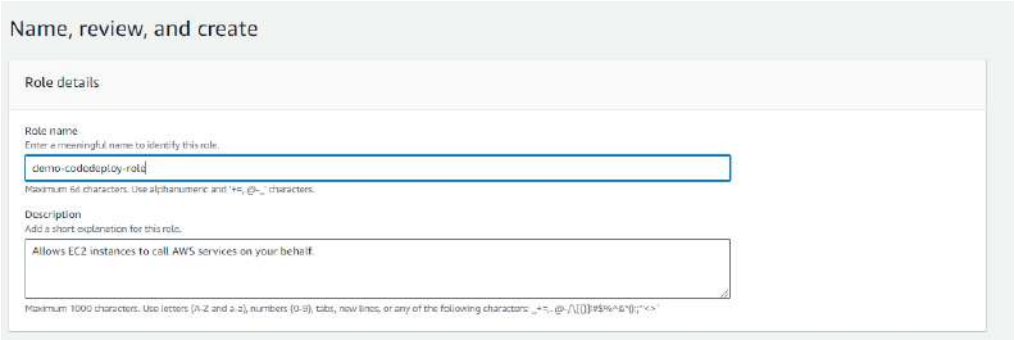


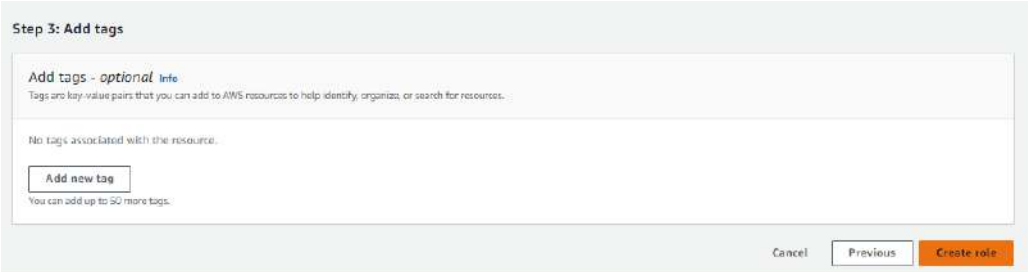


Step 2: In this step, you need to **Add permissions** and Click on **Next**.

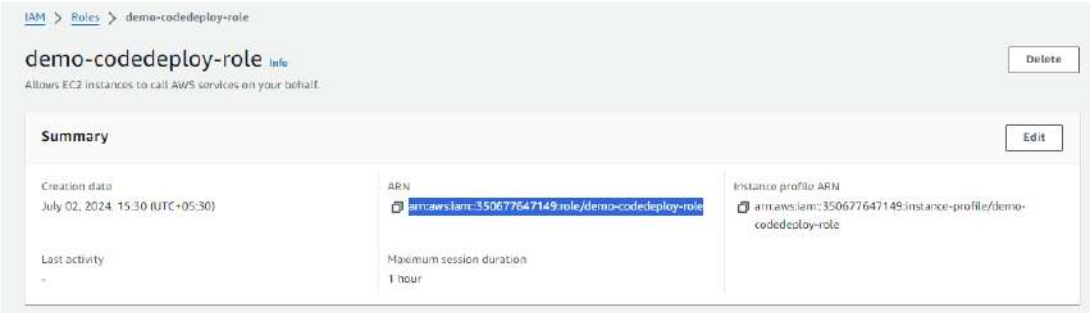


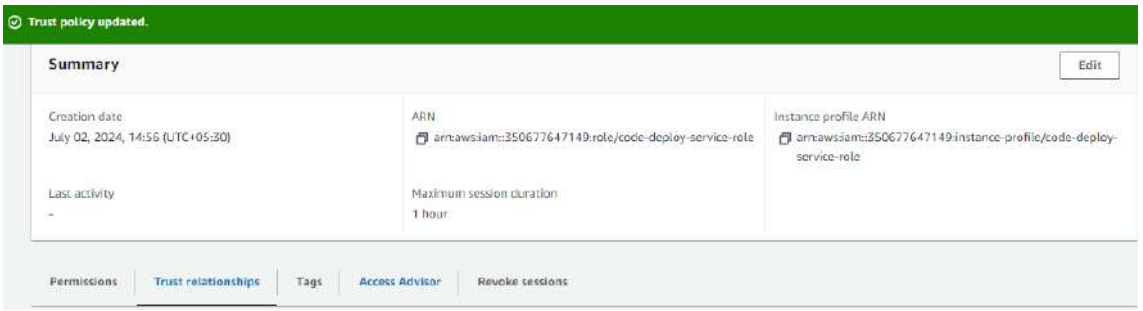
Step 3: In **Name, review, and create**, give the **Role name** and Click on **Create role.**

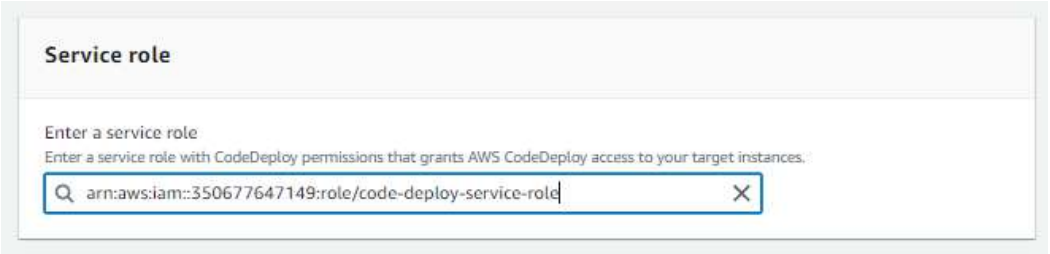


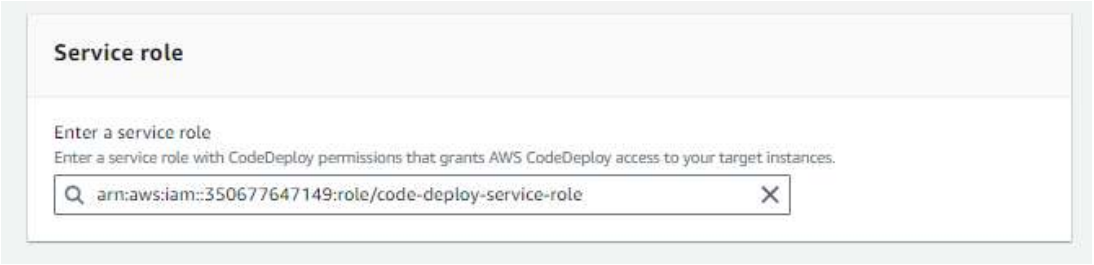


➢ Now Click on the created Role and Copy the **ARN** of the created role and paste it in the **Service role.**



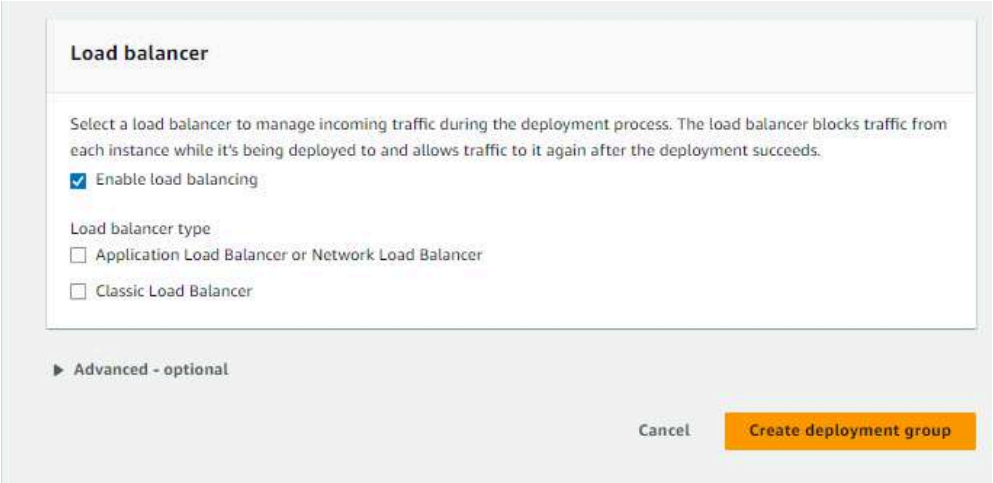




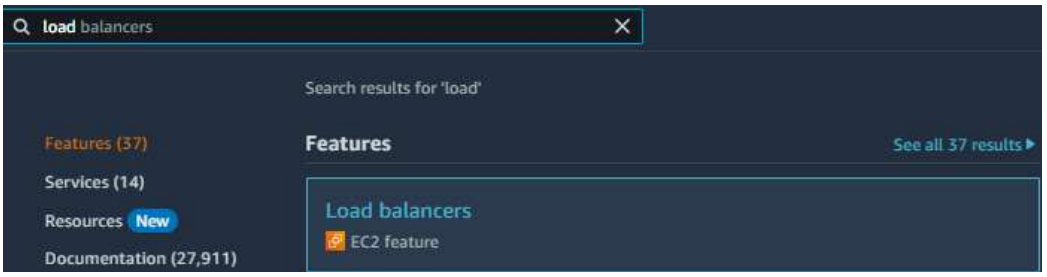


Step 7: Launch ELB to distribute the load

7.1 To distribute the load we need to select a **Load balancer type**, to select the load balancer type, first we need to **create a load balancer.**



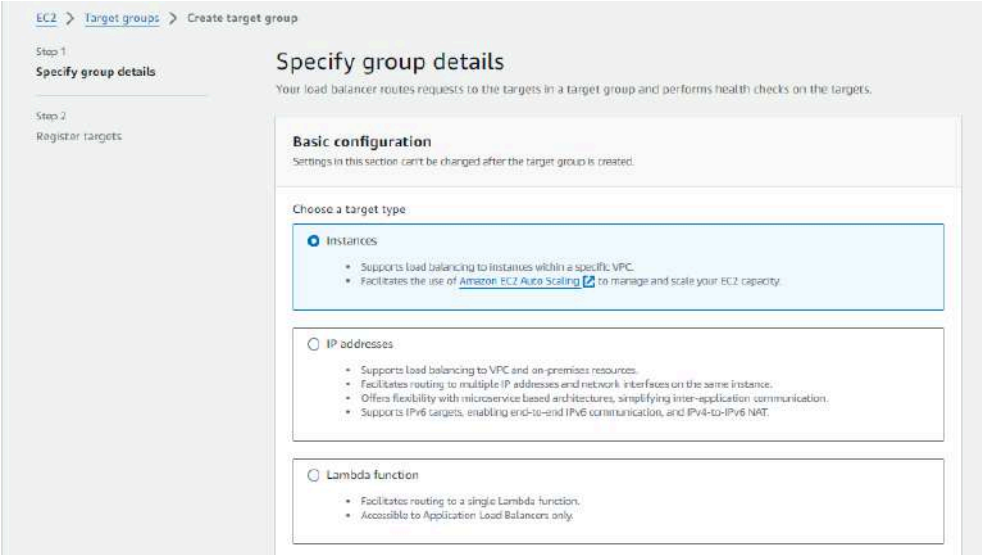
7.2 To create a load balancer, we need to search **load balancer** in AWS Console, Click on **Load balancer** to redirect to load balancer page, then Click on **target groups** .



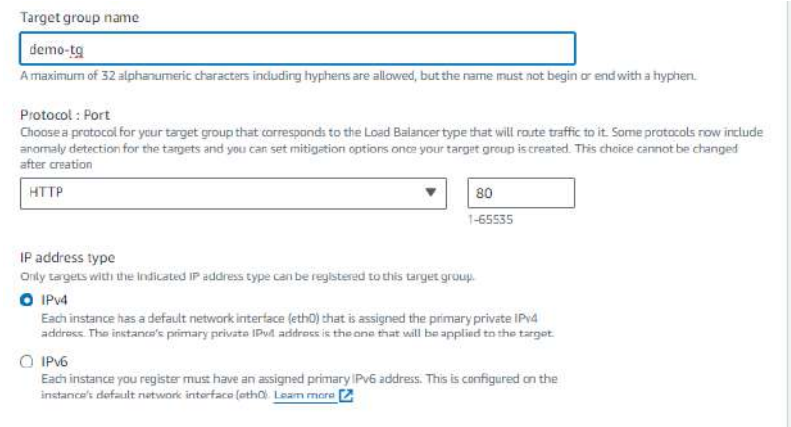
7.3 Now Click on **Create target group**

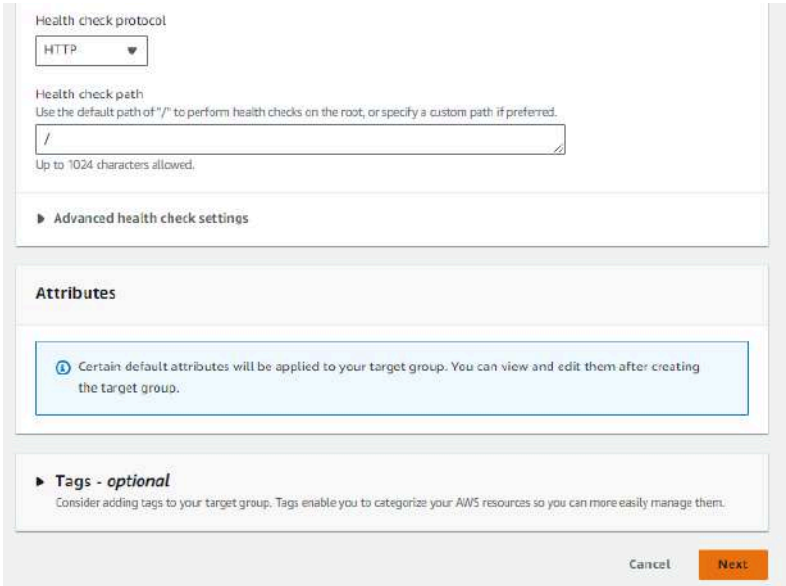
A target group is a collection of endpoints (such as EC2 instances, IP addresses, or Lambda functions) that a load balancer routes traffic to. Target groups are used to direct the traffic from the load balancer to specific targets based on rules and conditions.

7.4 To specify the target group, where the application traffic should be distributed from the load balancer, to do that you need to **Choose a target group type.**

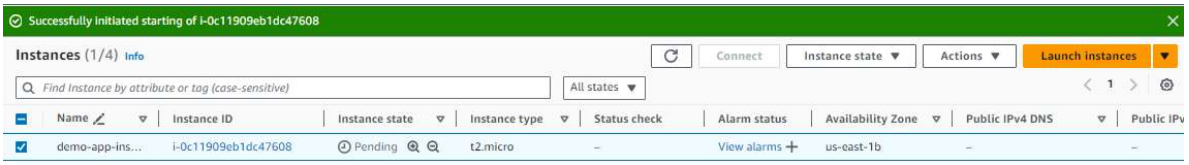


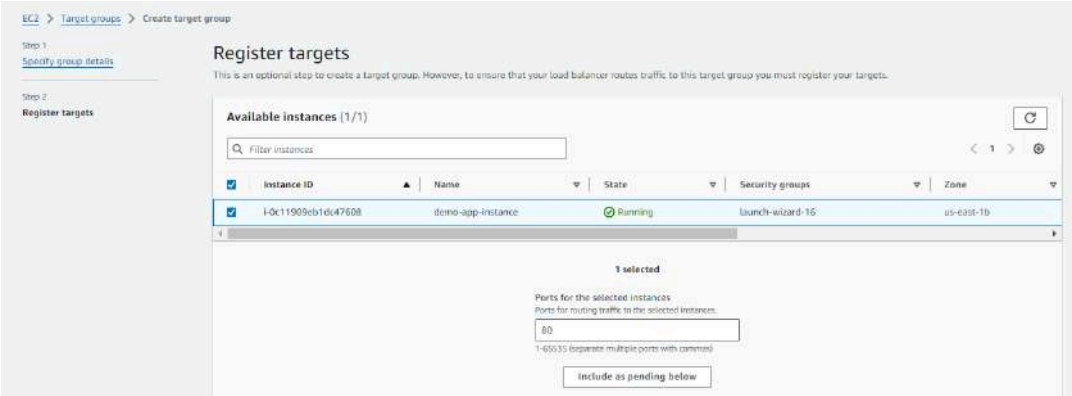
7.5 Give the **Target group name**, then Select **IP address type** as **IPV4** and Click on **Next**.

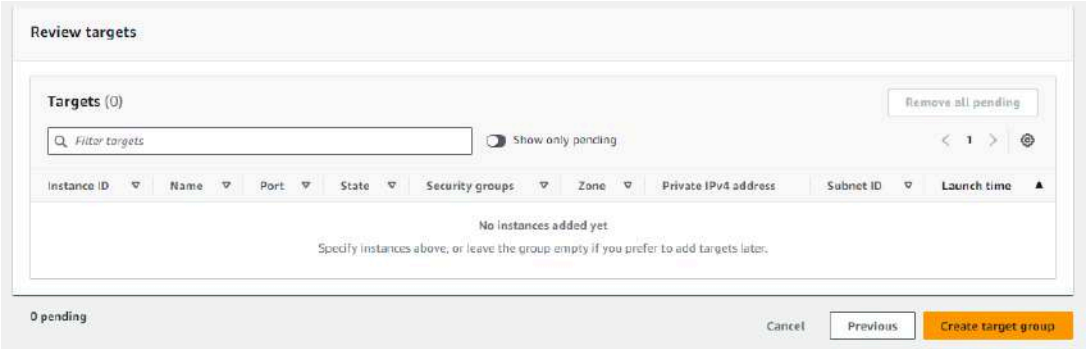


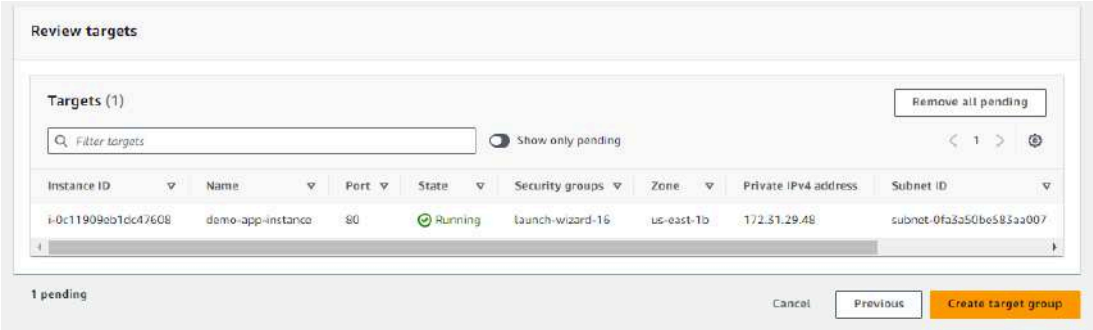


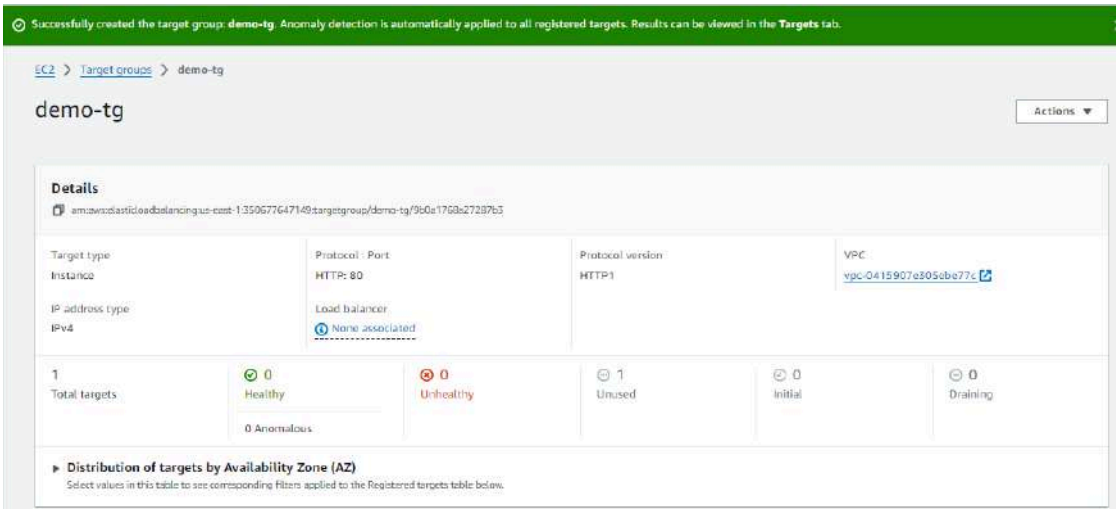
7.6 To register the created instance, Click on **Created Instance**, then **Target groups,** then **Create target group**, In **Register targets**, select the **created EC2 Instance**, then Click on **Include as pending below,** and wait for some time to see the registered instance in **Review targets,** then click on **Create target group.**

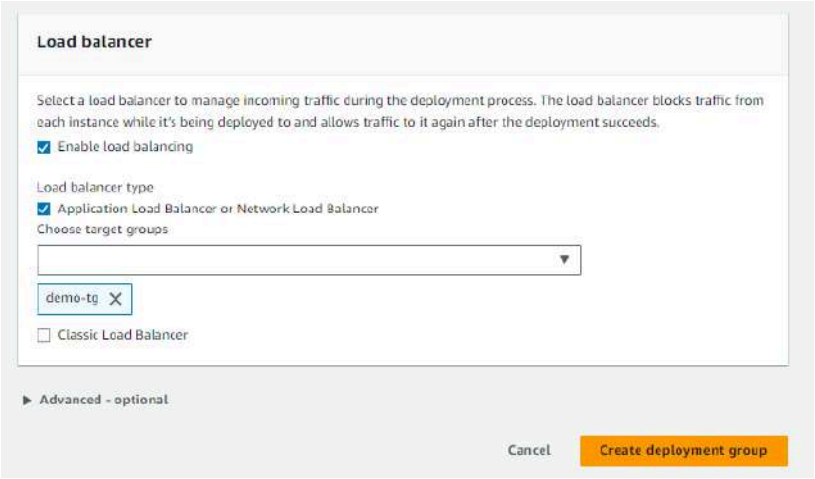


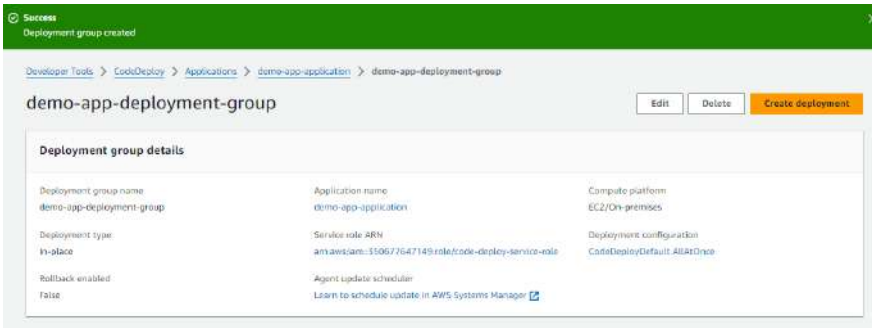

EC2 > Target groups > Create target group
Step 1
Specify group details
Step 2
Register targets
Register targets
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.
Available instances (1/1)
Filter instances
Instance ID
Name
State
Security groups
Zone
i-0c11909eb1dc47608
demo-app-instance
Running
launch-wizard-16
us-east-1b
1 selected
Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with commas)
Include as pending below


Review targets
Targets (0)
Remove all pending
Filter targets
Show only pending
Instance ID
Name
Port
State
Security groups
Zone
Private IPv4 address
Subnet ID
Launch time
No instances added yet
Specify instances above, or leave the group empty if you prefer to add targets later.
0 pending
Cancel
Previous
Create target group


Review targets
Targets (1)
Remove all pending
Filter targets
Show only pending
Instance ID
Name
Port
State
Security groups
Zone
Private IPv4 address
Subnet ID
i-0c11909eb1dc47608
demo-app-instance
80
Running
launch-wizard-16
us-east-1b
172.31.29.48
subnet-0fa3a50be583aa007
1 pending
Cancel
Previous
Create target group

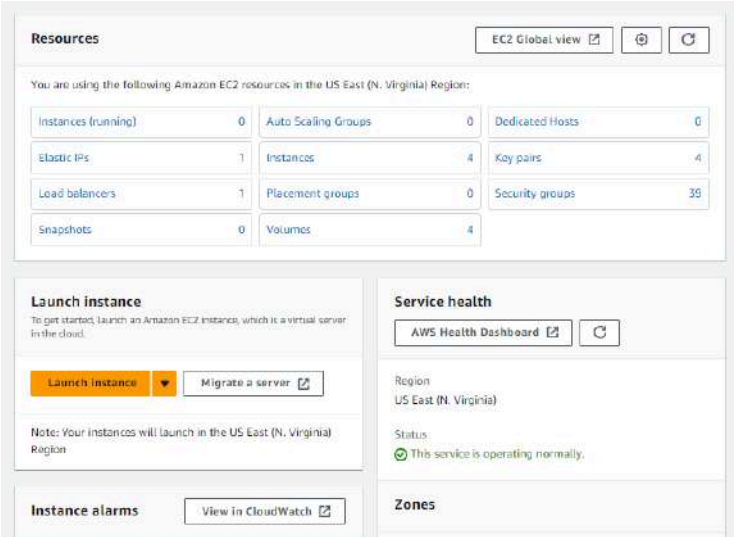

Successfully created the target group: demo-tg. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.
EC2 > Target groups > demo-tg
demo-tg
Actions
Details
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
IP address type
IPv4
Load balancer
None associated
1
Total targets
0
Healthy
0 Anomalous
0
Unhealthy
1
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

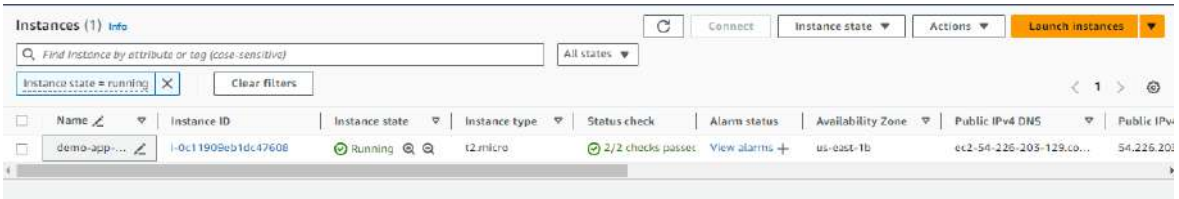


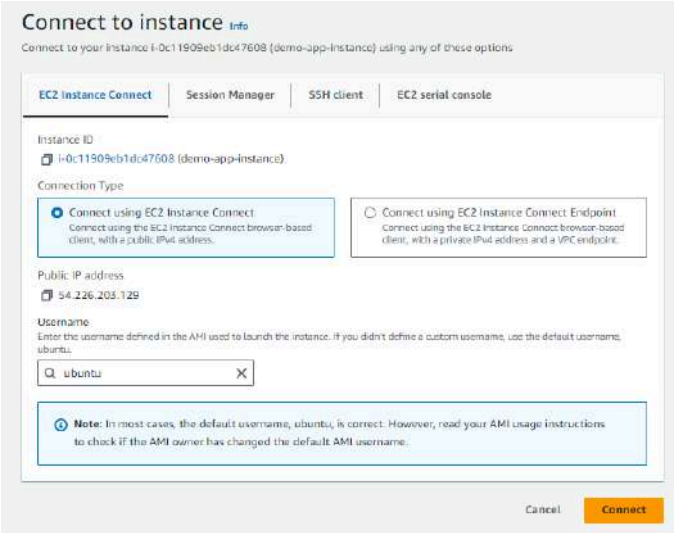


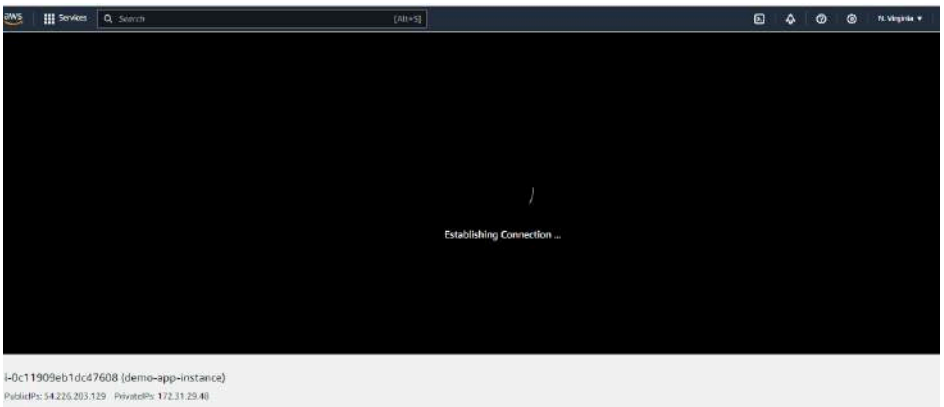
Step 8: Setting up code deploy agent on Ubuntu EC2 Instance

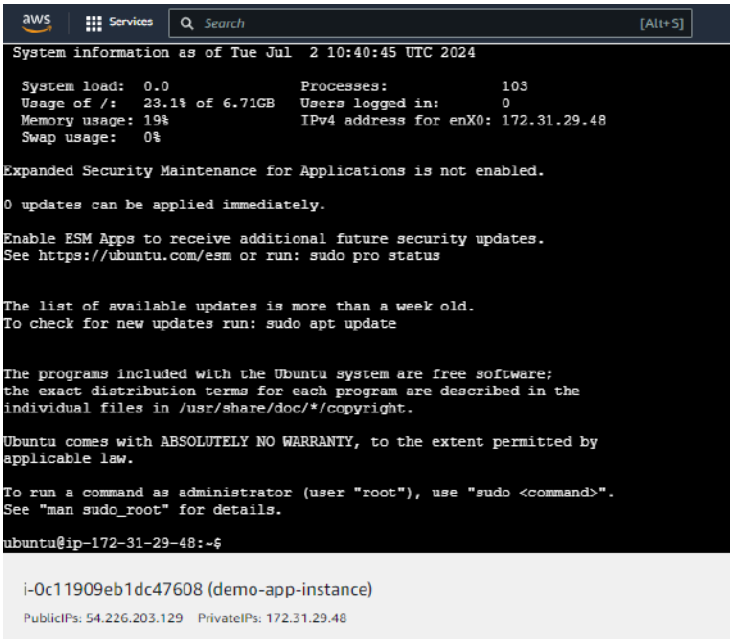
8.1 To install code deploy agent on Ubuntu EC2 Instance, Go to **EC2 Instance,** Click on **instances,** then Click on **created EC2 Instance,** Now to connect to instance, Select **EC2 Instance Connect,** Click on **Connect**, You can see **connect is establishing** now.



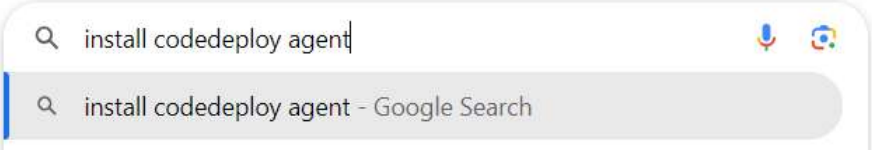

Instances (1) Info
Connect
Instance state
Actions
Launch instances
Find instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
demo-app-...
i-0c11909eb1dc47608
Running
t2.micro
2/2 checks passed
View alarms +
us-east-1b
ec2-54-226-203-129.co...

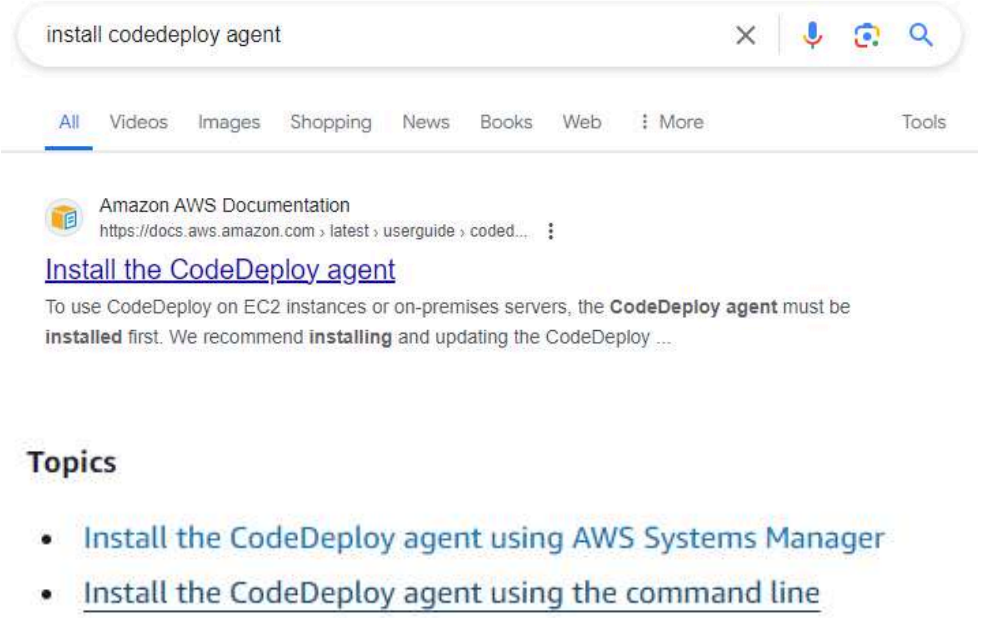

Connect to instance Info
Connect to your instance i-0c11909eb1dc47608 (demo-app-instance) using any of these options
EC2 Instance Connect
Session Manager
SSH client
EC2 serial console
Instance ID
i-0c11909eb1dc47608 (demo-app-instance)
Connection Type
Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 address.
Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.
Public IP address
54.226.203.129
Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.
ubuntu
Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.
Cancel
Connect


Services
Search
[Alt+S]
N. Virginia
Establishing Connection ...
i-0c11909eb1dc47608 (demo-app-instance)
PublicIPs: 54.226.203.129 PrivateIPs: 172.31.29.40



8.2 Now to Install Codedeploy agent, search on URL as **Install codedeploy agent,** Now Click on Install codedeploy agent, Now Click on **Install the codedeploy agent using the Command line,** Now Click on **Install the codedeploy agent for Ubuntu server**, to install the Codedeploy agent on Ubuntu server.





Install the CodeDeploy agent using the command line

PDF | RSS

> **Note**
> We recommend installing the CodeDeploy agent with AWS Systems Manager to be able to configure scheduled updates of the agent. For more information, see Install the CodeDeploy agent using AWS Systems Manager.

Use the following topics to install and run the CodeDeploy agent using the command line.

Topics

- Install the CodeDeploy agent for Amazon Linux or RHEL
- Install the CodeDeploy agent for Ubuntu Server
- Install the CodeDeploy agent for Windows Server

8.3 To **Install codedeploy agent on EC2 Ubuntu**, Follow the commands.

1. To install the CodeDeploy agent on Ubuntu Server
2. Enter the following commands, one after the other:

Install the CodeDeploy agent for Ubuntu Server

PDF | RSS

> **Note**
> We recommend installing the CodeDeploy agent with AWS Systems Manager to be able to configure scheduled updates of the agent. For more information, see Install the CodeDeploy agent using AWS Systems Manager.

To install the CodeDeploy agent on Ubuntu Server

1. Sign in to the instance.
2. Enter the following commands, one after the other:

```
sudo apt update
```

- **The following commands are used to install codedeploy agent on EC2 Ubuntu and Troubleshooting commands and install codedeploy agent on EC2 Ubuntu**

```
Auto (SQL)
sudo apt update
sudo apt install ruby-full
sudo apt --fix-broken install
sudo apt-get remove ruby-full
sudo apt update
sudo apt-get install -y software-properties-common
sudo apt-add-repository ppa:brightbox/ruby-ng
sudo apt update
sudo apt-get install -y ruby3.0
sudo apt-get install codedeploy-agent
sudo apt update
sudo apt install ruby-full
sudo apt install wget
chmod +x ./install
sudo ./install auto
sudo apt update
sudo apt install ruby-full
sudo apt install wget
cd /home/ubuntu
wget https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
la -al install
chmod +x ./install
sudo ./install auto
systemctl status codedeploy-agent
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt update
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
89 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-29-48:~$
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt install ruby-full
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
89 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-29-48:~$ sudo apt install ruby-full
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ruby-full is already the newest version (1:3.2~ubuntu1).
You might want to run 'apt --fix-broken install' to correct these.
The following packages have unmet dependencies:
 codedeploy-agent : Depends: ruby3.0 but it is not installable
E: Unmet dependencies. Try 'apt --fix-broken install' with no packages (or specify a solution).
ubuntu@ip-172-31-29-48:~$
```

Troubleshooting:

- Here we want to download codedeploy-agent, but here we don't have latest one, so to install latest one follow the commands:

```
ubuntu@ip-172-31-29-48:~$ sudo apt --fix-broken install
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt --fix-broken install
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Correcting dependencies... Done
The following packages will be REMOVED:
  codedeploy-agent
0 upgraded, 0 newly installed, 1 to remove and 89 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
Do you want to continue? [Y/n] y
(Reading database ... 91167 files and directories currently installed.)
Removing codedeploy-agent (1.3.2-1902) ...
needrestart is being skipped since dpkg has failed
ubuntu@ip-172-31-29-48:~$
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt-get remove ruby-full
```

```
Reading state information... Done
The following packages were automatically installed and are no longer req
  fonts-lato javascript-common libgmp-dev libgmpxx4ldbl libjs-jquery libr
  ruby3.2-doc rubygems-integration unzip zip
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  ruby-full
0 upgraded, 0 newly installed, 1 to remove and 89 not upgraded.
After this operation, 12.3 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 88852 files and directories currently installed.)
Removing ruby-full (1:3.2~ubuntu1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-29-48:~$ sudo apt-get update
sudo apt-get install -y software-properties-common
sudo apt-add-repository ppa:brightbox/ruby-ng
sudo apt-get update
sudo apt-get install -y ruby3.0
```

```
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:5 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble InRelease
Err:6 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release
  404  Not Found [IP: 185.125.190.80 443]
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:5 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble InRelease
Err:6 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release
  404  Not Found [IP: 185.125.190.80 443]
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package ruby3.0 is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'ruby3.0' has no installation candidate
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt-get install codedeploy-agent
```

```
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-1008-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Jul  2 15:21:30 UTC 2024

  System load:  0.08               Processes:             106
  Usage of /:   30.2% of 6.71GB    Users logged in:       0
  Memory usage: 32%                IPv4 address for enX0: 172.31.29.48
  Swap usage:   0%

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

90 updates can be applied immediately.
55 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Tue Jul  2 12:54:18 2024 from 18.206.107.27
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt update
```

```
Expanded Security Maintenance for Applications is not enabled.

90 updates can be applied immediately.
55 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Tue Jul  2 12:54:18 2024 from 18.206.107.27
ubuntu@ip-172-31-29-48:~$ sudo apt update
Warning: The unit file, source configuration file or drop-ins of apt-news.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Ign:5 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble InRelease
Err:6 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release
  404  Not Found [IP: 185.125.190.80 443]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [189 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [48.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [139 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [27.0 kB]
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt install ruby-full
```

```
Last login: Tue Jul  2 12:54:18 2024 from 18.206.107.27
ubuntu@ip-172-31-29-48:~$ sudo apt update
Warning: The unit file, source configuration file or drop-ins of apt-news.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Warning: The unit file, source configuration file or drop-ins of esm-cache.service changed on disk. Run 'systemctl daemon-reload' to reload units.
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Ign:5 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble InRelease
Err:6 https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release
  404  Not Found [IP: 185.125.190.80 443]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [189 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [48.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [139 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [27.0 kB]
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
ubuntu@ip-172-31-29-48:~$ sudo apt install ruby-full
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ruby-full is already the newest version (1:3.2~ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 89 not upgraded.
ubuntu@ip-172-31-29-48:~$
```

```
ubuntu@ip-172-31-29-48:~$ sudo apt install wget
```

```
Reading package lists... Done
E: The repository 'https://ppa.launchpadcontent.net/brightbox/ruby-ng/ubuntu noble Release' does not have a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
ubuntu@ip-172-31-29-48:~$ sudo apt install ruby-full
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ruby-full is already the newest version (1:3.2~ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 89 not upgraded.
ubuntu@ip-172-31-29-48:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
0 upgraded, 0 newly installed, 0 to remove and 89 not upgraded.
ubuntu@ip-172-31-29-48:~$ cd /home/ubuntu
ubuntu@ip-172-31-29-48:~$ wget https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
--2024-07-02 15:27:38--  https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
Resolving aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com (aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com)... 52.217.192.98, 52.217.232.234, 16.182.71.226, ...
Connecting to aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com (aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com)|52.217.192.98|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 19045 (19K) []
Saving to: 'install'

install                                   100%[===================================================================================>]  18.60K  --.-KB/s    in 0s

2024-07-02 15:27:38 (139 MB/s) - 'install' saved [19045/19045]
```

```
2024-07-02 15:27:38 (139 MB/s) - 'install' saved [19045/19045]

ubuntu@ip-172-31-29-48:~$ chmod +X ./install
ubuntu@ip-172-31-29-48:~$ sudo ./install auto
```

- **sudo apt update**

```
 System information as of Tue Jul  2 15:37:58 UTC 2024

  System load:  0.14               Processes:             106
  Usage of /:   23.1% of 6.71GB    Users logged in:       0
  Memory usage: 19%                IPv4 address for enX0: 172.31.27.96
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-27-96:~$ sudo apt update
```

```
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [215 kB]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [58.6 kB]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [8564 B]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [112 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [40.8 kB]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [45.0 kB]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [112 B]
Get:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [139 kB]
Get:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [27.0 kB]
Get:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [14.1 kB]
Get:42 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [3608 B]
Get:43 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [212 B]
Get:44 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [116 B]
Get:45 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]
Get:46 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [112 B]
Get:47 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [8728 B]
Get:48 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [10.0 kB]
Get:49 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [17.6 kB]
Get:50 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [116 B]
Get:51 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:52 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:53 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:54 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 29.5 MB in 6s (5176 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
90 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-27-96:~$
```

- **sudo apt install ruby-full**

```
ubuntu@ip-172-31-27-96:~$ sudo apt install ruby-full
```

```
Get:46 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [112 B]
Get:47 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [8728 B]
Get:48 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [10.0 kB]
Get:49 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [17.6 kB]
Get:50 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [116 B]
Get:51 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:52 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:53 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:54 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 29.5 MB in 6s (5176 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
90 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-27-96:~$ sudo apt install ruby-full
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-lato javascript-common libgmp-dev libgmpxx4ldbl libjs-jquery libruby libruby3.2 rake ri ruby ruby-dev
  ruby3.2 ruby3.2-dev ruby3.2-doc rubygems-integration unzip zip
Suggested packages:
  apache2 | lighttpd | httpd gmp-doc libgmp10-doc libmpfr-dev bundler
The following NEW packages will be installed:
  fonts-lato javascript-common libgmp-dev libgmpxx4ldbl libjs-jquery libruby libruby3.2 rake ri ruby ruby-dev
  ruby-xmlrpc ruby3.2 ruby3.2-dev ruby3.2-doc rubygems-integration unzip zip
0 upgraded, 23 newly installed, 0 to remove and 90 not upgraded.
Need to get 12.1 MB of archives.
After this operation, 65.4 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
Setting up ruby-webrick (1.8.1-1) ...
Setting up libjs-jquery (3.6.1+dfsg+~3.5.14-1) ...
Setting up ruby-xmlrpc (0.3.2-2) ...
Setting up libgmp-dev:amd64 (2:6.3.0+dfsg-2ubuntu6) ...
Setting up ruby3.2 (3.2.3-1ubuntu0.24.04.1) ...
Setting up libruby:amd64 (1:3.2~ubuntu1) ...
Setting up ri (1:3.2~ubuntu1) ...
Setting up ruby (1:3.2~ubuntu1) ...
Setting up rake (13.0.6-3) ...
Setting up ruby-sdbm:amd64 (1.0.0-5build4) ...
Setting up libruby3.2:amd64 (3.2.3-1ubuntu0.24.04.1) ...
Setting up ruby-rubygems (3.4.20-1) ...
Setting up ruby3.2-dev:amd64 (3.2.3-1ubuntu0.24.04.1) ...
Setting up ruby-dev:amd64 (1:3.2~ubuntu1) ...
Setting up ruby-full (1:3.2~ubuntu1) ...
Processing triggers for libc-bin (2.39-0ubuntu8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-27-96:~$
```

- **sudo apt install wget**

```
ubuntu@ip-172-31-27-96:~$ sudo apt install wget
```

```
ubuntu@ip-172-31-27-96:~$ sudo apt install wget
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be upgraded:
  wget
1 upgraded, 0 newly installed, 0 to remove and 89 not upgraded.
Need to get 334 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 334 kB in 0s (13.5 MB/s)
(Reading database ... 88851 files and directories currently installed.)
Preparing to unpack .../wget_1.21.4-1ubuntu4.1_amd64.deb ...
Unpacking wget (1.21.4-1ubuntu4.1) over (1.21.4-1ubuntu4) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-27-96:~$
```

3. **Enter the following command:**

```
ubuntu@ip-172-31-27-96:~$ cd /home/ubuntu
```

/home/ubuntu represents the default user name for an Ubuntu Server instance. If your instance was created using a custom AMI, the AMI owner might have specified a different default username.

4. **Enter the following command:**

   wget https://bucket-name.s3.region-identifier.amazonaws.com/latest/install

- bucket-name is the name of the Amazon S3 bucket that contains the CodeDeploy Resource Kit files for your region, and region-identifier is the identifier for your region.

For my region

wget https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install

```
ubuntu@ip-172-31-27-96:~$ wget https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
```

```
Unpacking wget (1.21.4-1ubuntu4.1) over (1.21.4-1ubuntu4) ...
Setting up wget (1.21.4-1ubuntu4.1) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-27-96:~$ cd /home/ubuntu
ubuntu@ip-172-31-27-96:~$ wget https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
--2024-07-02 15:52:51--  https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/install
Resolving aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com (aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com)... 54.231.229.250, 54.231.232.194, 52.216.30.80, ...
Connecting to aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com (aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com)|54.231.229.250|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 19045 (19K) []
Saving to: 'install'

install                                   100%[===================================================================================>]  18.60K  --.-KB/s    in 0.001s

2024-07-02 15:52:51 (31.7 MB/s) - 'install' saved [19045/19045]

ubuntu@ip-172-31-27-96:~$
```

```
ubuntu@ip-172-31-27-96:~$ ls -al install
-rw-rw-r-- 1 ubuntu ubuntu 19045 Feb 22 18:24 install
ubuntu@ip-172-31-27-96:~$
```

5. **Enter the following command:**
6. **Do one of the following:**

- **To install the latest version of the CodeDeploy agent on any supported version of Ubuntu Server *except* 20.04:**

```
ubuntu@ip-172-31-27-96:~$ chmod +x ./install
ubuntu@ip-172-31-27-96:~$ sudo ./install auto
```

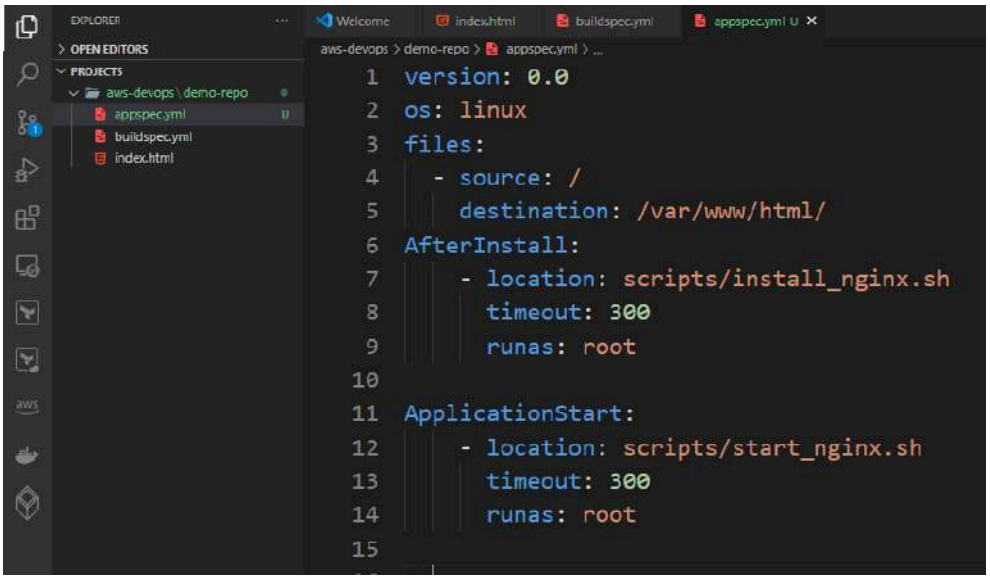
```
I, [2024-07-02T15:57:41.591800 #2366]  INFO -- : Checking EC2 metadata service for region information...
I, [2024-07-02T15:57:41.605471 #2366]  INFO -- : Checking AWS_DOMAIN environment variable for domain information...
I, [2024-07-02T15:57:41.605641 #2366]  INFO -- : Checking EC2 metadata service for domain information...
I, [2024-07-02T15:57:41.612178 #2366]  INFO -- : Downloading version file from bucket aws-codedeploy-us-east-1 and key latest/LATEST_VERSION...
I, [2024-07-02T15:57:41.612501 #2366]  INFO -- : Endpoint: https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/latest/LATEST_VERSION
dpkg-query: package 'codedeploy-agent' is not installed and no information is available
Use dpkg --info (= dpkg-deb --info) to examine archive files.
I, [2024-07-02T15:57:41.693139 #2366]  INFO -- : Running version No running version
I, [2024-07-02T15:57:41.696105 #2366]  INFO -- : Downloading package from bucket aws-codedeploy-us-east-1 and key releases/codedeploy-agent_1.7.0-
I, [2024-07-02T15:57:41.696401 #2366]  INFO -- : Endpoint: https://aws-codedeploy-us-east-1.s3.us-east-1.amazonaws.com/releases/codedeploy-agent_1
I, [2024-07-02T15:57:41.799803 #2366]  INFO -- : Executing `/usr/bin/gdebi -n -o Dpkg::Options::=--force-confdef -o Dpkg::Options::=--force-confol
tmp-20240702-2366-hcjb2b.deb`...
/usr/bin/gdebi:113: SyntaxWarning: invalid escape sequence '\S'
  c = findall("[[(](\S+)/\S+[])]", msg)[0].lower()
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Reading state information... Done
Selecting previously unselected package codedeploy-agent.
(Reading database ... 88923 files and directories currently installed.)
Preparing to unpack .../codedeploy-agent_1.7.0-92_all.tmp-20240702-2366-hcjb2b.deb ...
Unpacking codedeploy-agent (1.7.0-92) ...
Setting up codedeploy-agent (1.7.0-92) ...
codedeploy-agent.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable codedeploy-agent
codedeploy-agent.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install is-enabled codedeploy-agent
I, [2024-07-02T15:57:47.363302 #2366]  INFO -- : Update check complete.
I, [2024-07-02T15:57:47.364628 #2366]  INFO -- : Stopping updater.
ubuntu@ip-172-31-27-96:~$
```

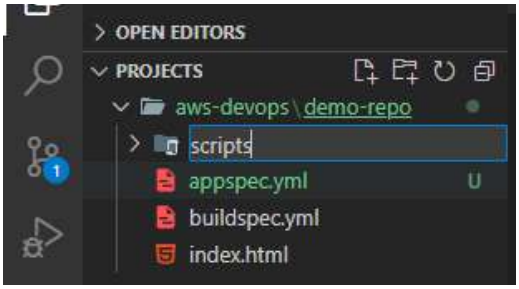
```
ubuntu@ip-172-31-27-96:~$ systemctl status codedeploy-agent
```

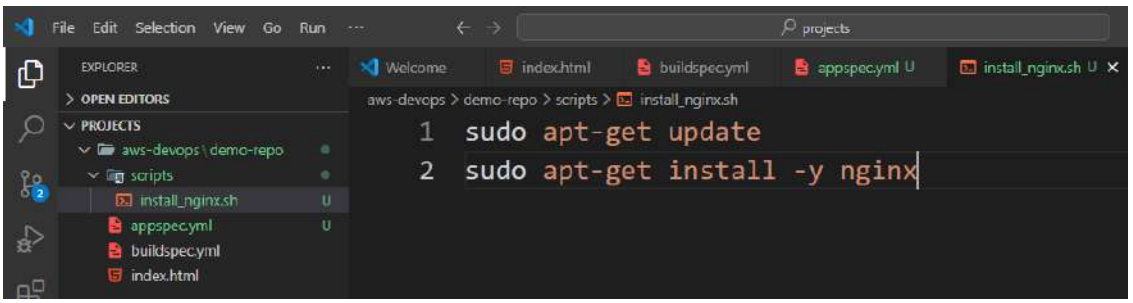
```
Reading state information... Done
Reading state information... Done
Selecting previously unselected package codedeploy-agent.
(Reading database ... 88923 files and directories currently installed.)
Preparing to unpack .../codedeploy-agent_1.7.0-92_all.tmp-20240702-2366-hcjb2b.deb ...
Unpacking codedeploy-agent (1.7.0-92) ...
Setting up codedeploy-agent (1.7.0-92) ...
codedeploy-agent.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable codedeploy-agent
codedeploy-agent.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install is-enabled codedeploy-agent
I, [2024-07-02T15:57:47.363302 #2366]  INFO -- : Update check complete.
I, [2024-07-02T15:57:47.364628 #2366]  INFO -- : Stopping updater.
ubuntu@ip-172-31-27-96:~$ systemctl status codedeploy-agent
● codedeploy-agent.service - LSB: AWS CodeDeploy Host Agent
     Loaded: loaded (/etc/init.d/codedeploy-agent; generated)
     Active: active (running) since Tue 2024-07-02 15:57:47 UTC; 1min 12s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 2655 ExecStart=/etc/init.d/codedeploy-agent start (code=exited, status=0/SUCCESS)
      Tasks: 3 (limit: 1130)
     Memory: 64.4M (peak: 64.6M)
        CPU: 1.106s
     CGroup: /system.slice/codedeploy-agent.service
             ├─2661 "codedeploy-agent: master 2661"
             └─2664 "codedeploy-agent: InstanceAgent::Plugins::CodeDeployPlugin::CommandPoller of master 2661"

Jul 02 15:57:46 ip-172-31-27-96 systemd[1]: Starting codedeploy-agent.service - LSB: AWS CodeDeploy Host Agent...
Jul 02 15:57:47 ip-172-31-27-96 codedeploy-agent[2655]: Starting codedeploy-agent:
Jul 02 15:57:47 ip-172-31-27-96 systemd[1]: Started codedeploy-agent.service - LSB: AWS CodeDeploy Host Agent.
ubuntu@ip-172-31-27-96:~$
```

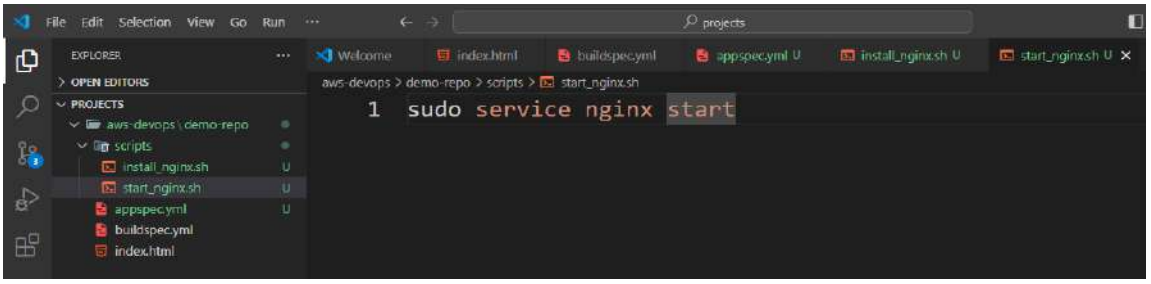
Step 9:To manage a deployment using appspecs.yml is used for CodeDeploy

9.1 To manage a deployment using appspecs.yml is used for CodeDeploy, to do that we need to write the code in any terminal.





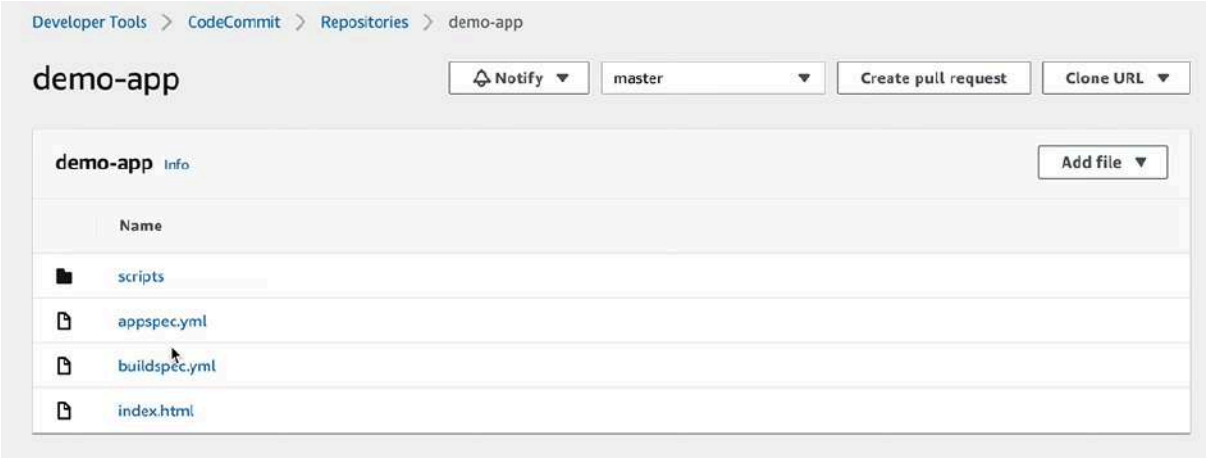




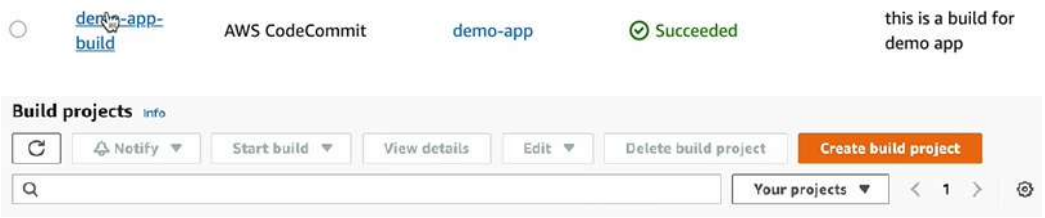

```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git add .
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git commit -m "added appspec"
[dev 8f2a7f5] added appspec
 3 files changed, 19 insertions(+)
 create mode 100644 appspec.yml
 create mode 100644 scripts/install_nginx.sh
 create mode 100644 scripts/start_nginx.sh
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo>
```

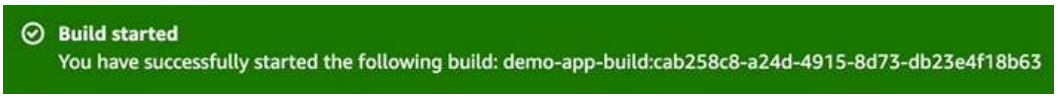
```
PS C:\Users\DELL\Desktop\projects\aws-devops\demo-repo> git push origin master
```

9.2 After pushing the appspec.yml file,the pushed files can be seen in CodeCommit.

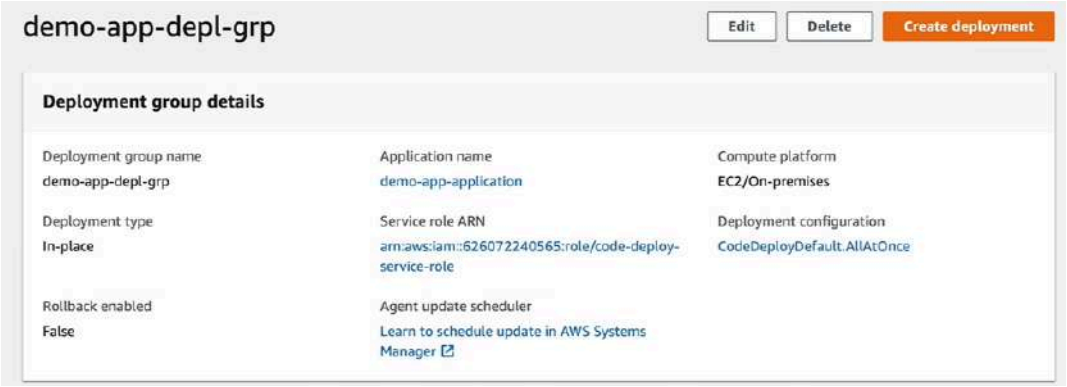


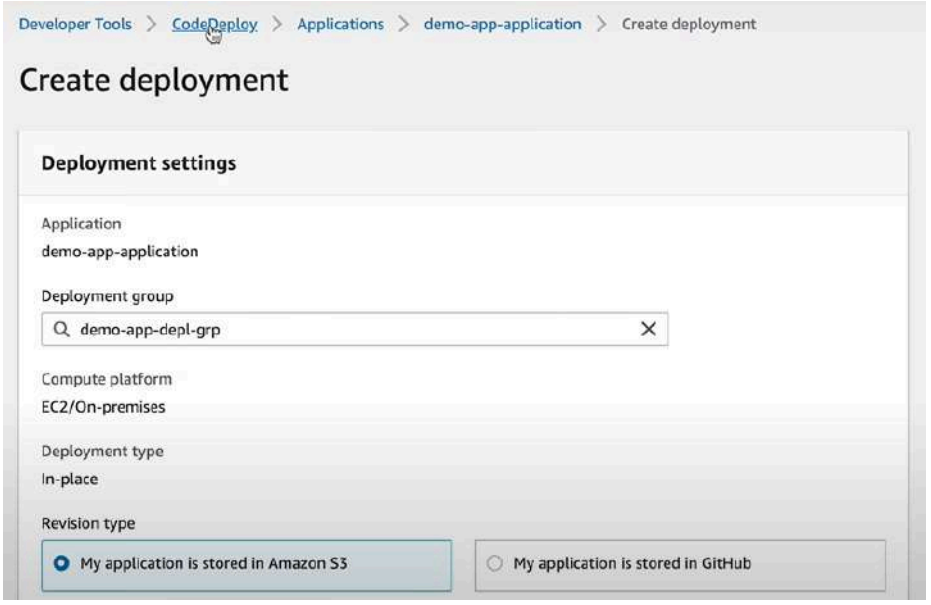
9.3 Now go to **CodeBuild,** then **Build projects,** Select the **Build project** and Click on **Create on build project,** then click on **Create build Project** to start the building the project, then **Build will be started.**



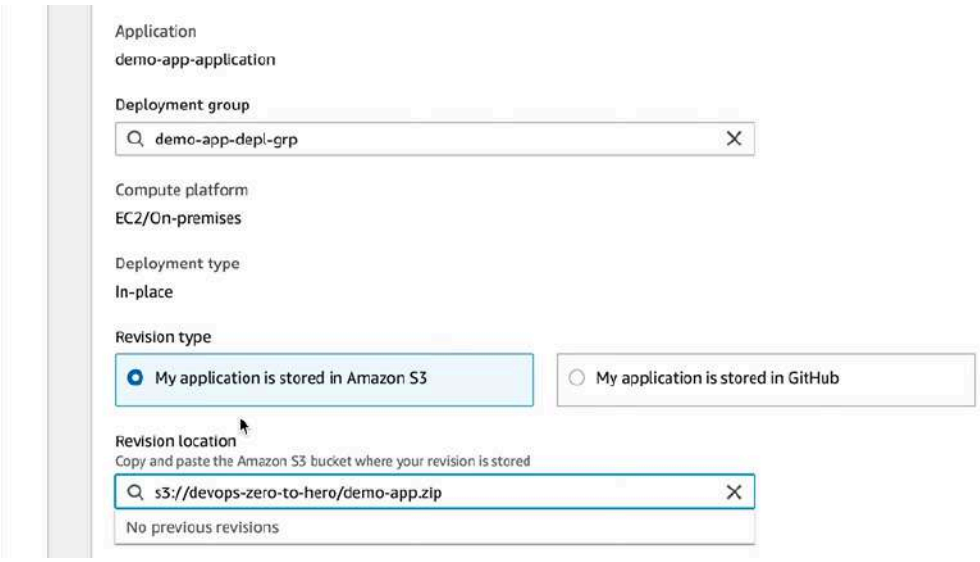


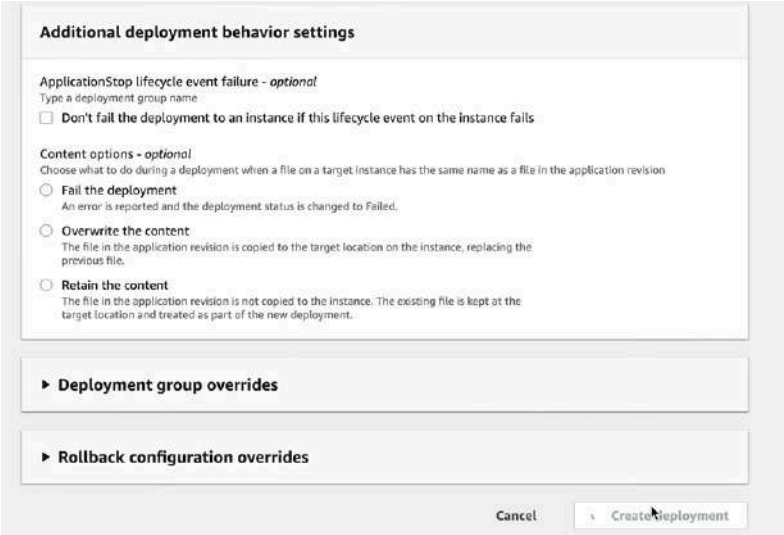
9.4 Now we have go to **deploy** the created application, for that go to **CodeDeploy,** then Click on **Application,** then Click on **Create deployment group.**

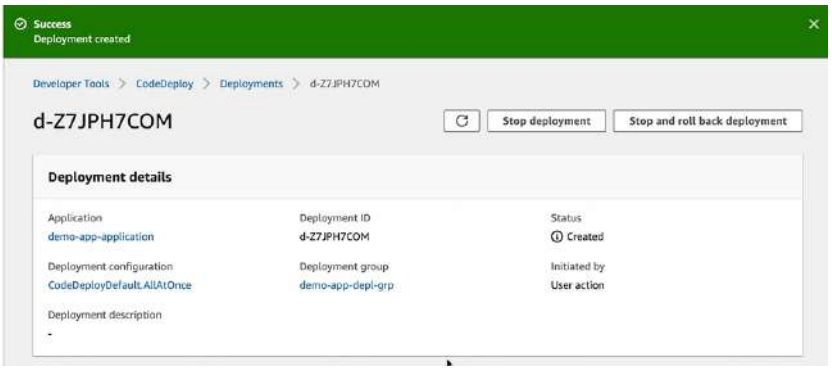



Developer Tools > CodeDeploy > Applications > demo-app-application > Create deployment
Create deployment
Deployment settings
Application
demo-app-application
Deployment group
demo-app-depl-grp
Compute platform
EC2/On-premises
Deployment type
In-place
Revision type
My application is stored in Amazon S3
My application is stored in GitHub

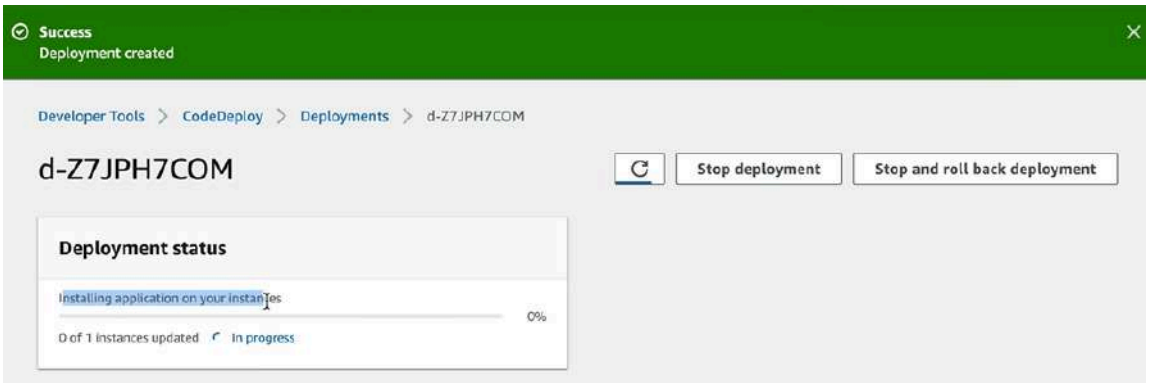
Amazon S3 > Buckets > devops-zero-to-hero > demo-app.zip
S3 URI copied
demo-app.zip Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions

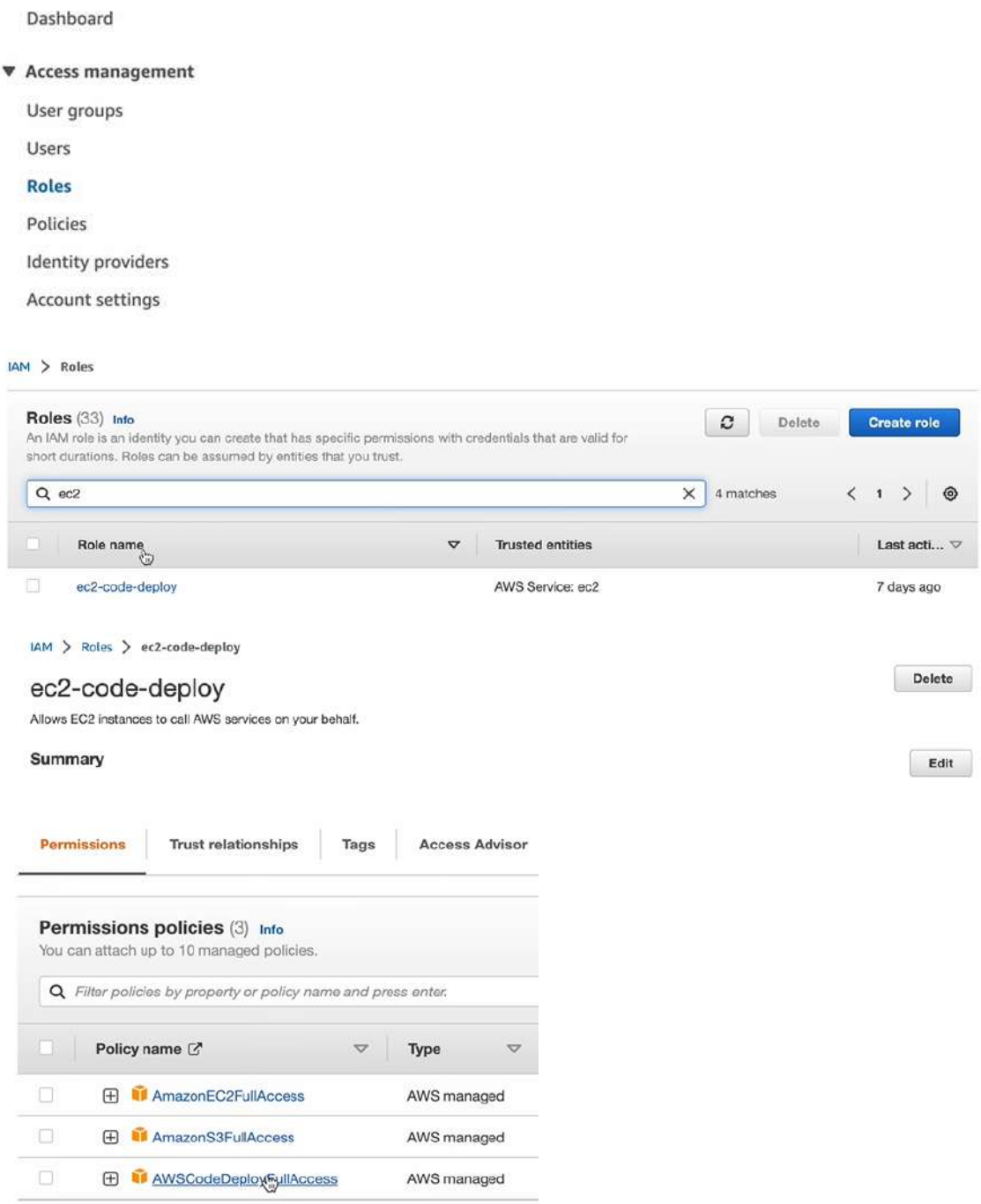

Application
demo-app-application
Deployment group
demo-app-depl-grp
Compute platform
EC2/On-premises
Deployment type
In-place
Revision type
My application is stored in Amazon S3
My application is stored in GitHub
Revision location
Copy and paste the Amazon S3 bucket where your revision is stored
s3://devops-zero-to-hero/demo-app.zip
No previous revisions





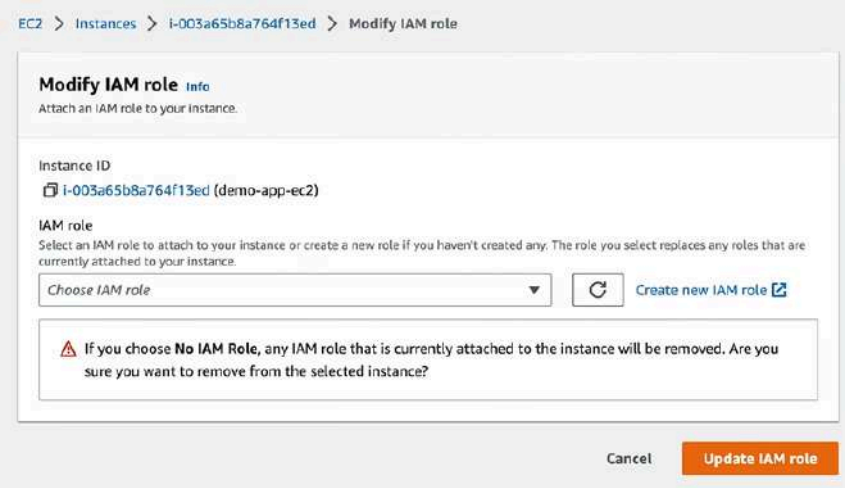
9.5 Deployment Status : Installing application on your instance we be in pending state, because we need to attach **EC2-code-deploy Role,**



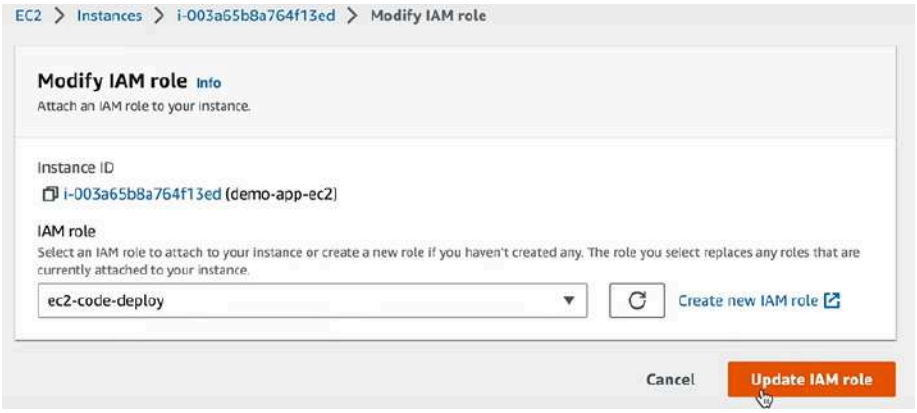


9.6 Installing applications on your instance is pending because EC2 instance does not have permission to connect with Amazon S3 and to connect with CodeDeploy.

- To give **instance permissions**, go to **Created EC2 Instance** and select the **instance,** then go to **Actions,** now Select on **security,** now Click on **Modify IAM Role.**



9.7 In **Choose IAM role,** Select **EC2-Code-Deploy,** then click on **Update IAM role.**

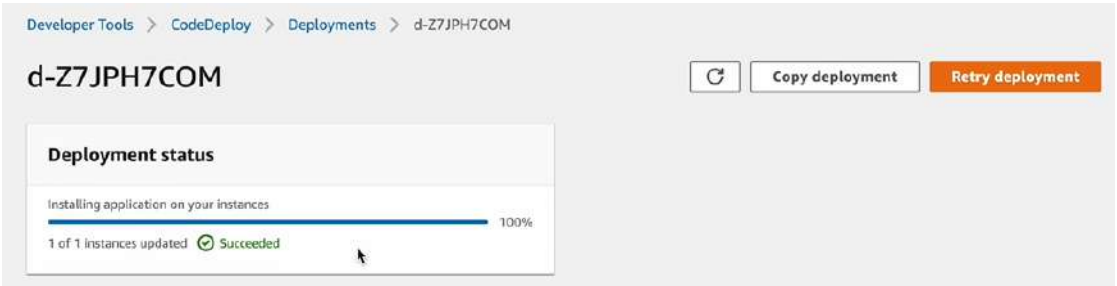


9.8 After the updating of IAM Role, **connect to EC2 Instance** and we have to give some commands, because we have changed some IAM permissions, so we have to restart the instance, Restarting will be taking some time, After restarting the codedeploy agent to check the status use the command.

- Sudo Service Codedeploy-agent restart

```
sudo service codedeploy-agent restart
```

```
sudo service codedeploy-agent status
```



my demo-repo

hello this is rohith

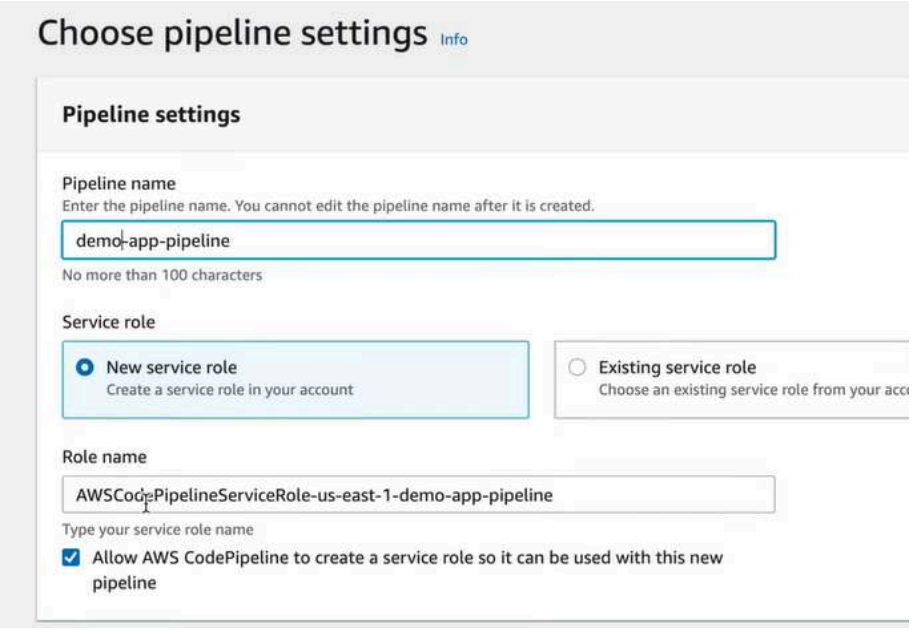
Step 10: AWS CodePipeline to streamline and automate the software release process, enabling faster delivery of features and updates.

- Using Codepipeline code will be Commit, then Commited Code will be built and then BuildCode will be Deployed for the users.

10.1 To create a pipeline goto AWS Codepipeline, Click on Create pipeline.

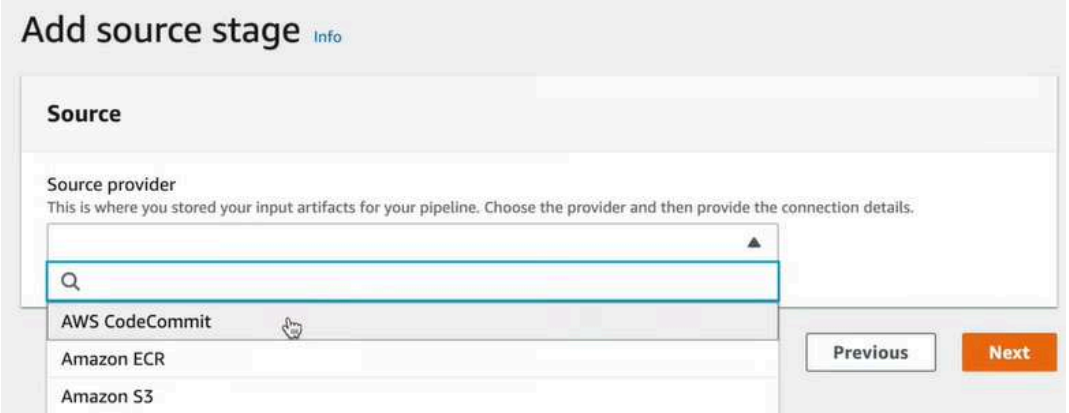
Step 1: Choose Pipeline Settings

- ➢ In **Choose pipeline settings,** give name to the **pipeline name**, which will automatically take the code then commit, then build, finally deploy it, then select the **role name,** then Click on **Next.**

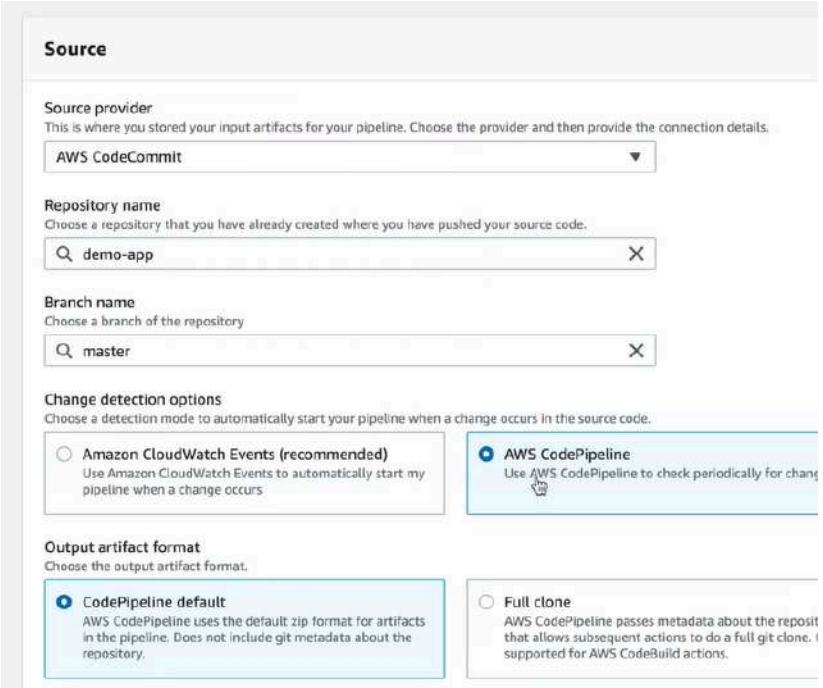


Step2 : Add source stage

➢ Add the source provider, from where the code is taken.

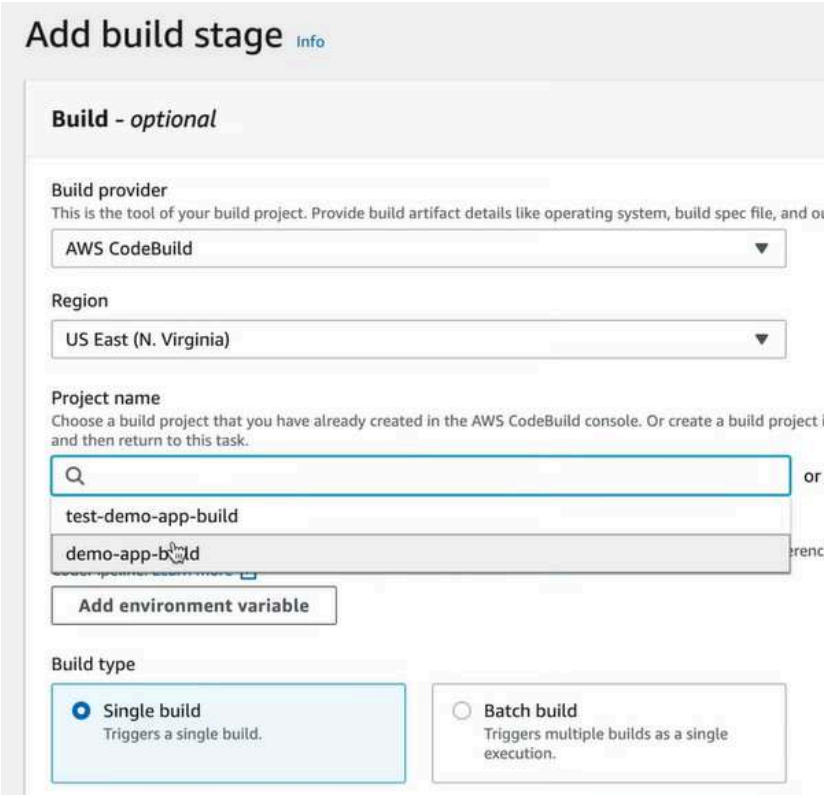


➢ In **Repository name,** select the repository which you have created, then select **Branch name** as **Master,** in **change detection options,** Select **AWS CodePipeline,** which is used to detect the changes in the CodeCommit and Push it into Codepipeline, Then Click on **Next.**



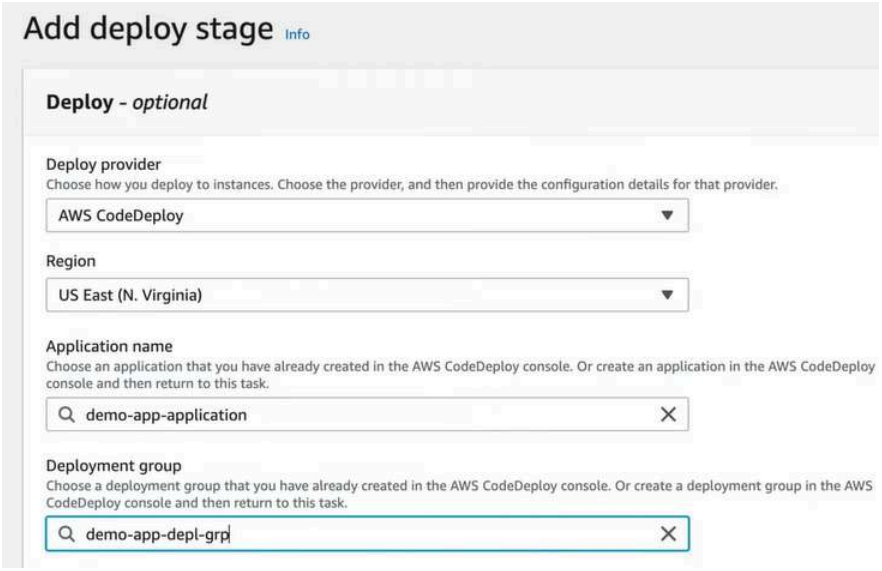
Step 3: Add build stage

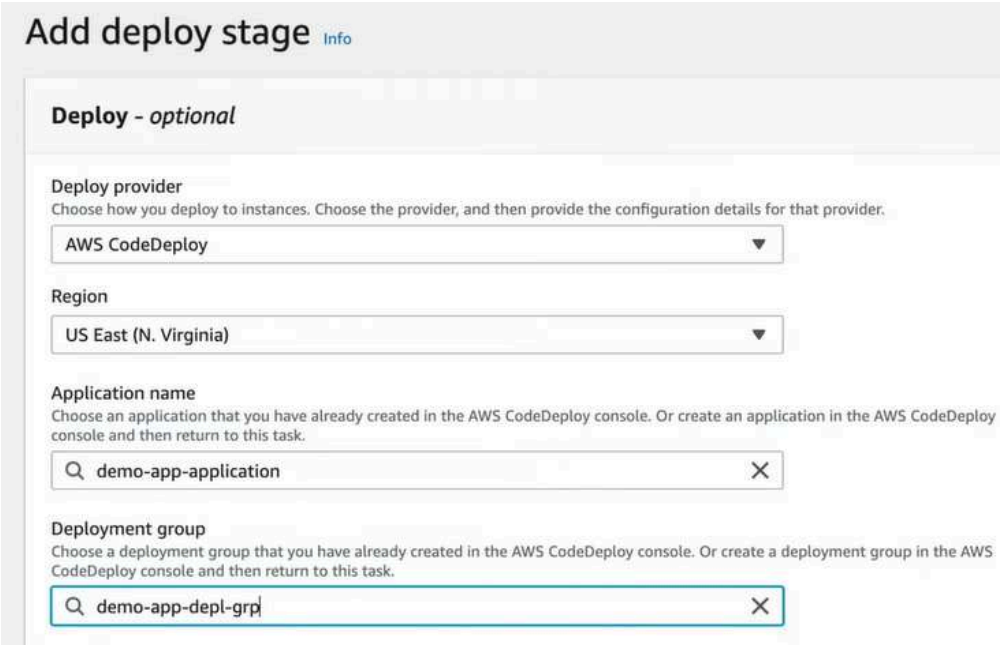
- In **Build provider,** Choose **AWS Codebuild** in **project name sel**ect the project name, then Click on **Next.**



Step 4: Add deploy stage

- In **Deploy provider,** select the **provider name,** Now Select the **application name ,** then Select the **Deployment group,** now Click on **Next.**





Step 5: Review

Review Info

Step 1: Choose pipeline settings

Pipeline settings

Pipeline name
demo-app-pipeline

Artifact location
codepipeline-us-east-1-355945017600

Service role name
AWSCodePipelineServiceRole-us-east-1-demo-app-pipeline

Step 2: Add source stage

Source action provider

Source action provider
AWS CodeCommit

RepositoryName
demo-app

BranchName
master

PollForSourceChanges
true

OutputArtifactFormat
CODE_ZIP

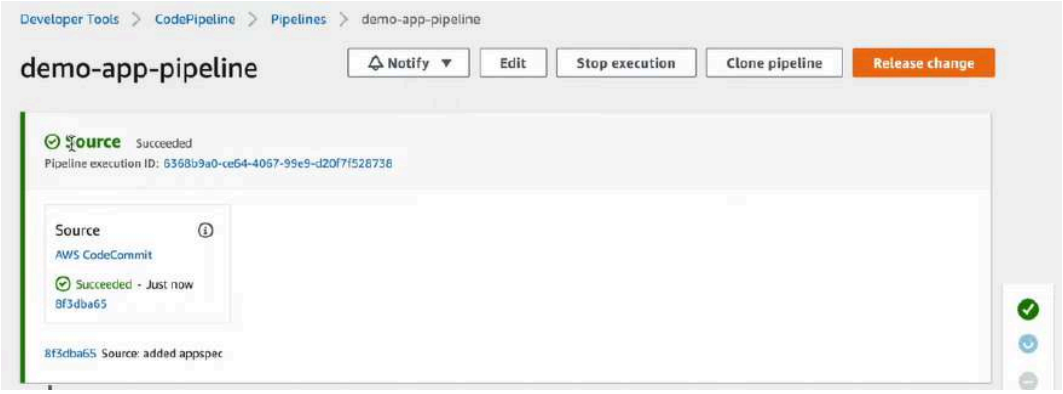
Step 3: Add build stage

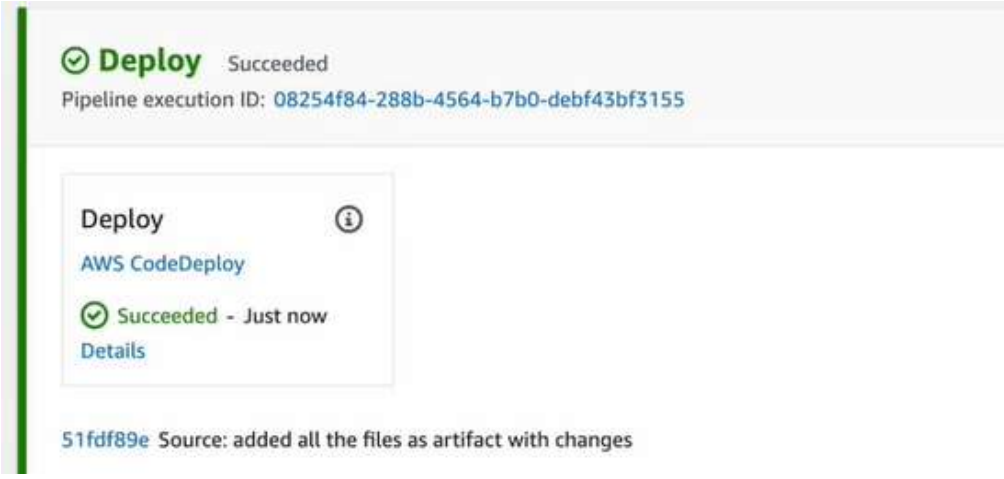
Build action provider

Build action provider
AWS CodeBuild

ProjectName
demo-app-build

10.2 After the pipeline is created Successfully, CodeCommit, CodeBuild, CodeDeploy will be started Automatically.





Conclusion:

- By using the AWS CodePipeline which is a powerful tool for automating the software release process, enabling development teams to deliver high-quality software faster and more reliably.